

TRANSIENT IMAGING

Computational Imaging – Spring 2024

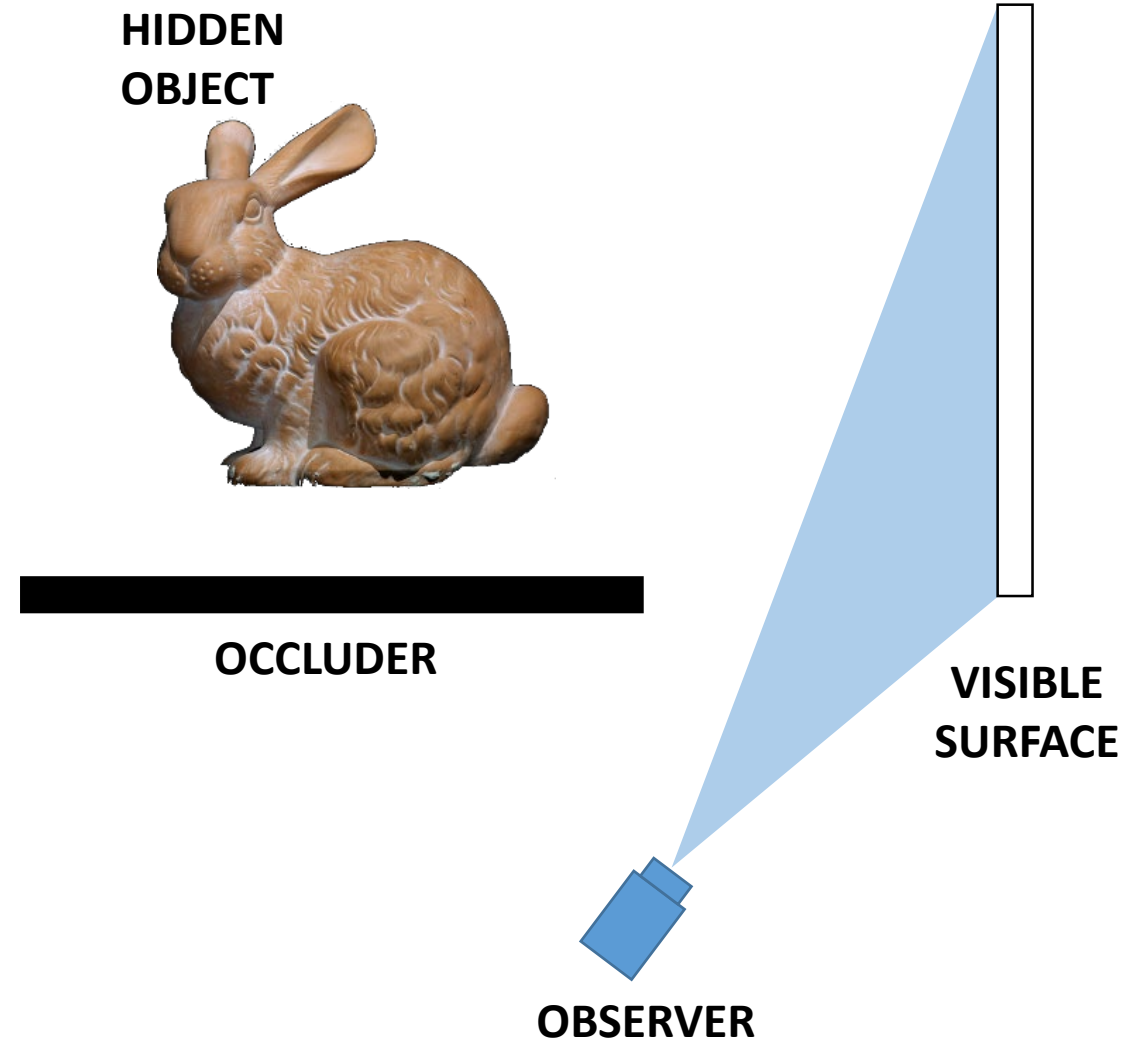
Julio Marco, Ana Serrano, Albert Redo-Sanchez

Master in Robotics, Graphics and Computer Vision – Universidad de Zaragoza

Non-Line-Of-Sight imaging

What is Non-Line-Of-Sight (NLOS) imaging?

Seeing scenes totally or partially occluded from us

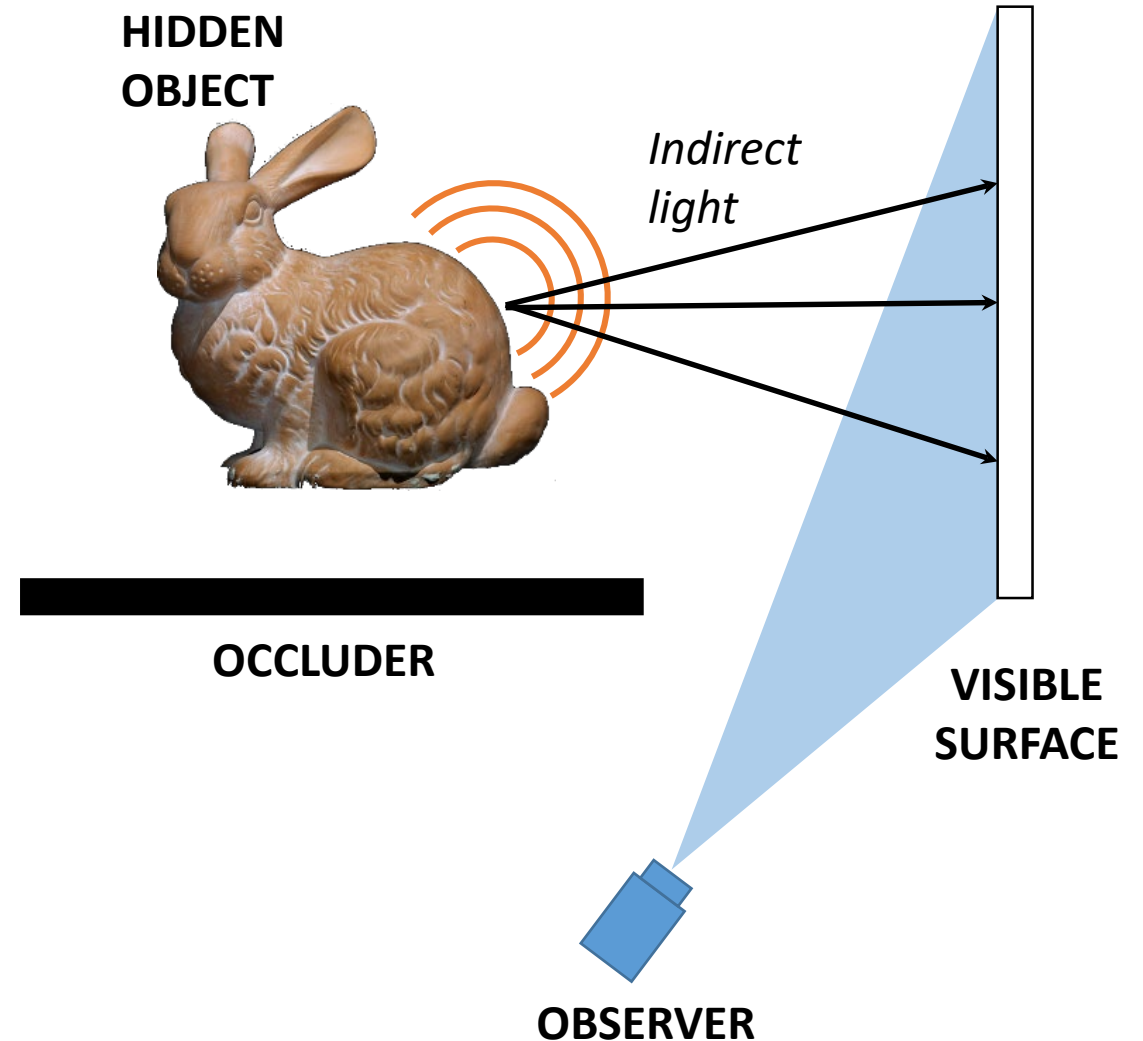


What is NLOS imaging?

Seeing scenes totally or partially occluded from us

Key principle:

Analyze indirect illumination measured at visible elements



What is NLOS imaging?

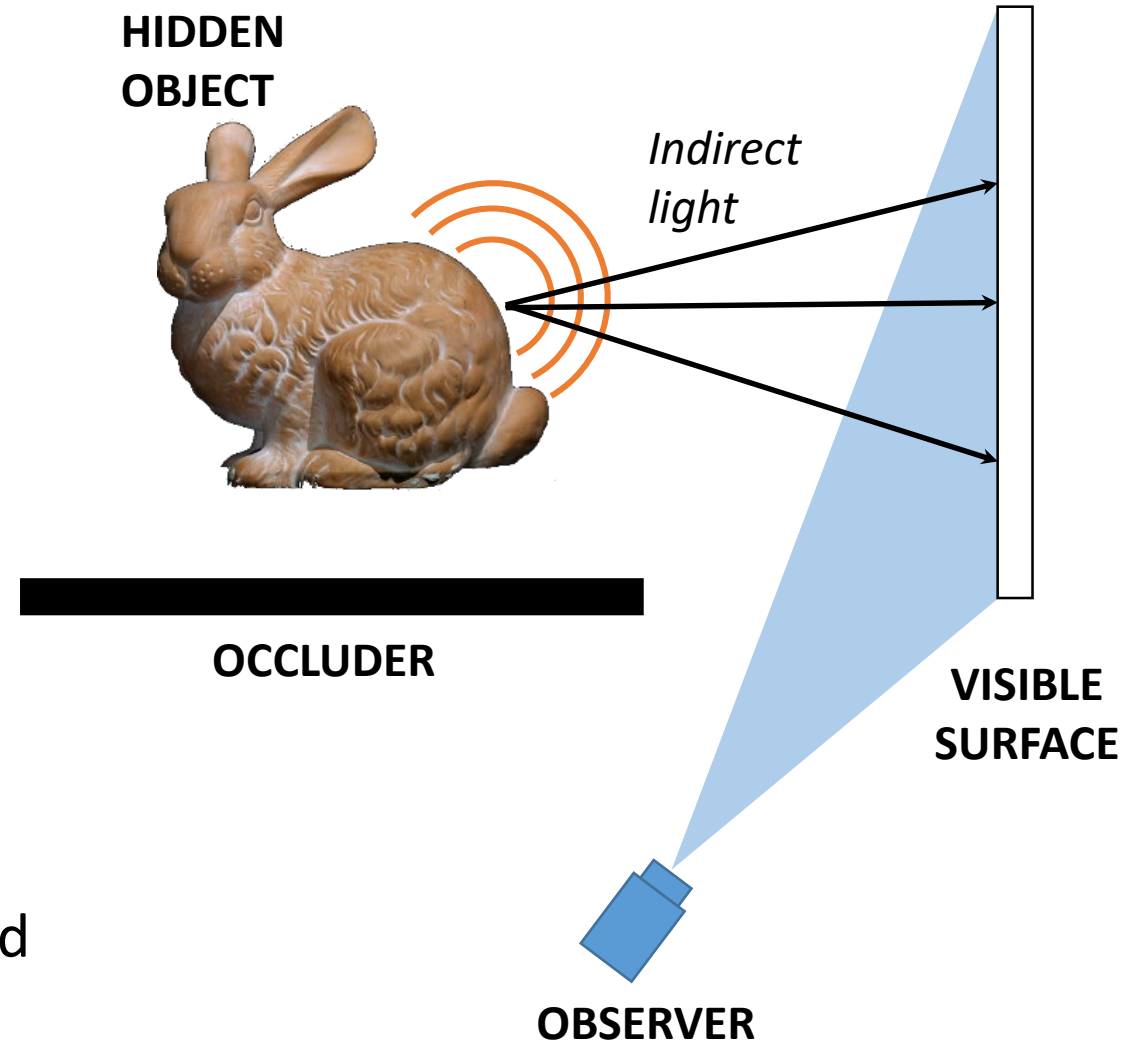
Seeing scenes totally or partially occluded from us

Key principle:

Analyze indirect illumination measured at visible elements

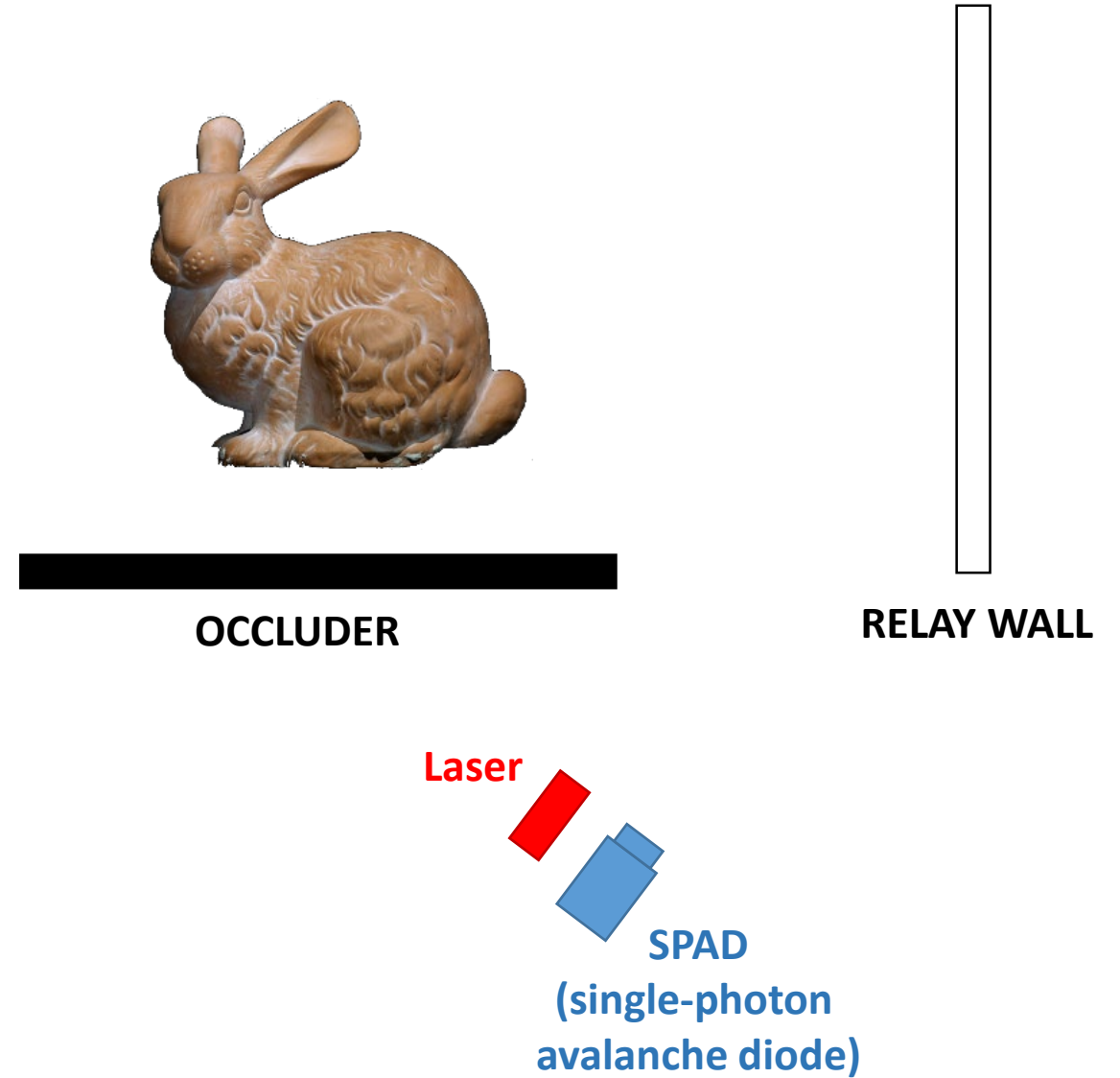
Many techniques:

- Time-resolved vs. steady-state
- Active vs. passive systems
- Wave domain, spatial domain, gradient-based



Time-gated NLOS imaging

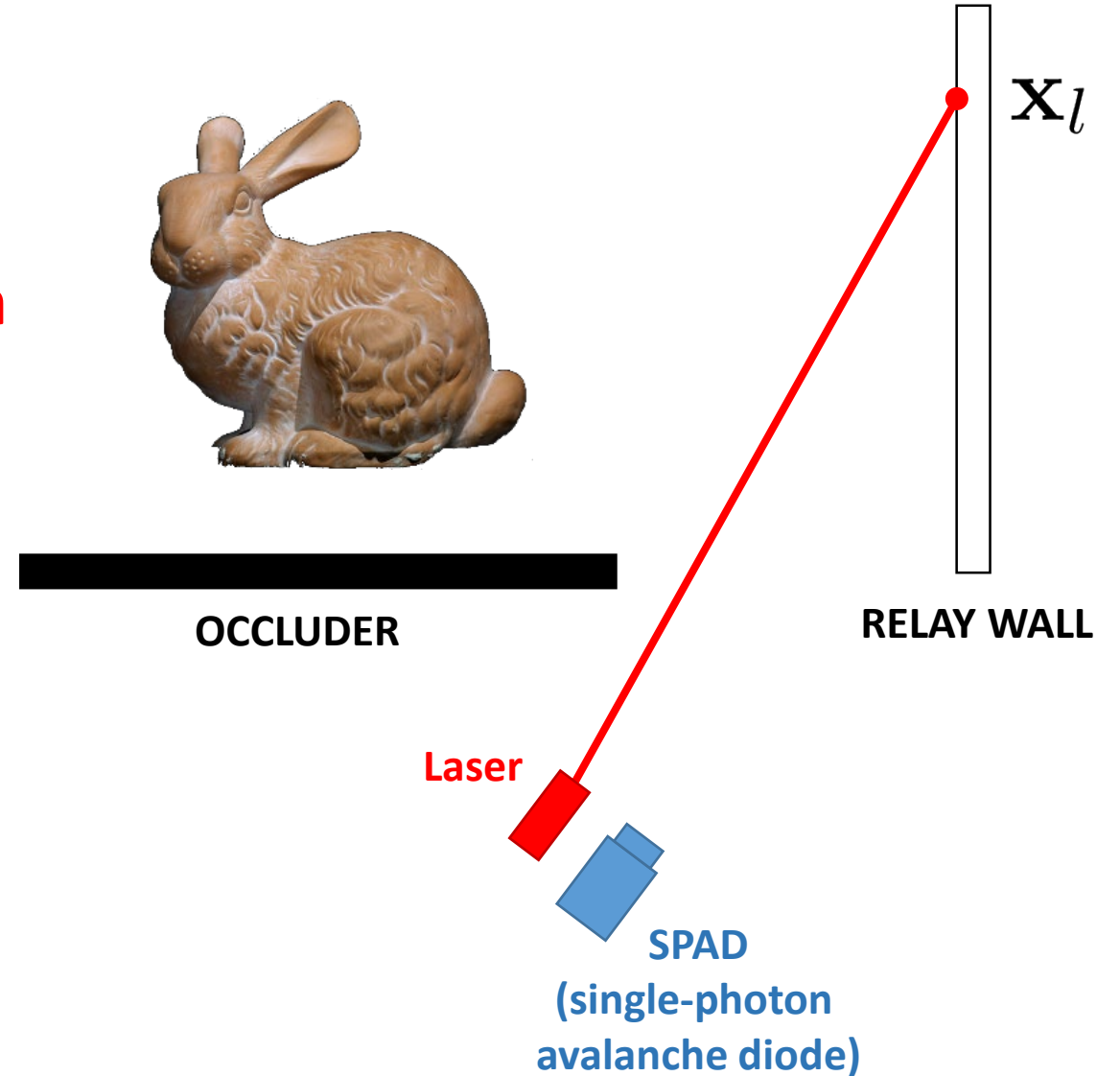
Usual capture setup:



Time-gated NLOS imaging

Usual capture setup:

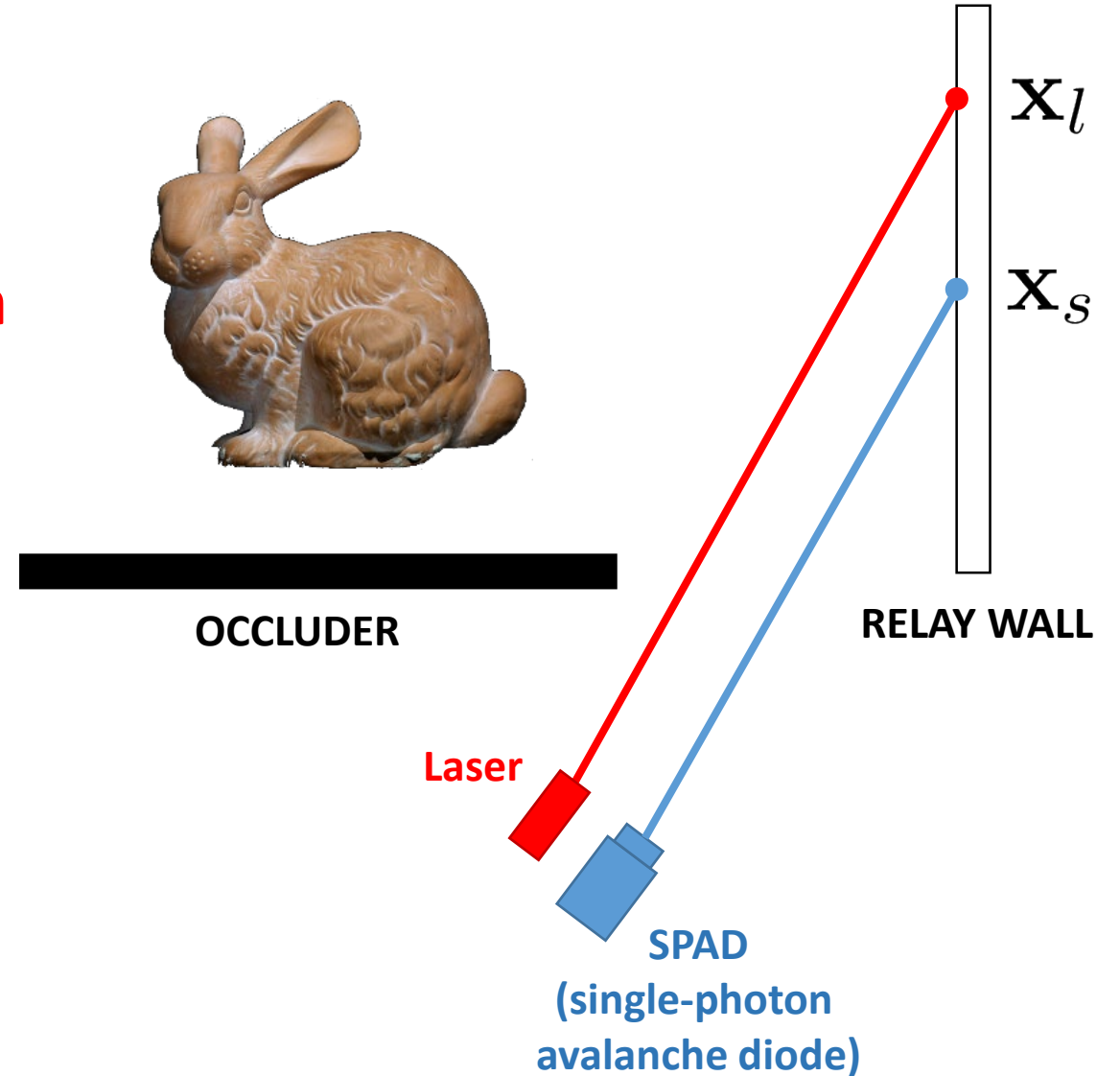
- **Laser emitting ultra-short illumination pulse at $\mathbf{x}_l$**



Time-gated NLOS imaging

Usual capture setup:

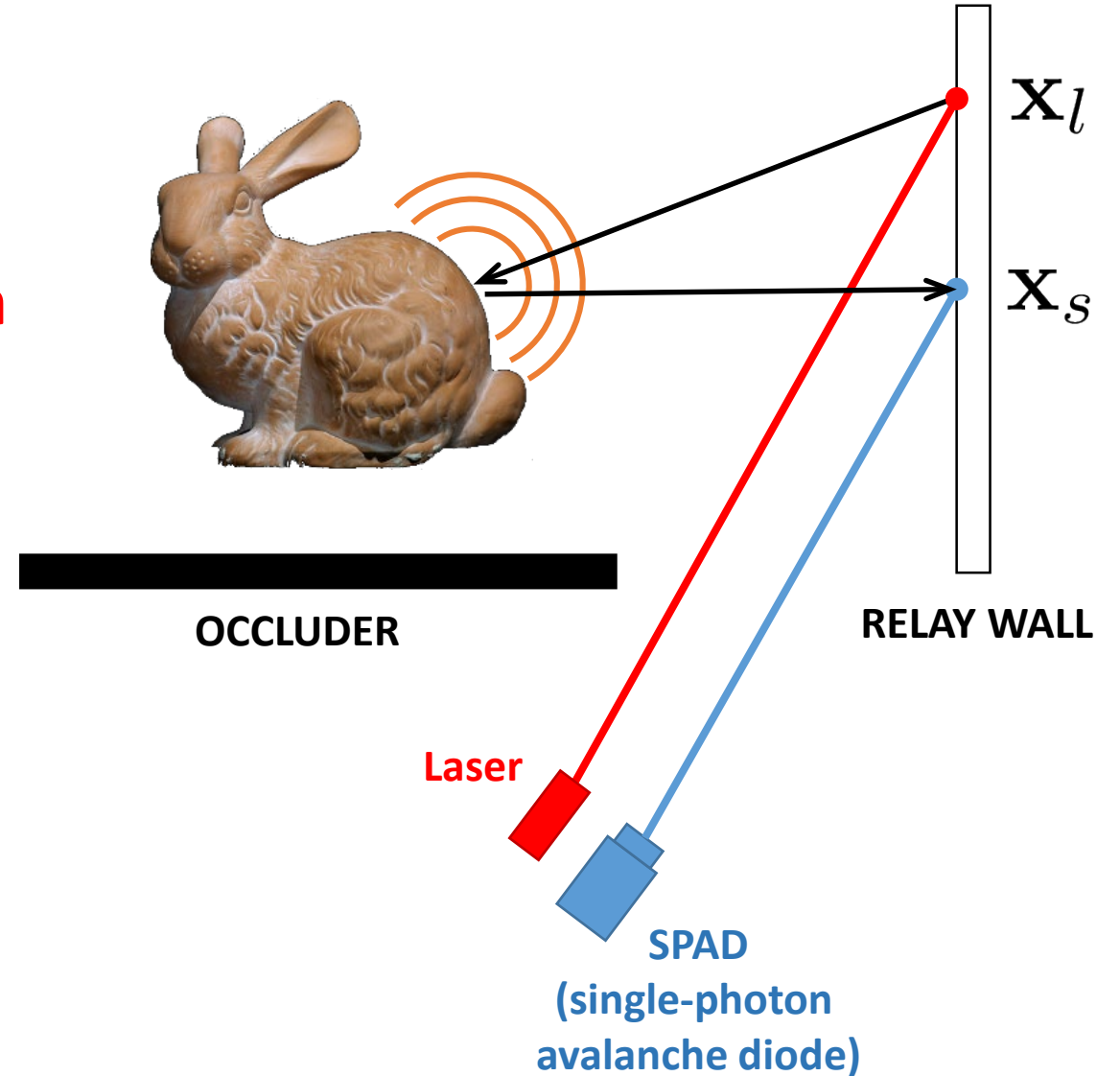
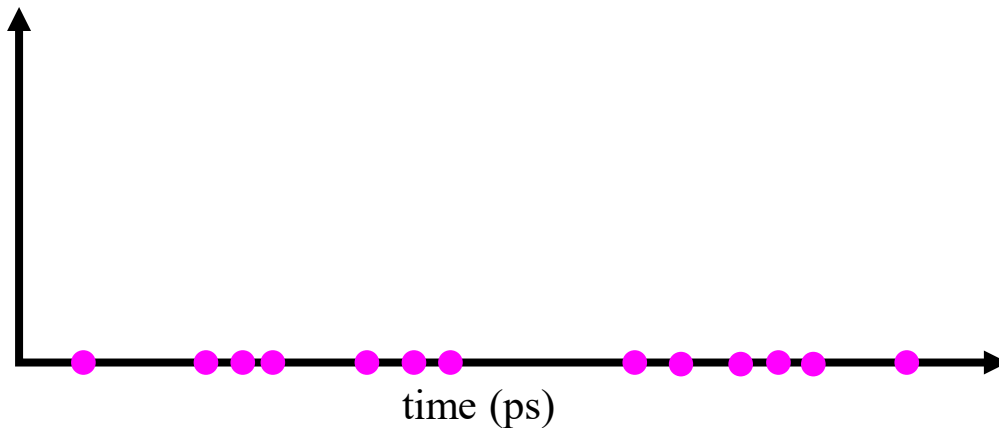
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- **SPAD measuring time of flight of arriving photons at $\mathbf{x}_s$**



Time-gated NLOS imaging

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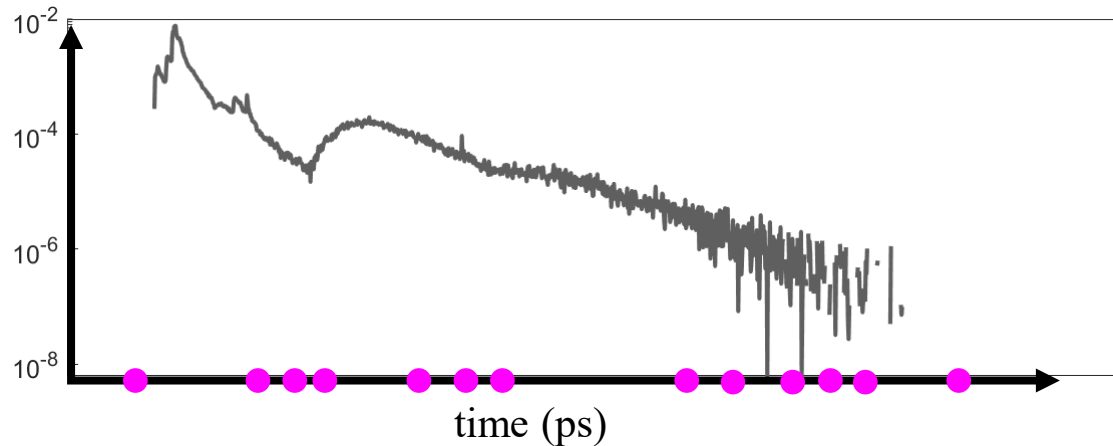
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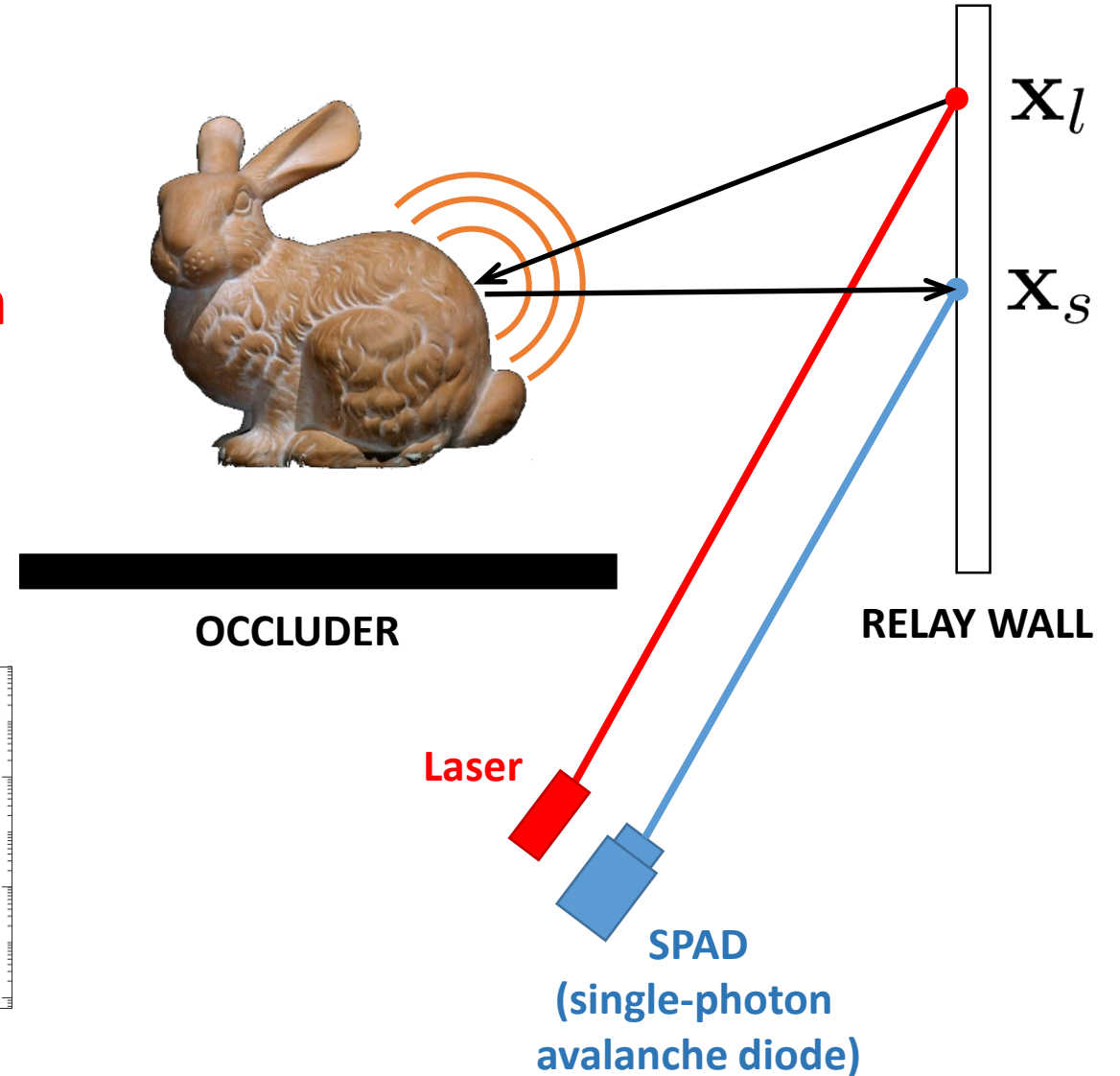
Time-gated NLOS imaging

Usual capture setup:

- **Laser emitting ultra-short illumination pulse at $\mathbf{x}_l$**
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Impulse response function $H(\mathbf{x}_l, \mathbf{x}_s, t)$

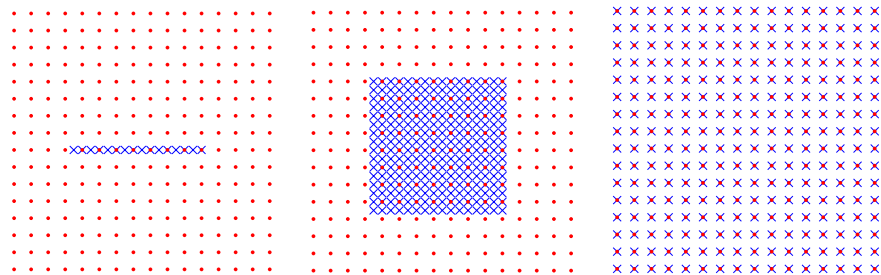


Time-gated NLOS imaging

Usual capture setup:

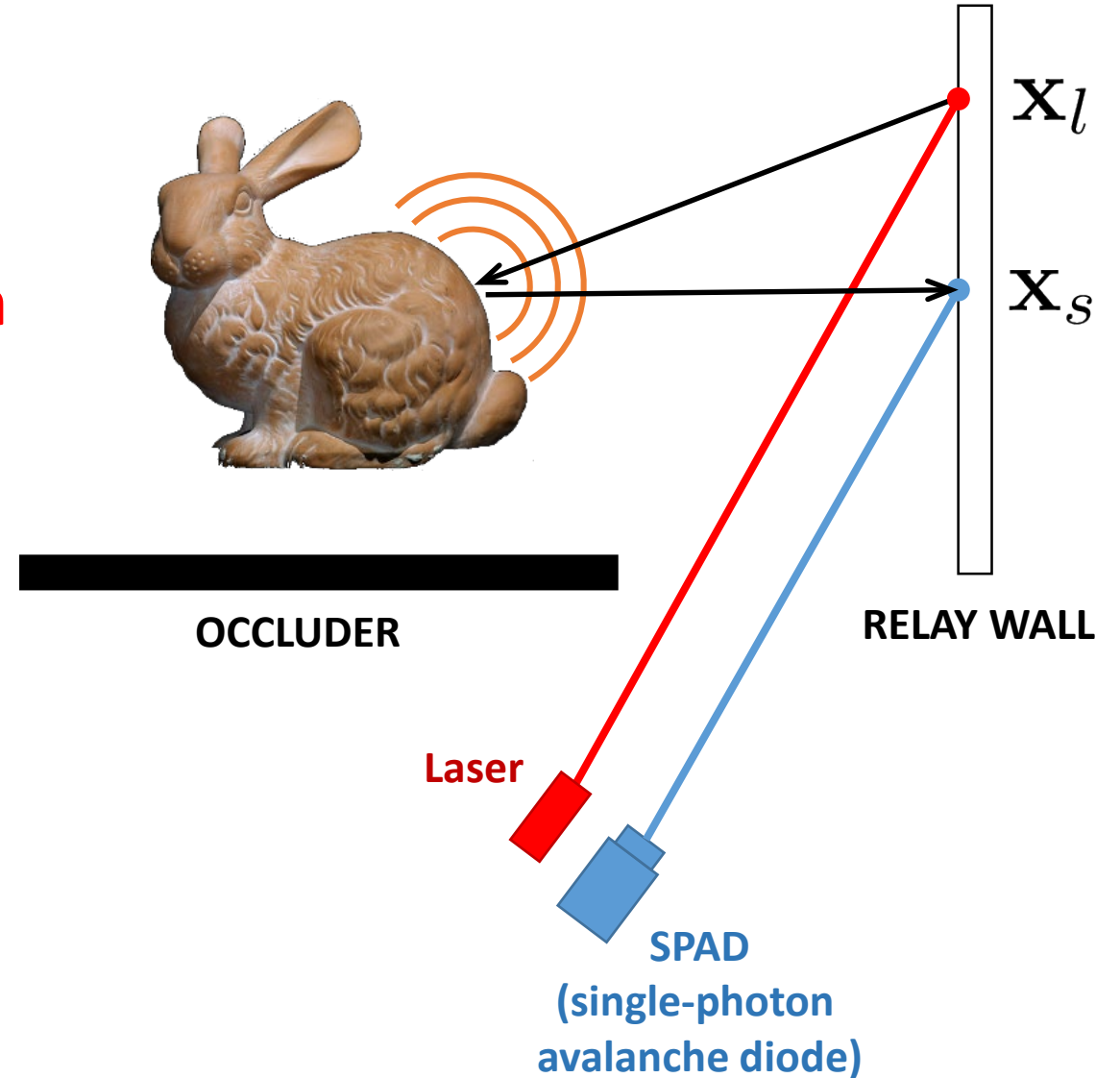
- **Laser emitting ultra-short illumination pulse at $\mathbf{x}_l$**
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Scan SPAD and/or laser on the relay



● L Laser positions $\mathbf{x}_l$

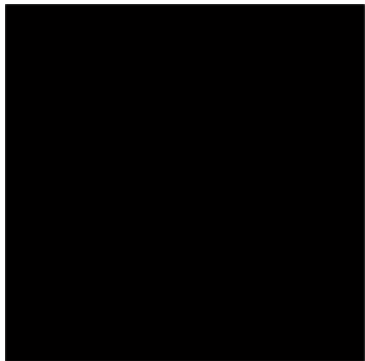
× S SPAD positions $\mathbf{x}_s$



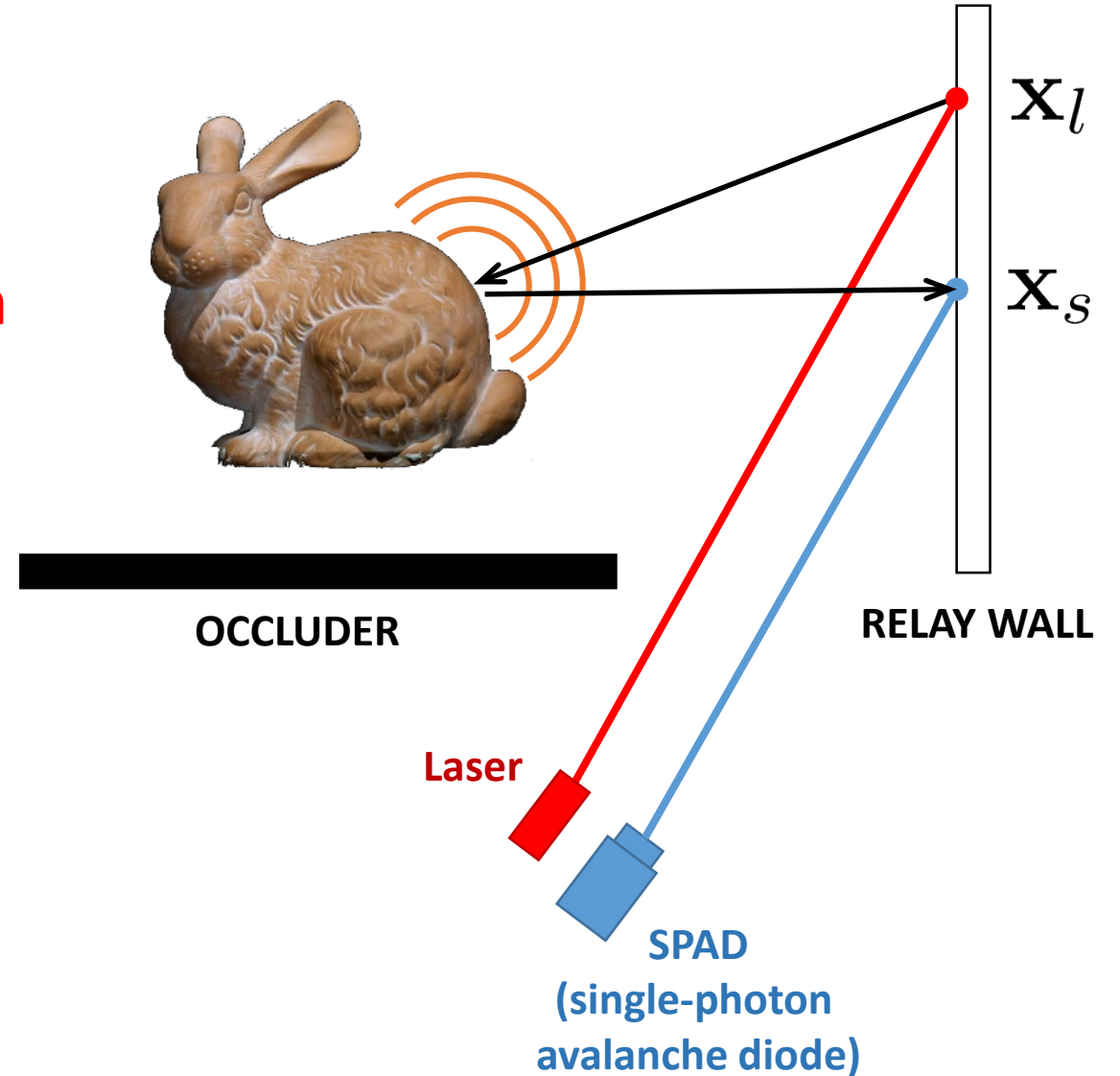
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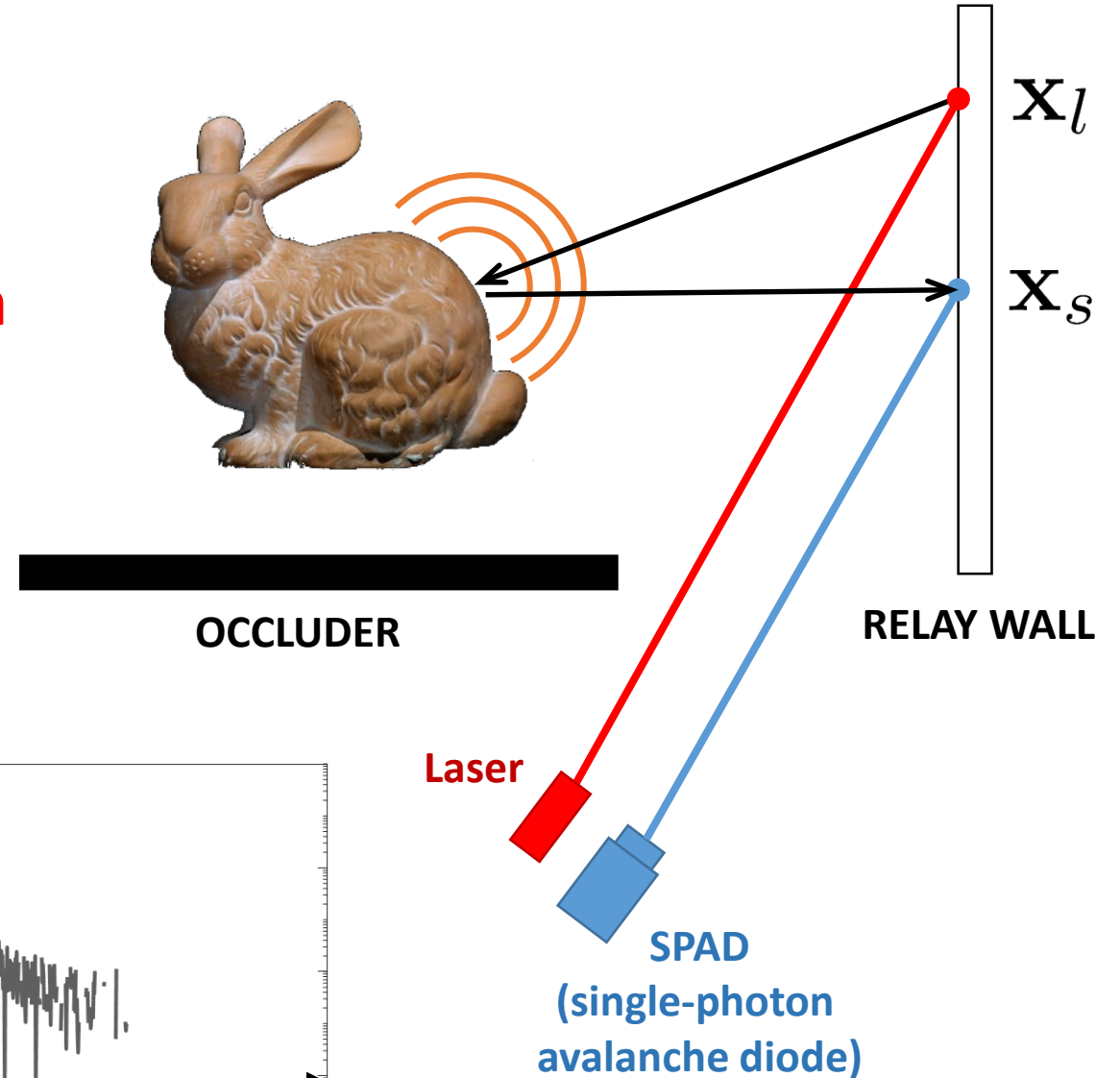
Transport at wall



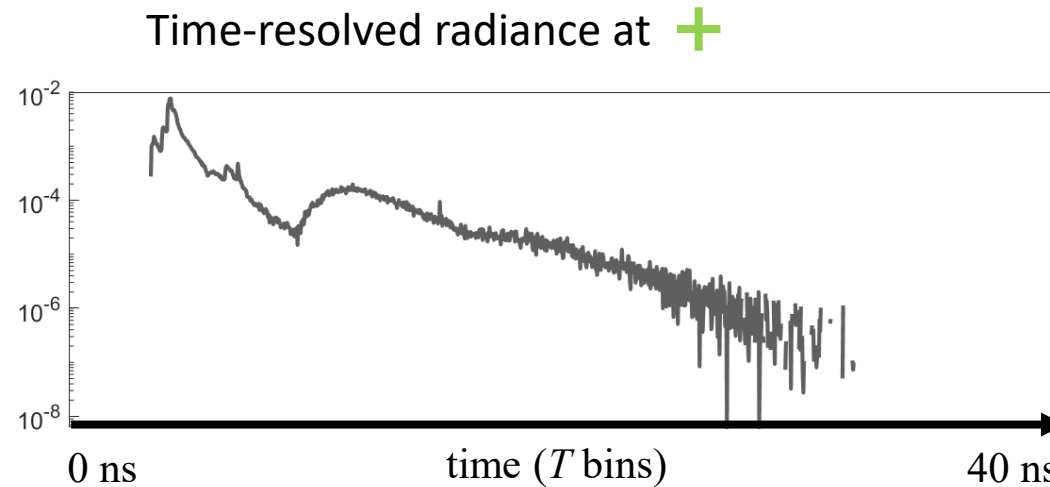
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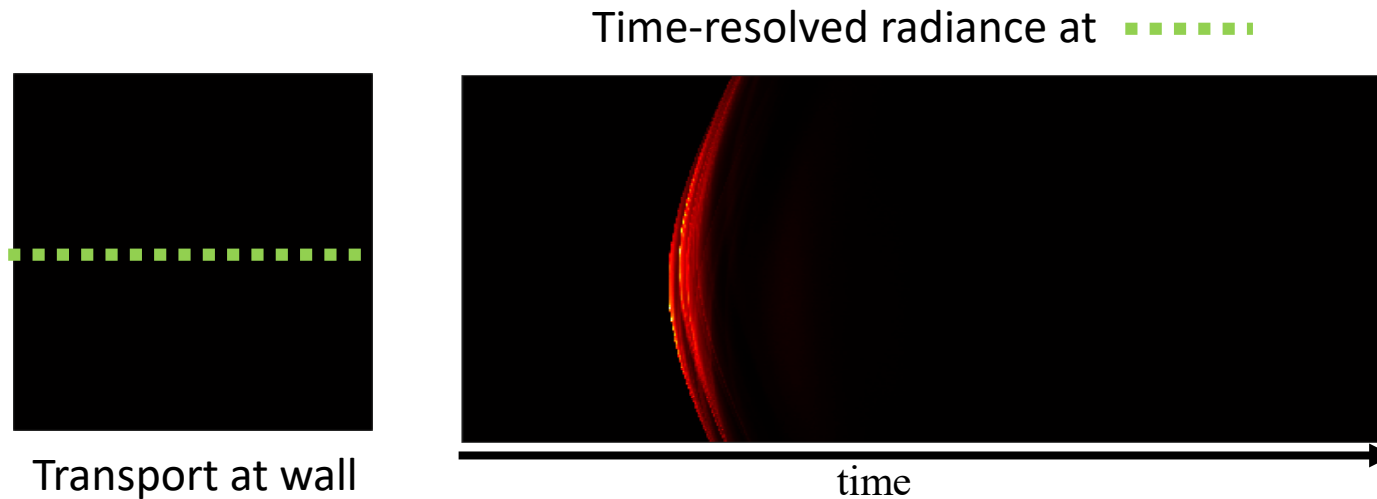
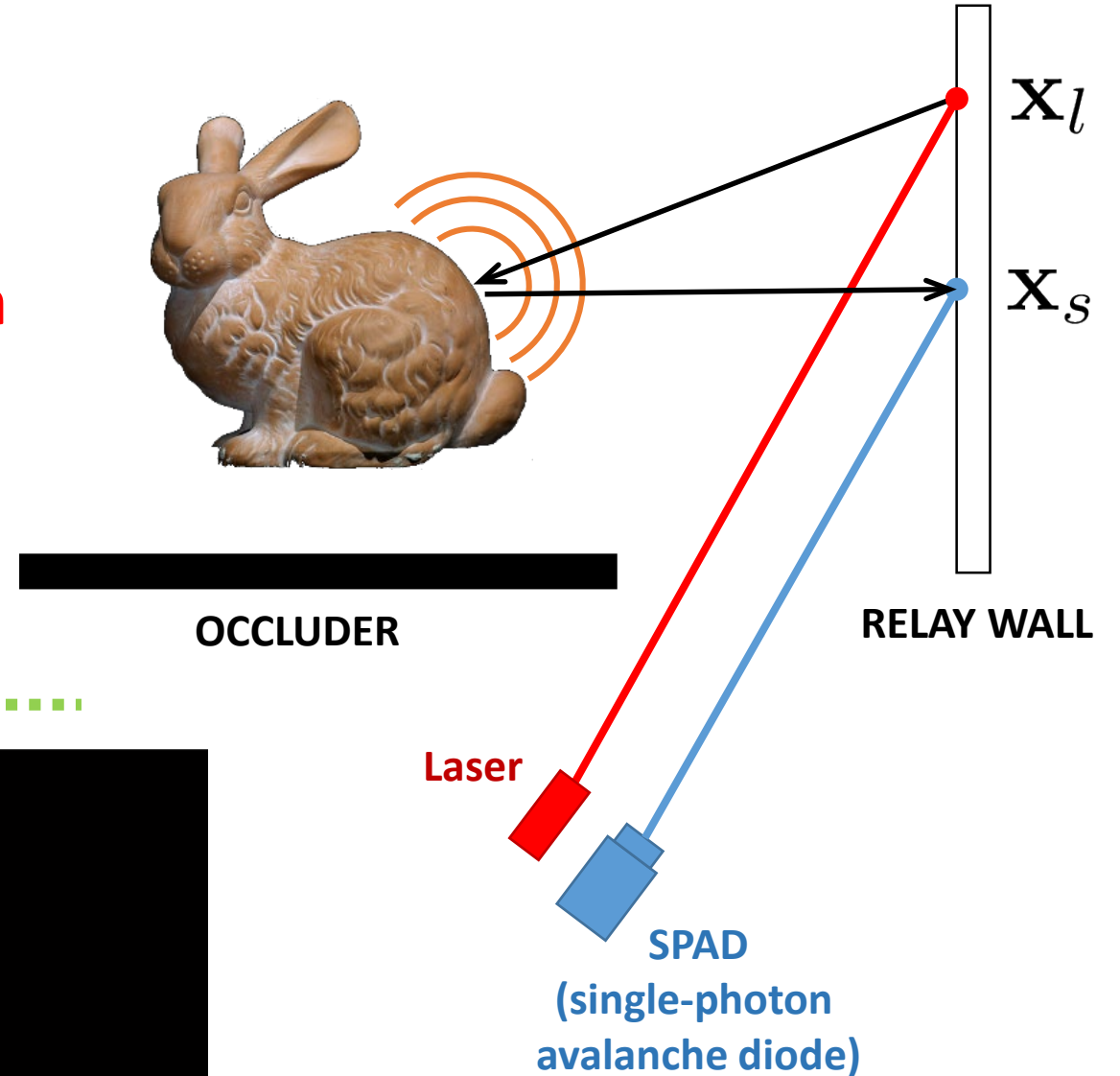
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Time-gated NLOS imaging

Usual capture setup:

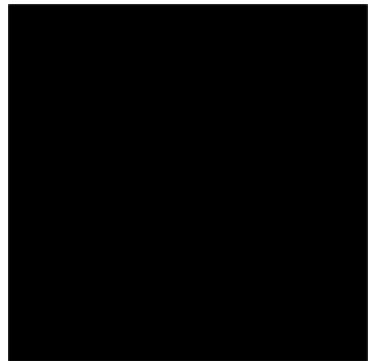
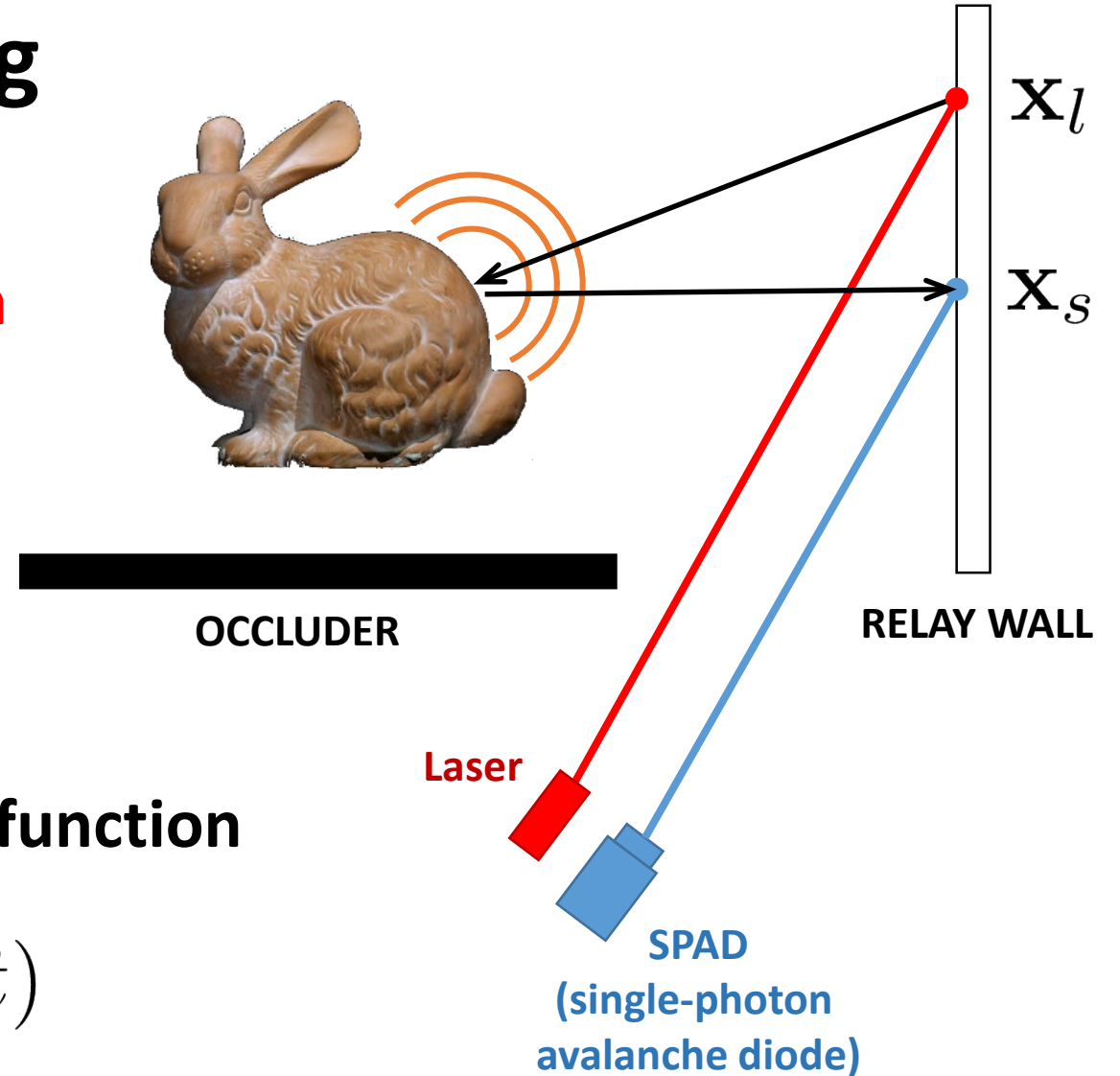
- **Laser emitting ultra-short illumination pulse at $\mathbf{x}_l$**
- **SPAD measuring time of flight of arriving photons at $\mathbf{x}_s$**



Time-resolved NLOS imaging

Usual capture setup:

- **Laser emitting ultra-short illumination pulse at $\mathbf{x}_l$**
- **SPAD measuring time of flight of arriving photons at $\mathbf{x}_s$**



Transport at wall

Impulse response function

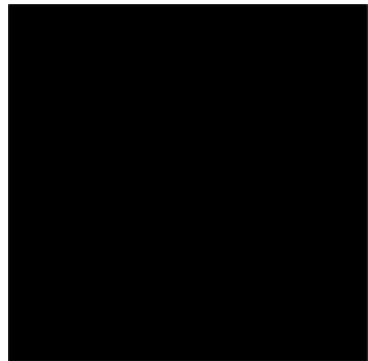
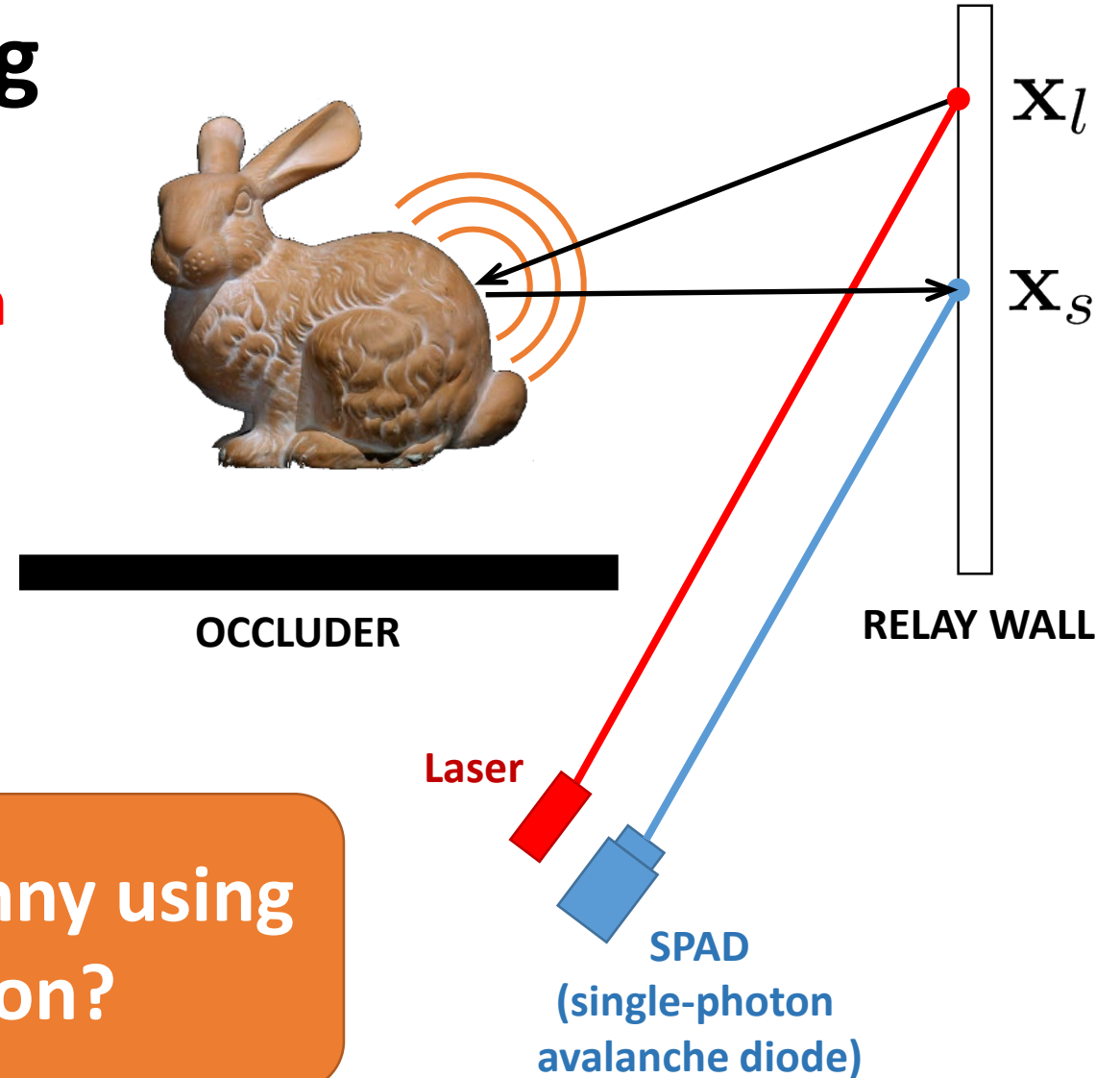
$$H(\mathbf{x}_l, \mathbf{x}_s, t)$$

Size: $L \times S \times T$

Time-resolved NLOS imaging

Usual capture setup:

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Transport at wall

How to see the bunny using this information?

NLOS imaging: Backprojection methods

[\[Velten et al. 2012\]](#)

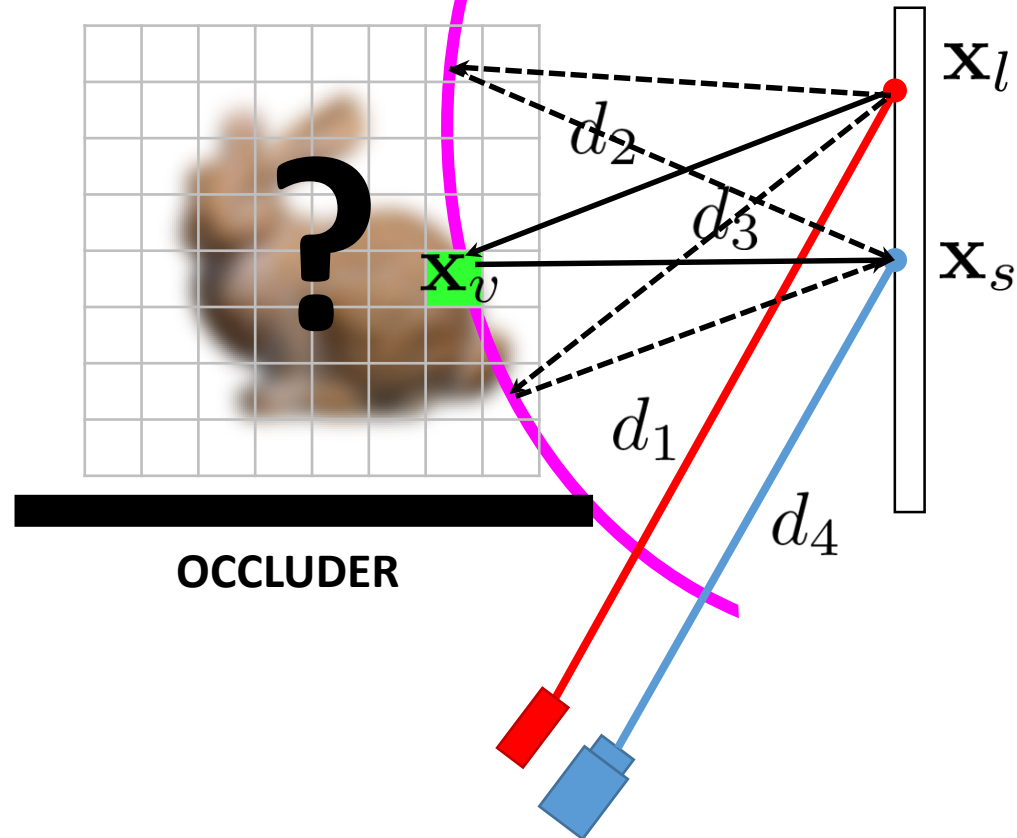
[\[Laurenzis and Velten 2014\]](#)

[\[Buttafava et al. 2015\]](#)

[\[Arellano et al. 2017\]](#)

[\[La Manna et al. 2018\]](#)

Backprojection methods



Triangulate position of hidden voxels

3rd-bounce propagation time:

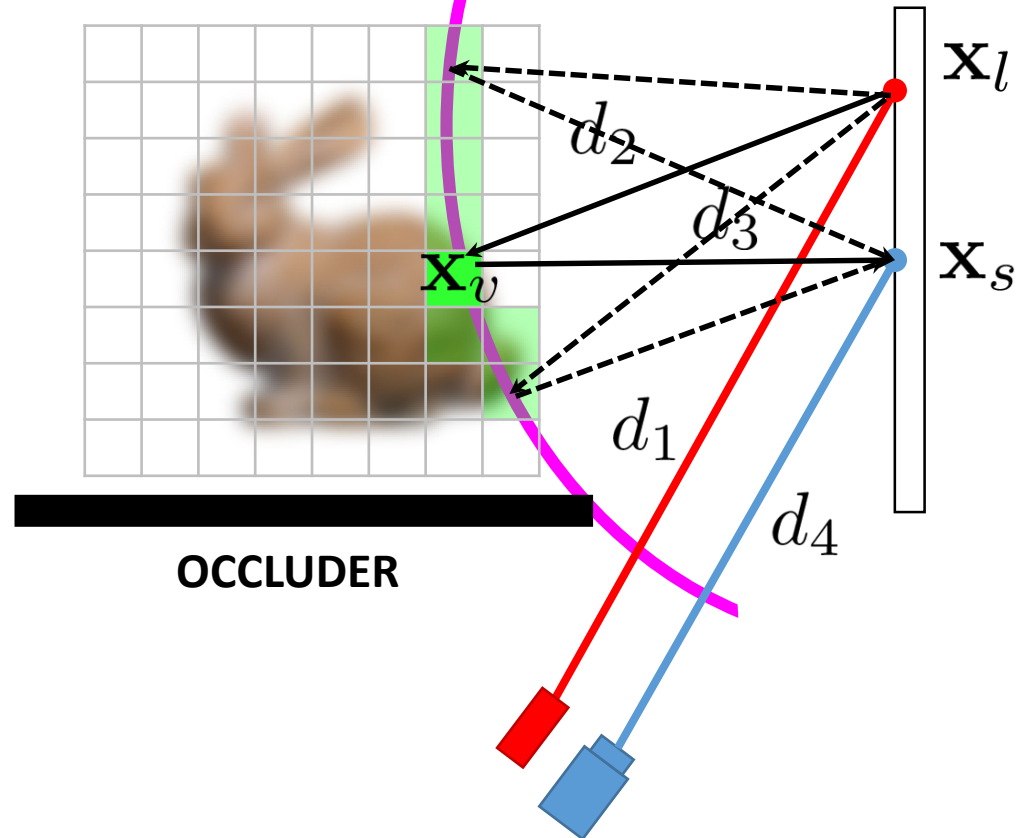
$$t_{lv} = \frac{d_1 + d_2}{c}, \quad t_{sv} = \frac{d_3 + d_4}{c}$$

Estimate scene f as

$$f(\mathbf{x}_v) = \iint_S \int_L H(\mathbf{x}_l, \mathbf{x}_s, t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

PROBLEM: Spatial ambiguities

Backprojection methods



Triangulate position of hidden voxels

3rd-bounce propagation time:

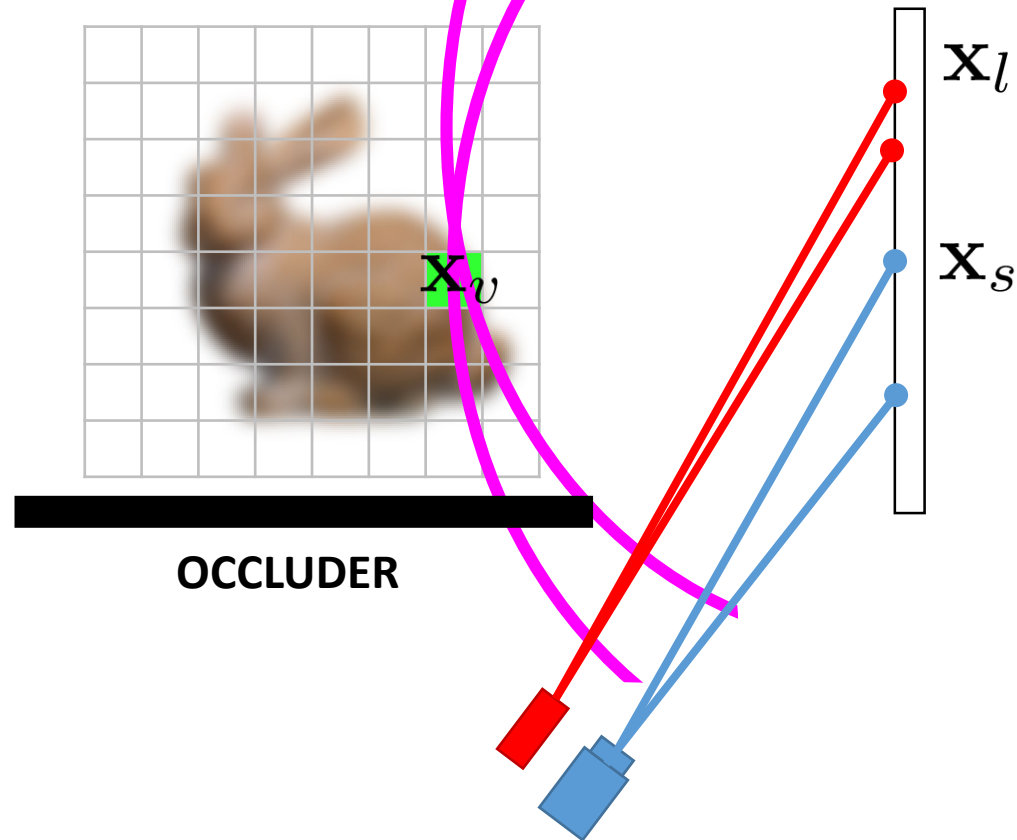
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PROBLEM: Spatial ambiguities

Backprojection methods



Triangulate position of hidden voxels

3rd-bounce propagation time:

$$t_{lv} = \frac{d_1 + d_2}{c}, \quad t_{sv} = \frac{d_3 + d_4}{c}$$

Filter f or H to enhance high frequencies:

$$f(\mathbf{x}_v) = K(\mathbf{x}_v) * \int_S \int_L H(\mathbf{x}_l, \mathbf{x}_s, t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

$$f(\mathbf{x}_v) = \int_S \int_L \mathbf{H}'(\mathbf{x}_l, \mathbf{x}_s, t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

$\xrightarrow{\hspace{1cm}} H' = H * K$

Backprojection flavors

- Iterative methods, error backprojection
- Change the filtering:
 - Laplacian
 - Laplacian of Gaussian
 - Optimized by least squares (Gram operators)
- Confocal-only methods: Faster, convolutional.
 - Light Cone Transform
 - fk-migration: Exact inverse, wave-based.
- All of them based on the same operation:

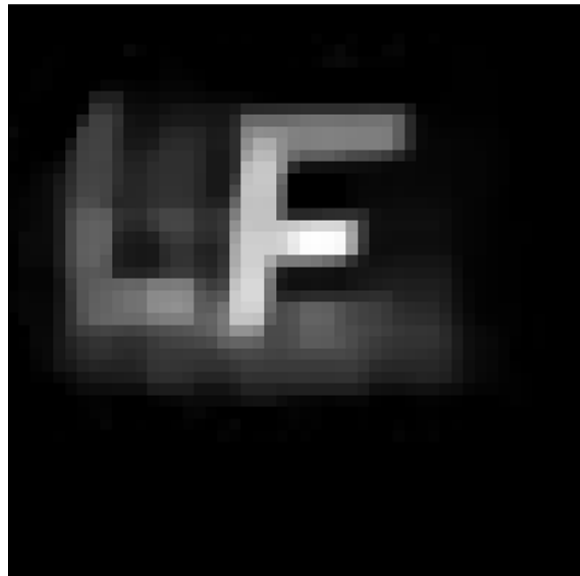
**Backproject (i.e. invert) light
transport to the hidden scene**

$$f(\mathbf{x}_v) = \iint_S \int_L H'(\mathbf{x}_l, \mathbf{x}_s, t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

Backprojection results



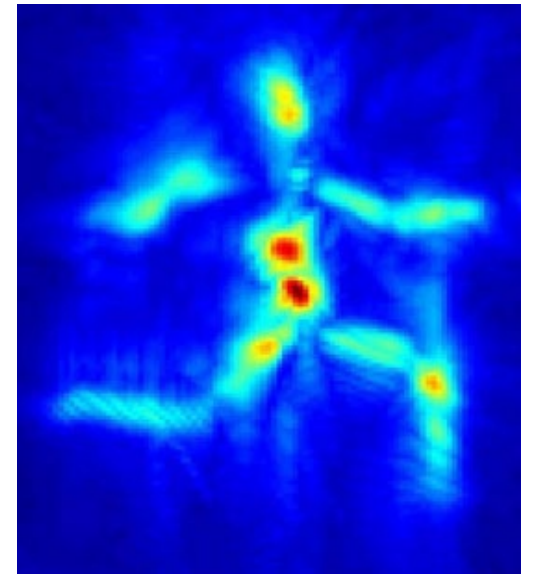
Hidden scene



Reconstructed image



Hidden scene



Reconstructed image

Backprojection results

- Good at: Simple scenes, isolated objects



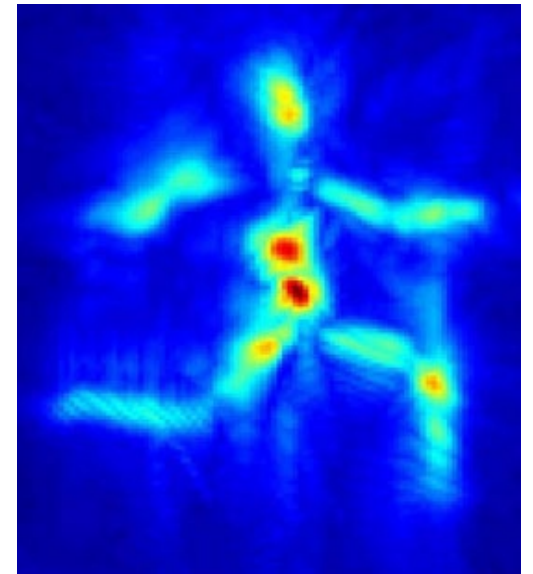
Hidden scene



Reconstructed image



Hidden scene



Reconstructed image

Backprojection results

- Good at: Simple scenes, isolated objects
- **Challenges: Complex geometry, global illumination, large capture baseline**



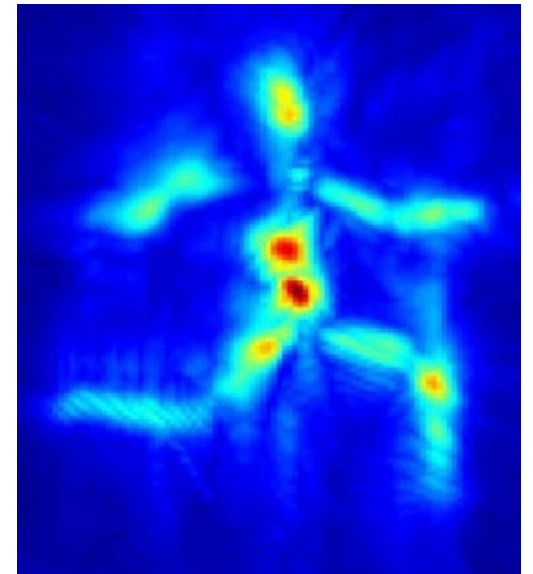
Hidden scene



Reconstructed image



Hidden scene



Reconstructed image



Questions

NLOS imaging: Virtual wave propagation

[\[Liu et al. 2019\]](#) (Nature)

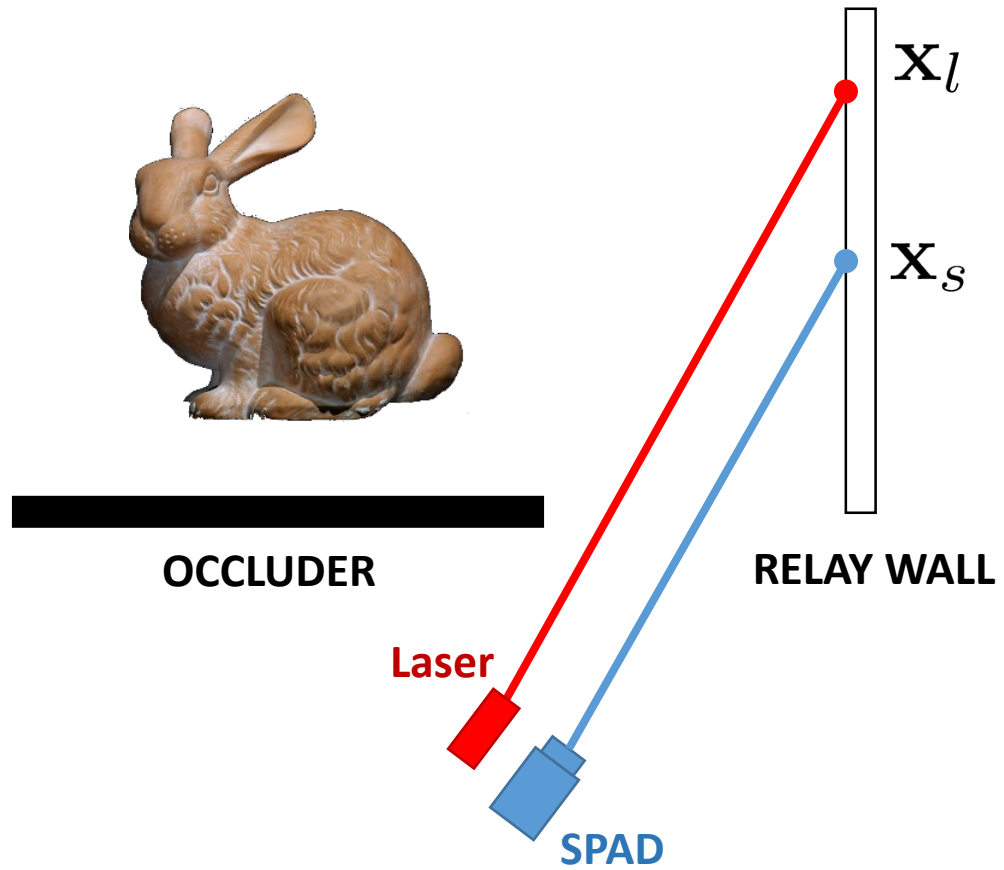
[\[Lindell et al. 2019\]](#) (SIGGRAPH)

[\[Liu et al. 2020a\]](#) (Nature Communications)

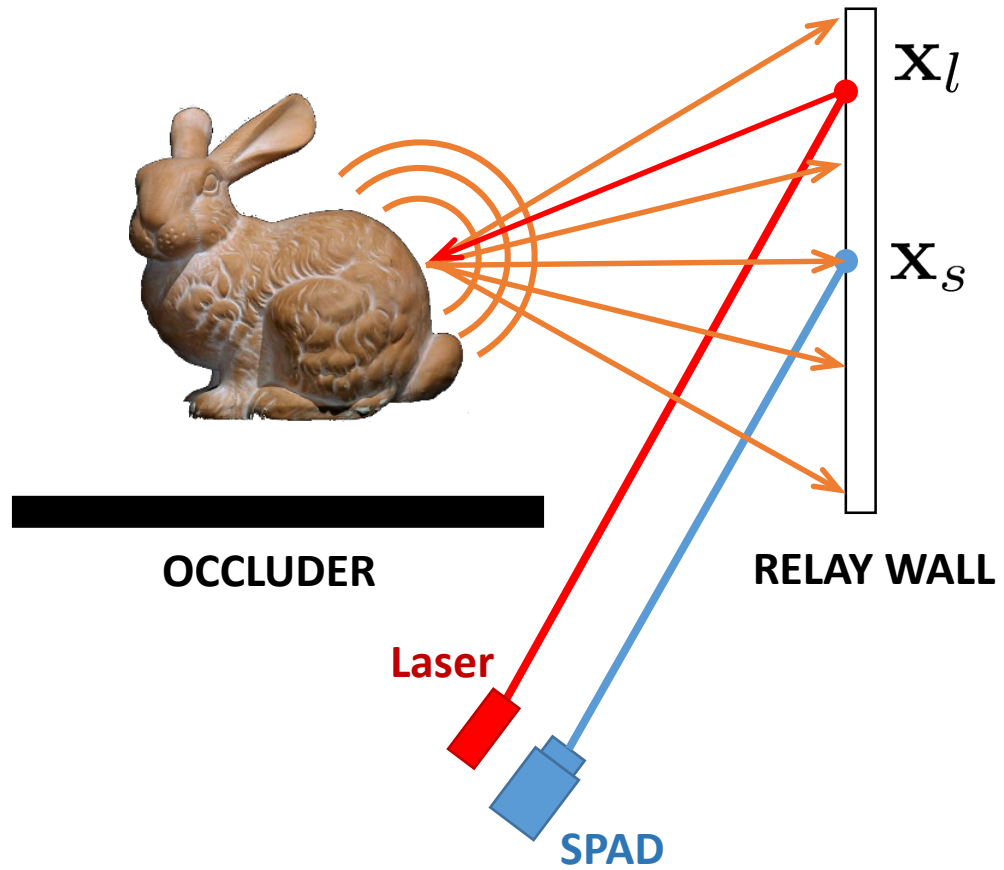
[\[Liu et al. 2020b\]](#) (ICCP)

[\[Guillén et al. 2020\]](#) (ICASSP)

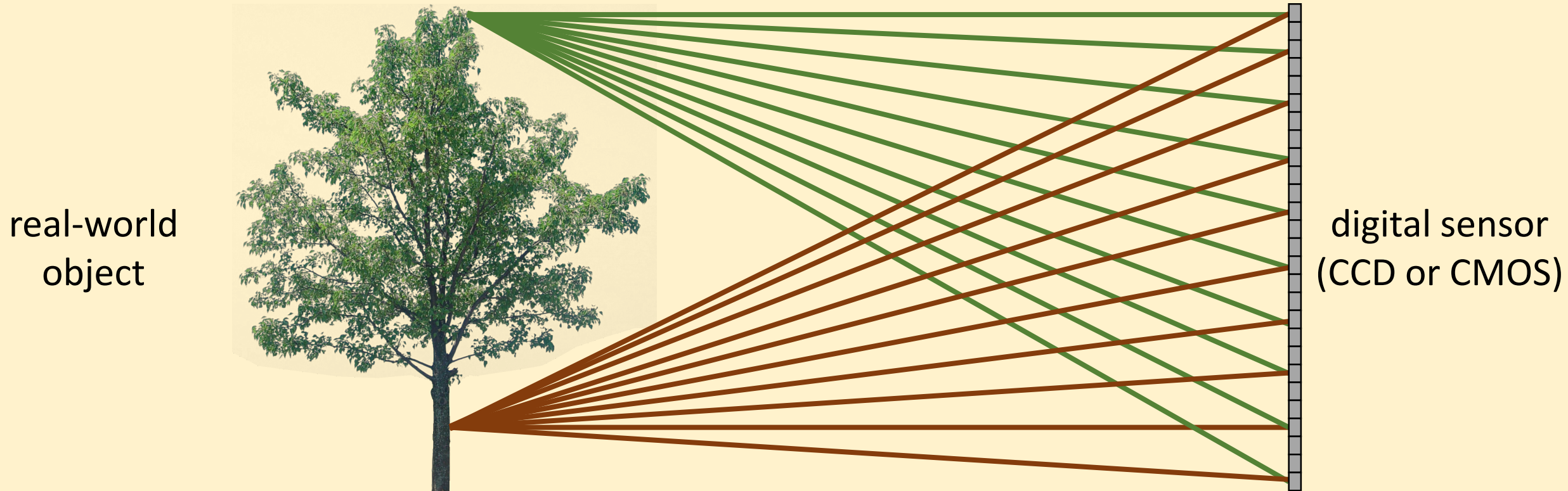
NLOS imaging



NLOS imaging



Lecture 1 brush-up: Bare-sensor imaging



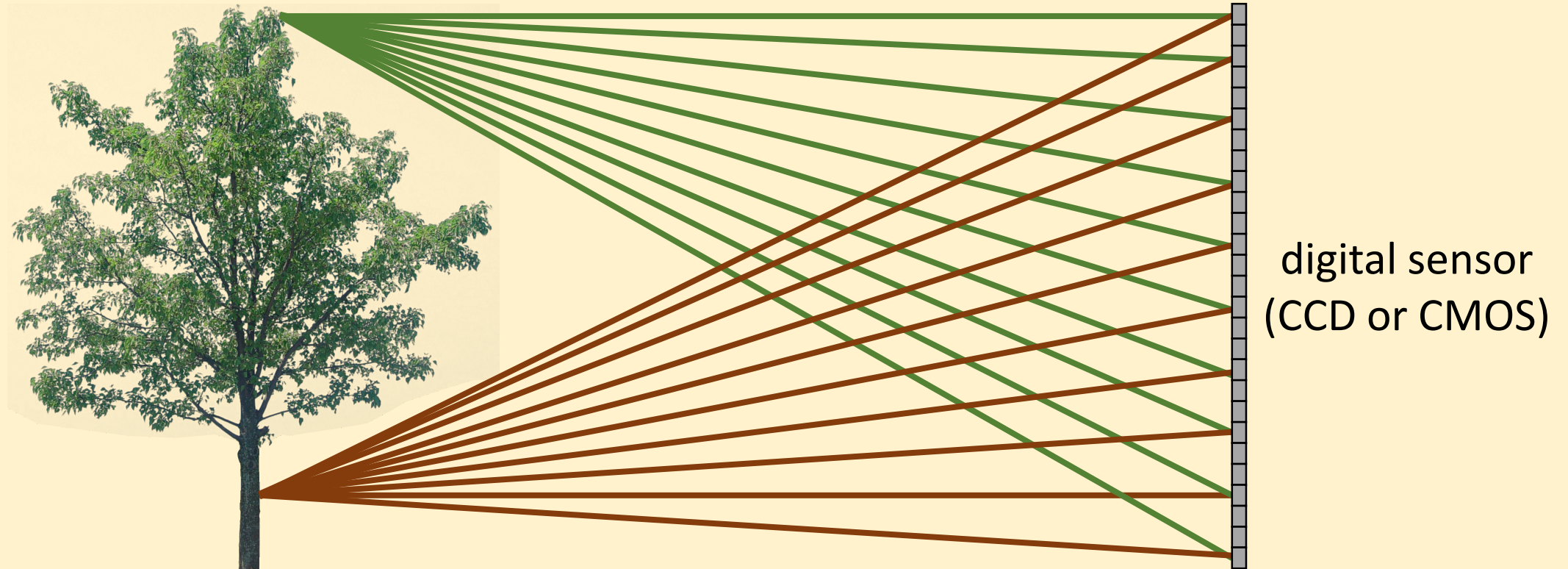
All scene points contribute to all sensor pixels

Lecture 1 brush-up: Bare-sensor imaging



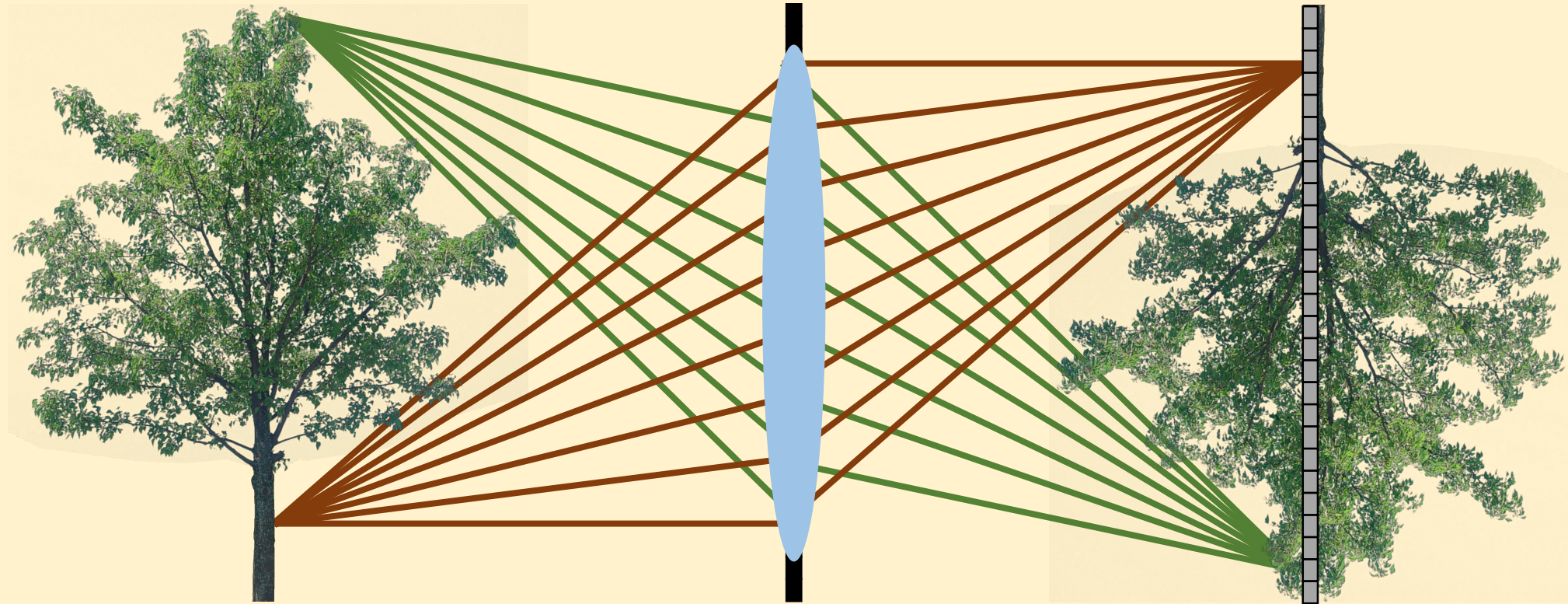
All scene points contribute to all sensor pixels

Lecture 1 brush-up: Bare-sensor imaging



All scene points contribute to all sensor pixels

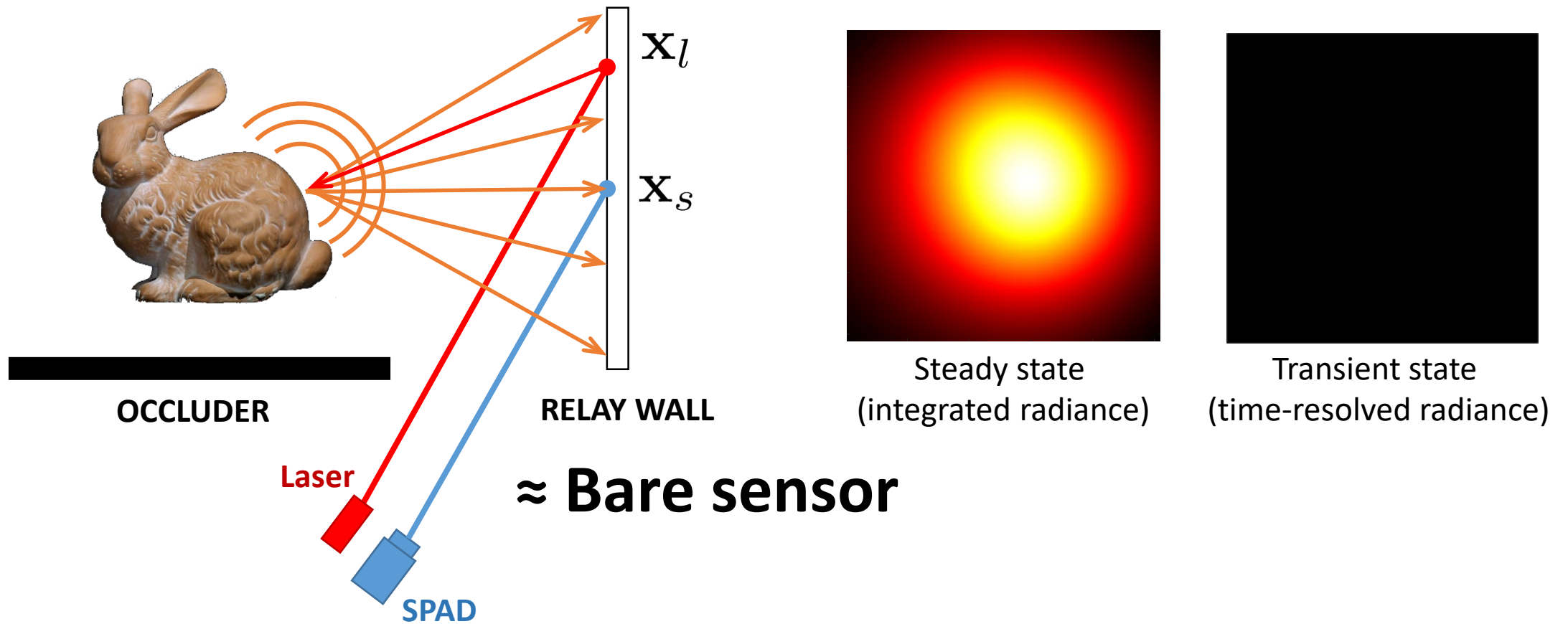
Lecture 1 brush-up: Use lenses to focus onto a sensor



Lenses map “bundles” of rays from points on the scene to the sensor.

NLOS imaging

- Relay wall behaves like a bare sensor



Backprojection vs. imaging optics

Backprojection
(primary space)

$$f(\mathbf{x}_v) = \int_S \int_L H'(\mathbf{x}_l, \mathbf{x}_s, t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

Looking around
corners

[Velten et al. 2012]

Backprojection vs. imaging optics

Backprojection
(primary space)

Looking around
corners

[Velten et al. 2012]

$$f(\mathbf{x}_v, t) = \int_S \int_L H'(\mathbf{x}_l, \mathbf{x}_s, t + t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

(at $t = 0$)

Backprojection vs. imaging optics

Backprojection
(primary space)

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(at $t = 0$)

Looking around
corners

[Velten et al. 2012]

Fourier transform (time)



Backprojection
(Fourier space)

$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

Backprojection vs. imaging optics

Backprojection
(primary space)

$$f(\mathbf{x}_v, t) = \iint_{S \ L} H'(\mathbf{x}_l, \mathbf{x}_s, t + t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

(at $t = 0$)

Looking around
corners

[Velten et al. 2012]

Fourier transform (time)

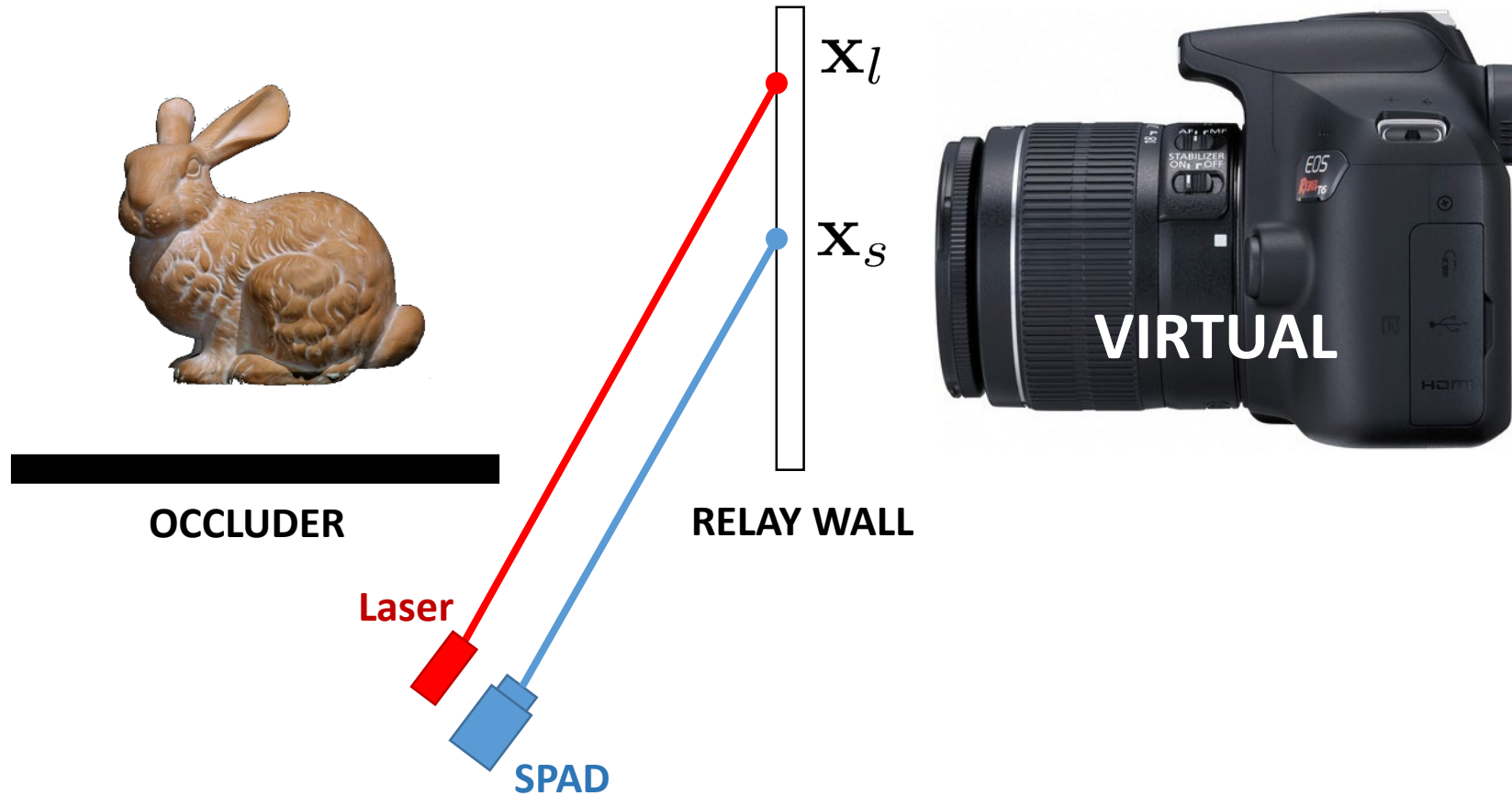
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Resemble lens propagation operators

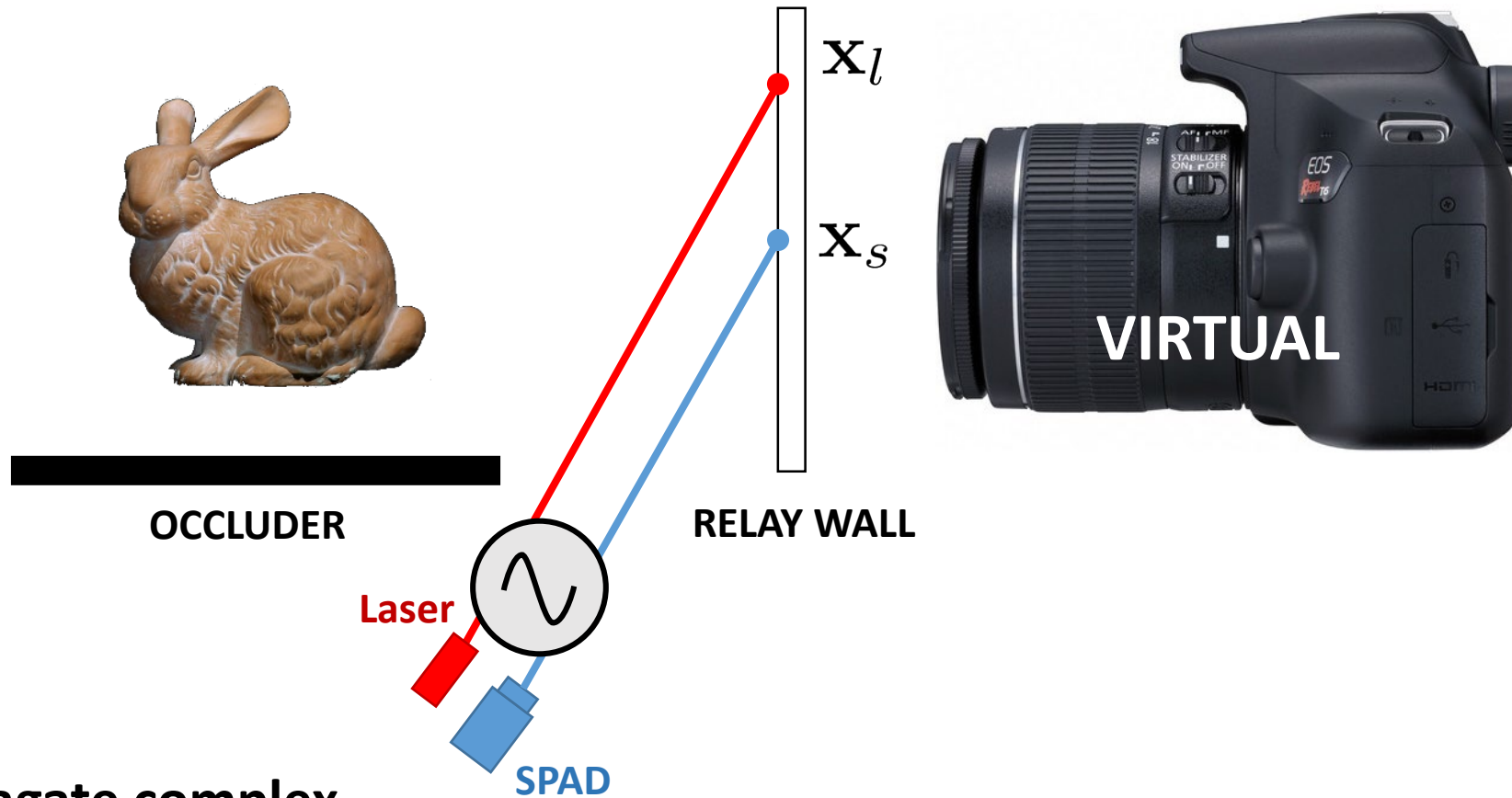
NLOS imaging: Phasor-fields [\[Liu et al. 2019\]](#)

- Key idea: FFT to turn the relay wall into **virtual cameras and light lenses**



NLOS imaging: Phasor-fields [\[Liu et al. 2019\]](#)

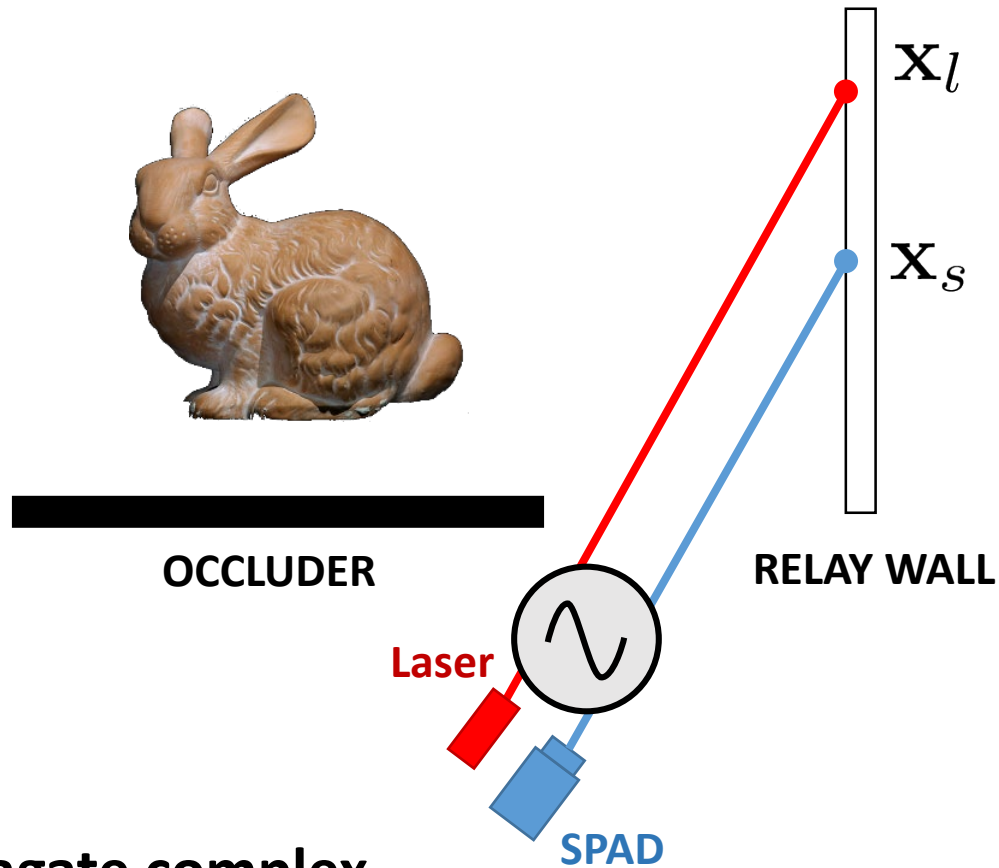
- Key idea: FFT to turn the relay wall into **virtual cameras and light lenses**



Propagate complex
waves instead of scalar radiance

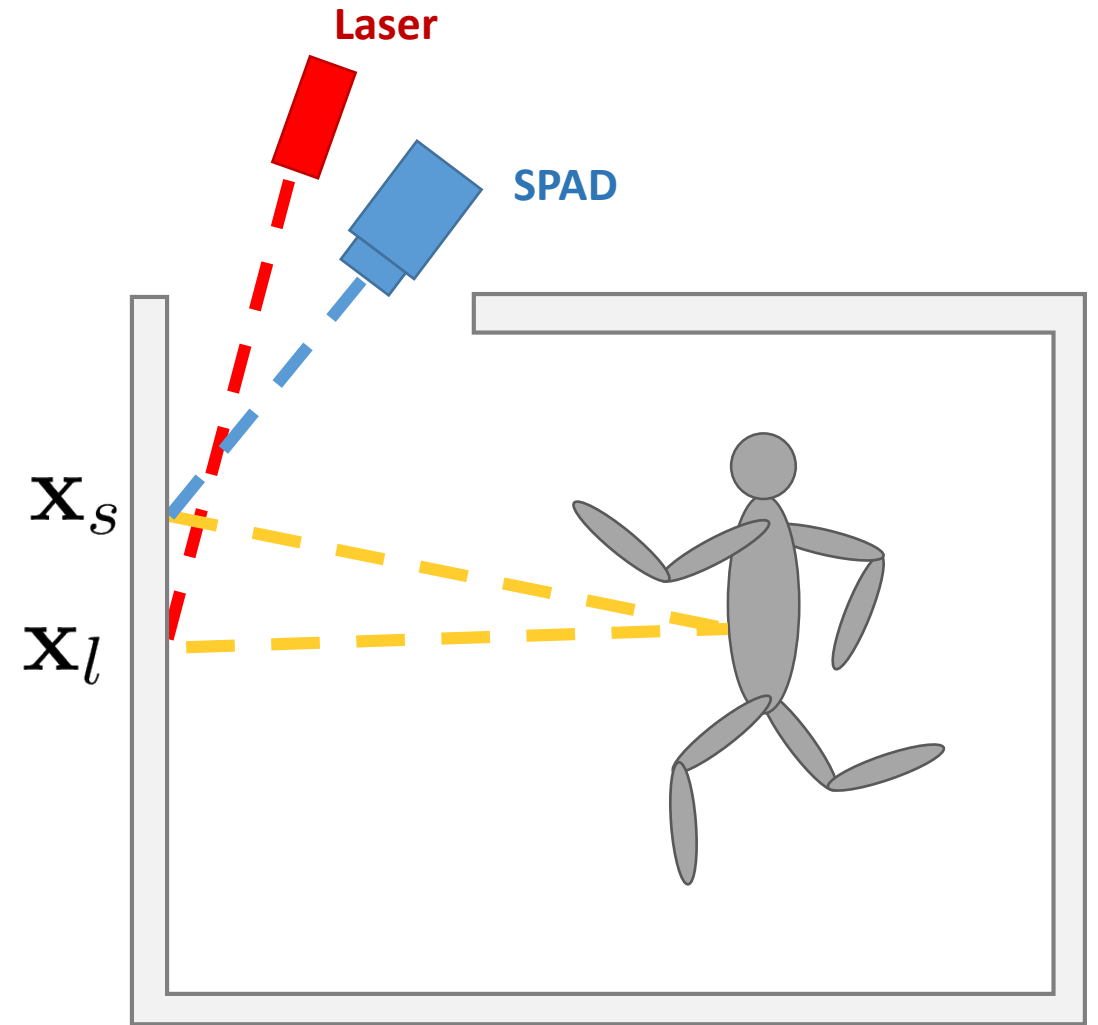
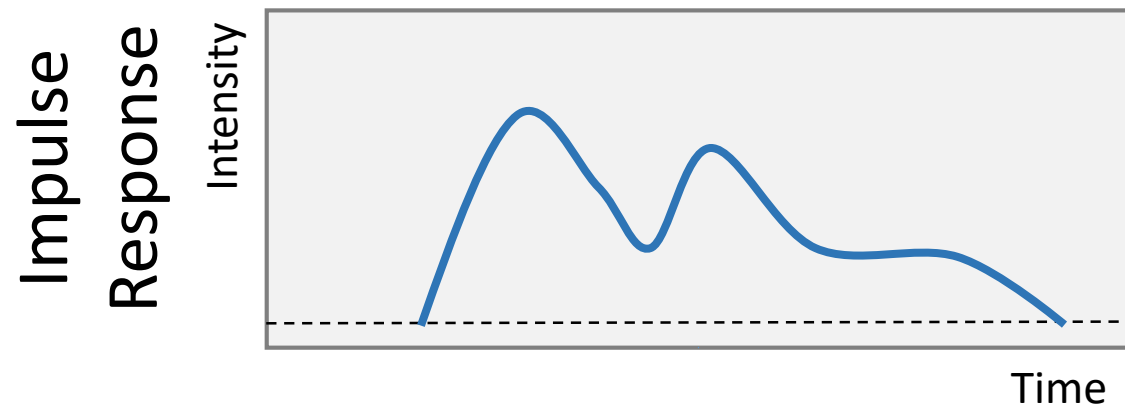
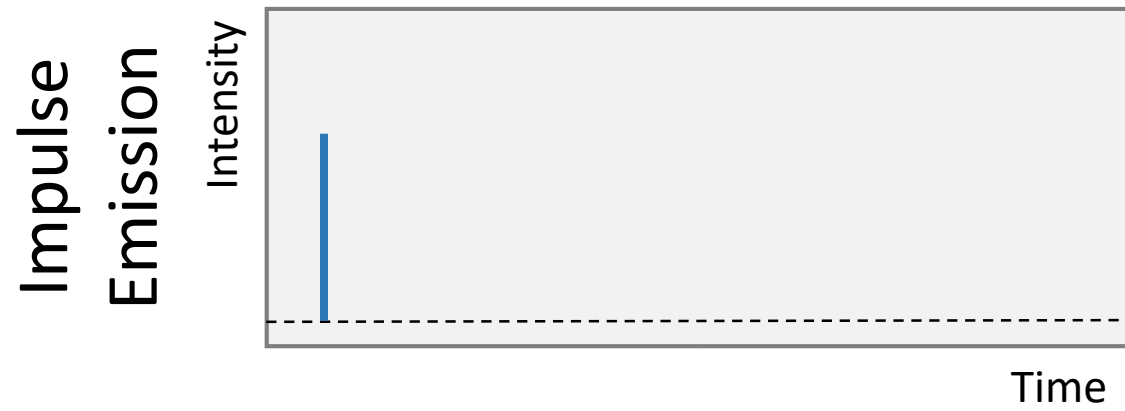
NLOS imaging: Phasor-fields [\[Liu et al. 2019\]](#)

- Key idea: FFT to turn the relay wall into **virtual cameras and light lenses**



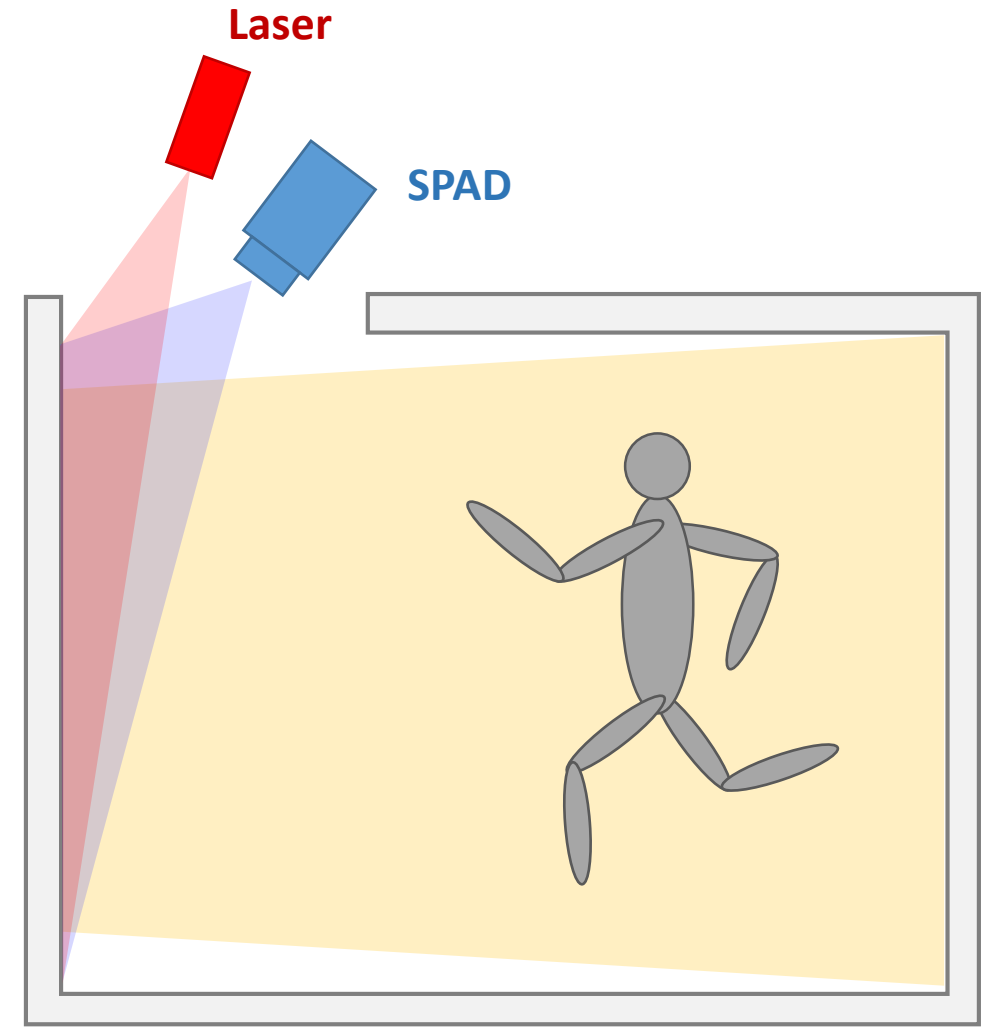
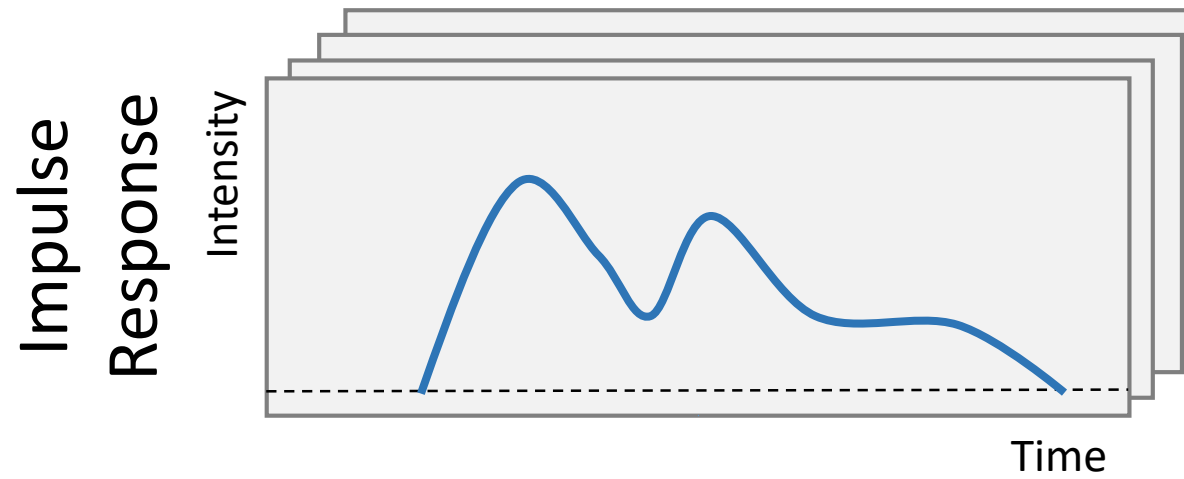
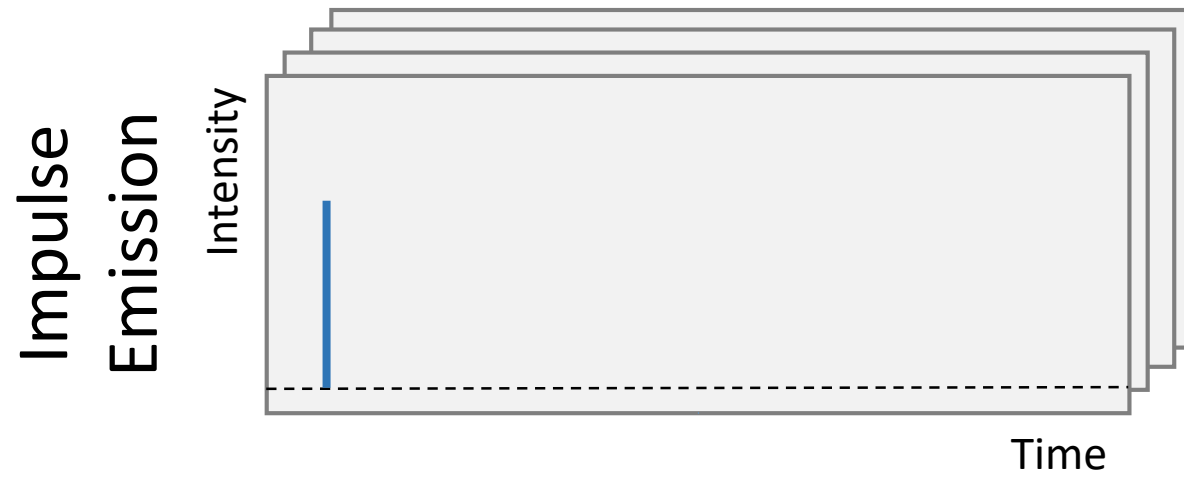
Propagate complex waves instead of scalar radiance

- Relay wall → Virtual lens aperture
- Directly observing the hidden scene
- Defined by classic optics models
- Leverage LOS imaging methods



CAPTURED IMPULSE RESPONSE

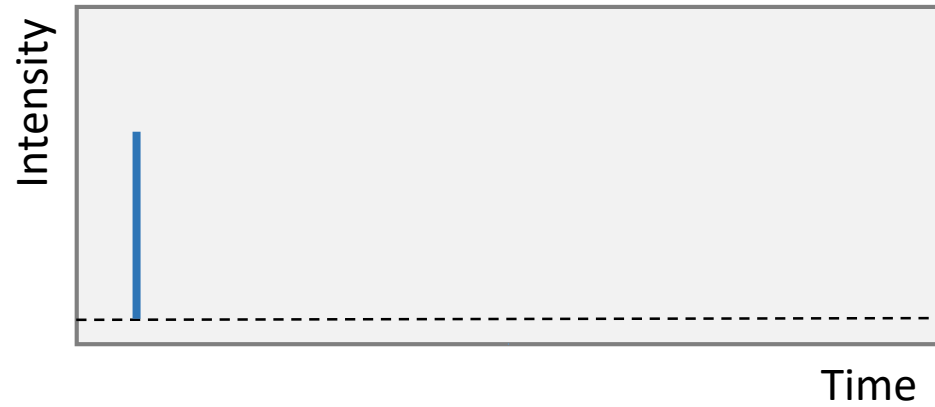
$$H(\mathbf{x}_l, \mathbf{x}_s, t)$$



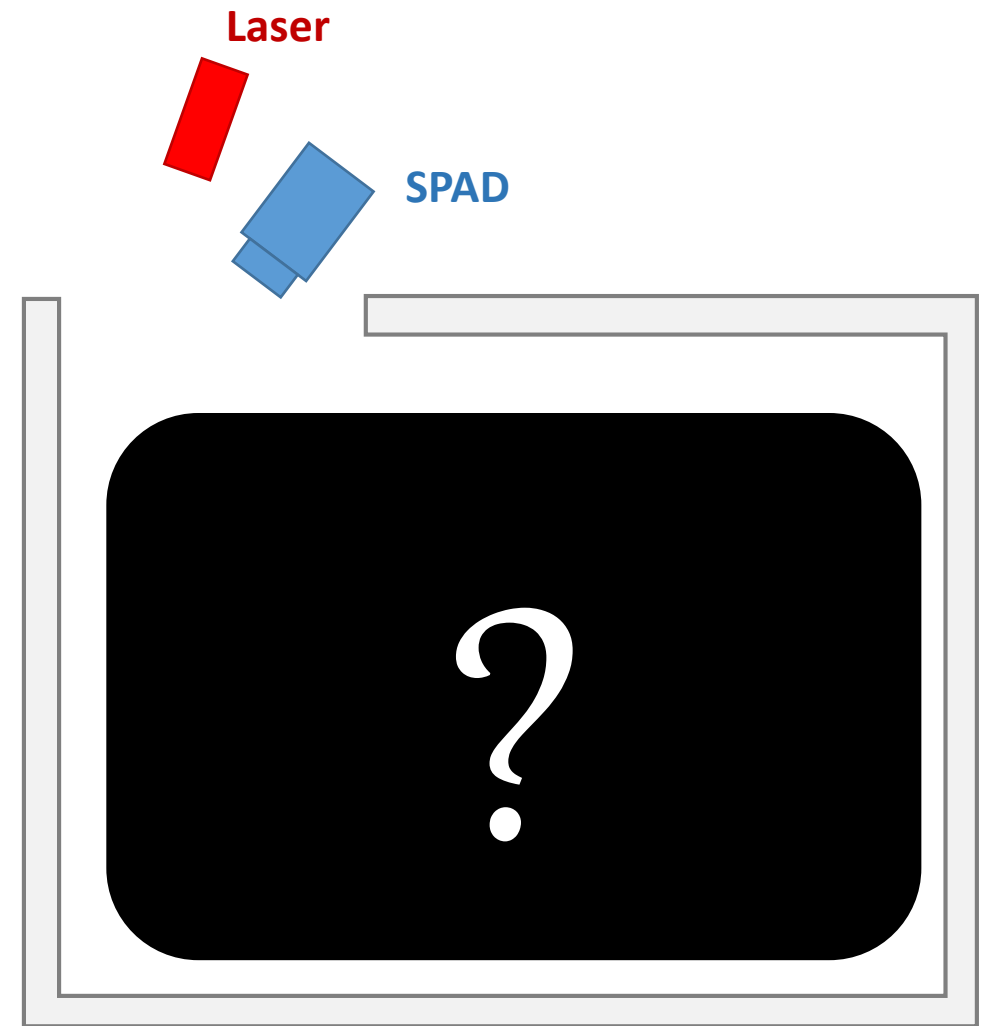
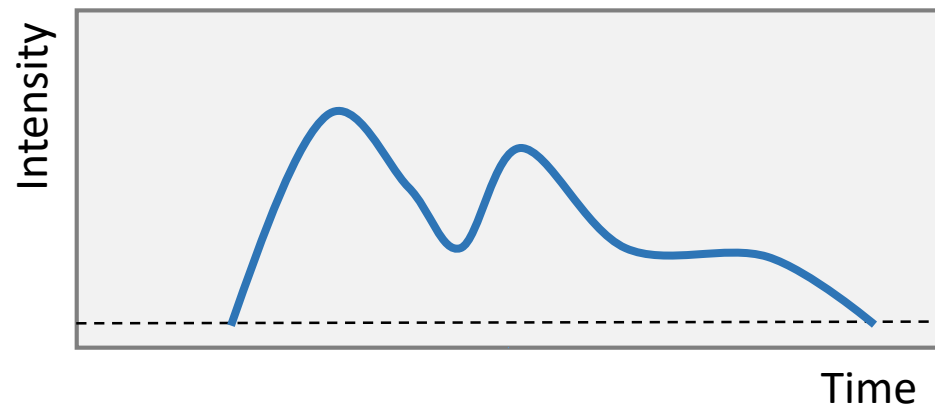
CAPTURED IMPULSE RESPONSE

$$H(\mathbf{x}_l, \mathbf{x}_s, t)$$

Impulse
Emission

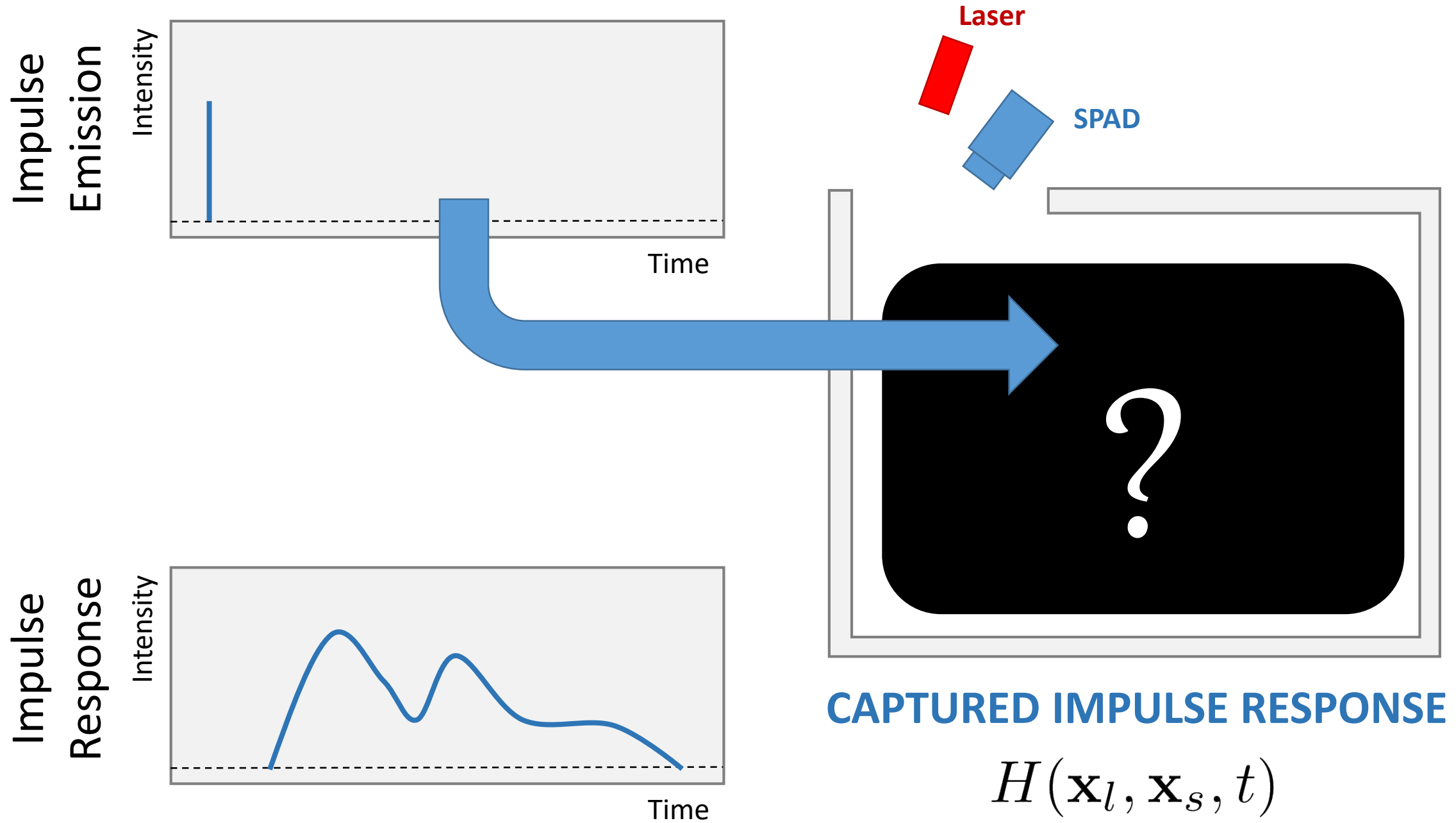


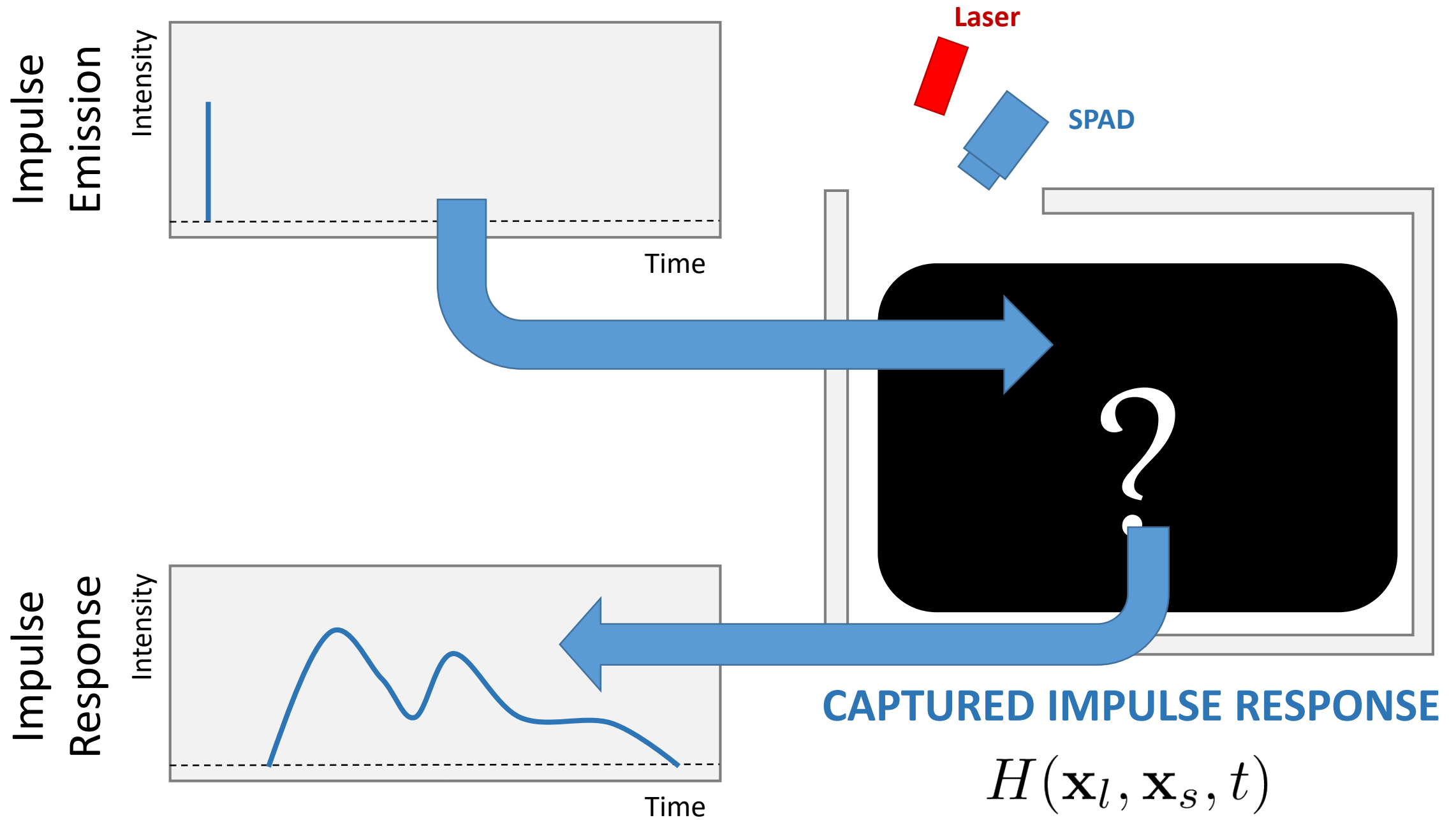
Impulse
Response



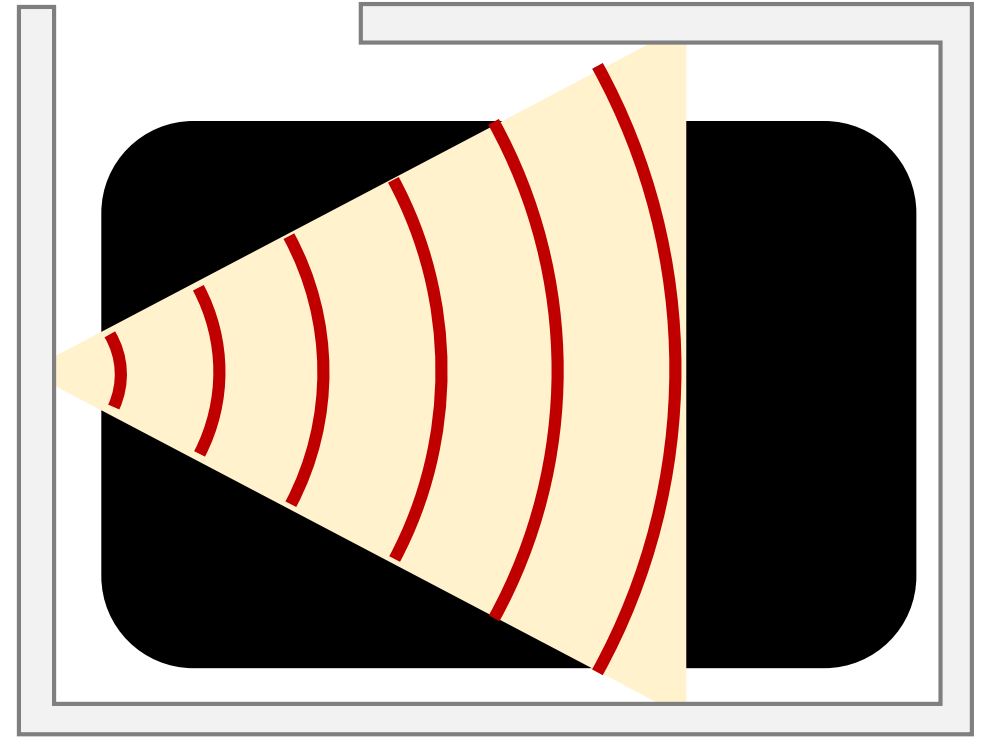
CAPTURED IMPULSE RESPONSE

$$H(\mathbf{x}_l, \mathbf{x}_s, t)$$



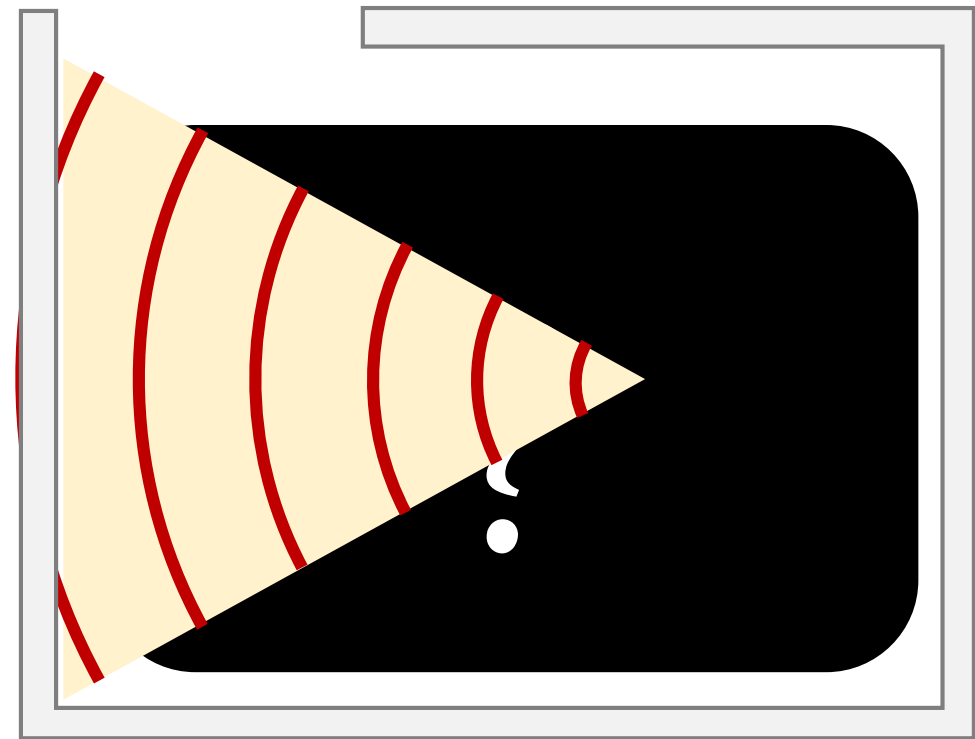
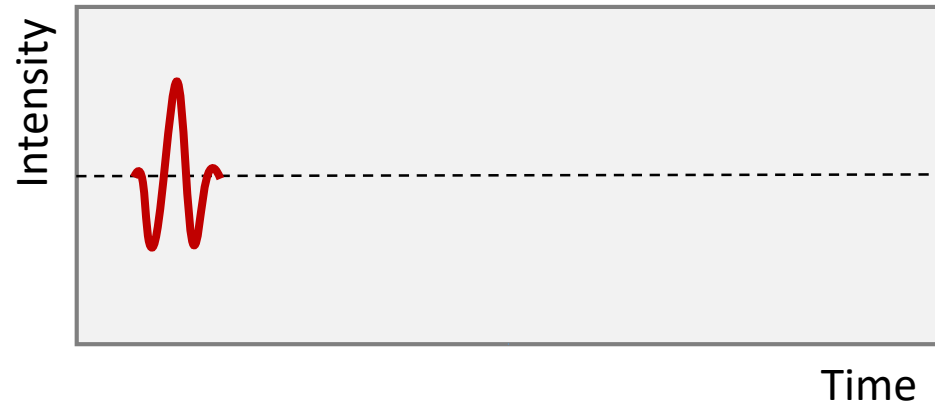


Emission

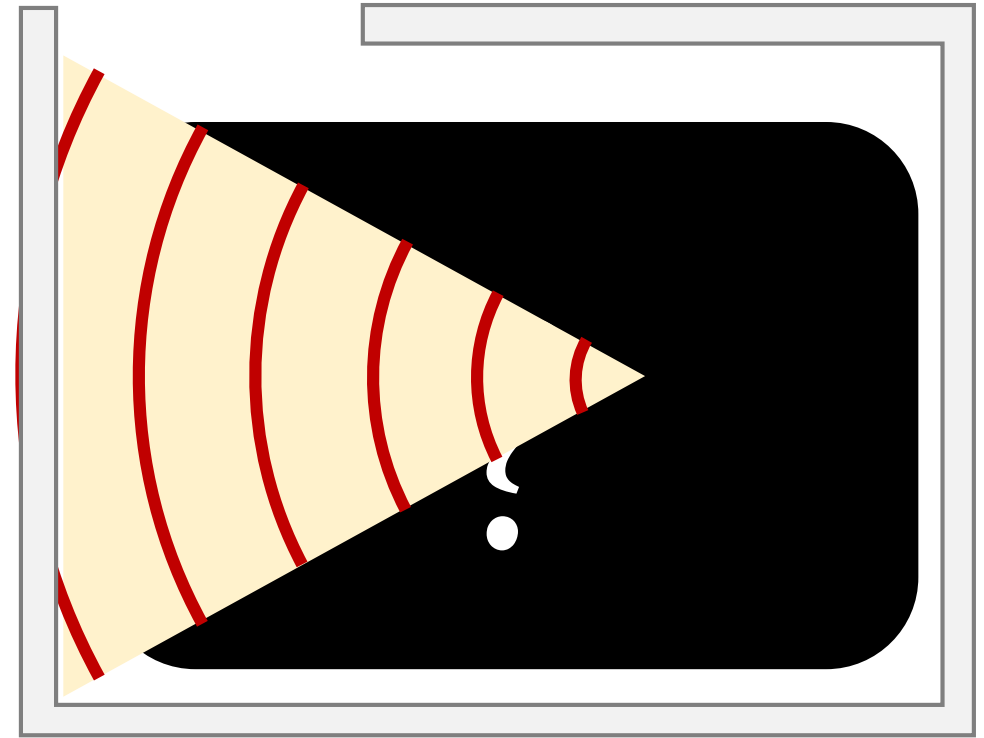
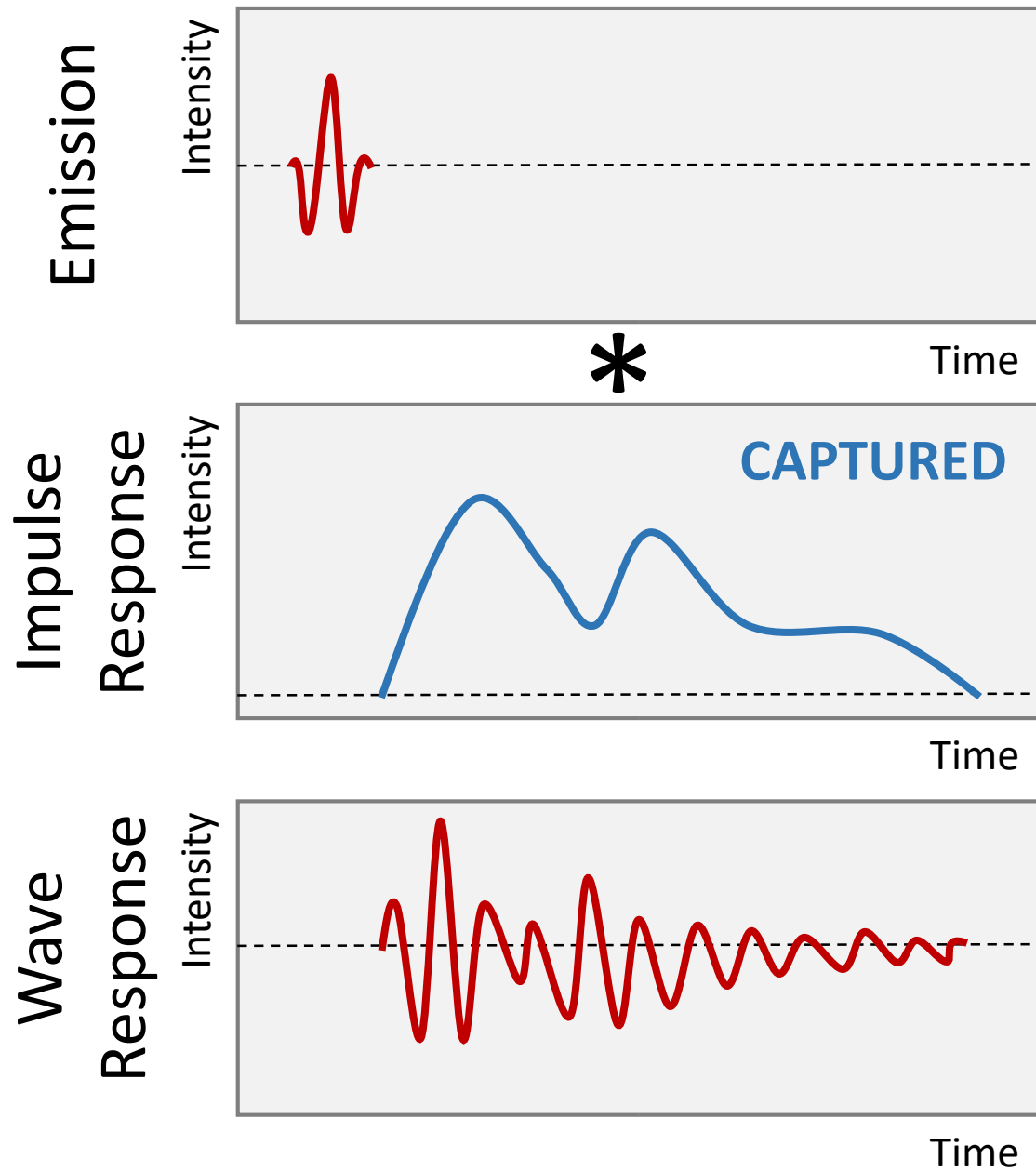


**VIRTUAL WAVE
PROPAGATION**

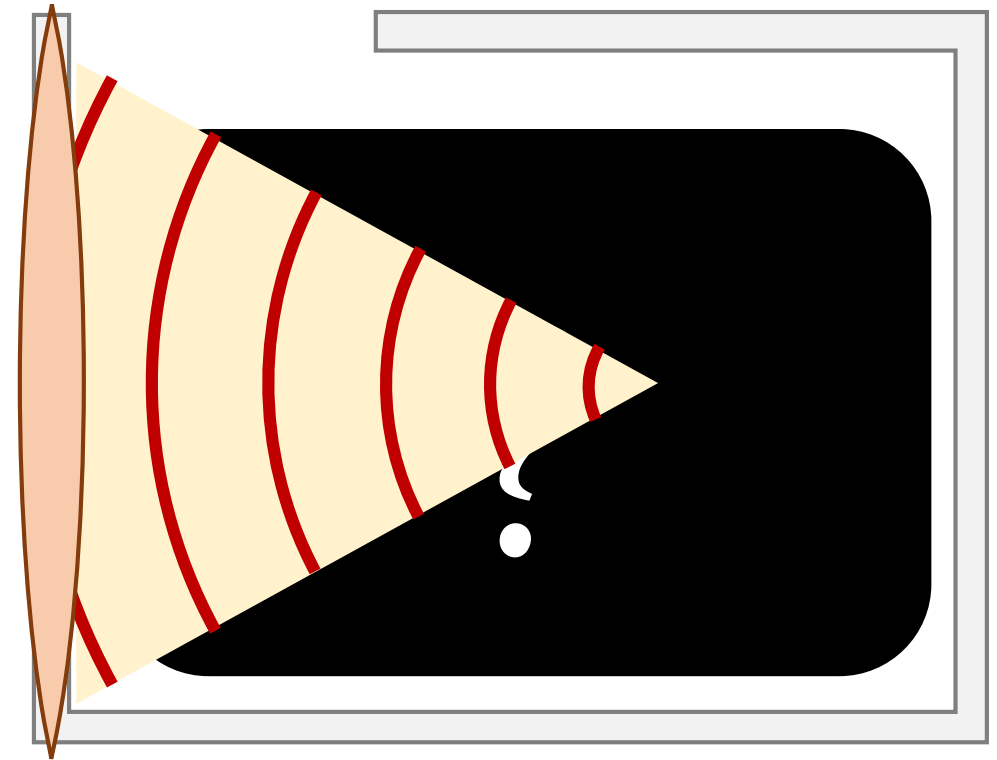
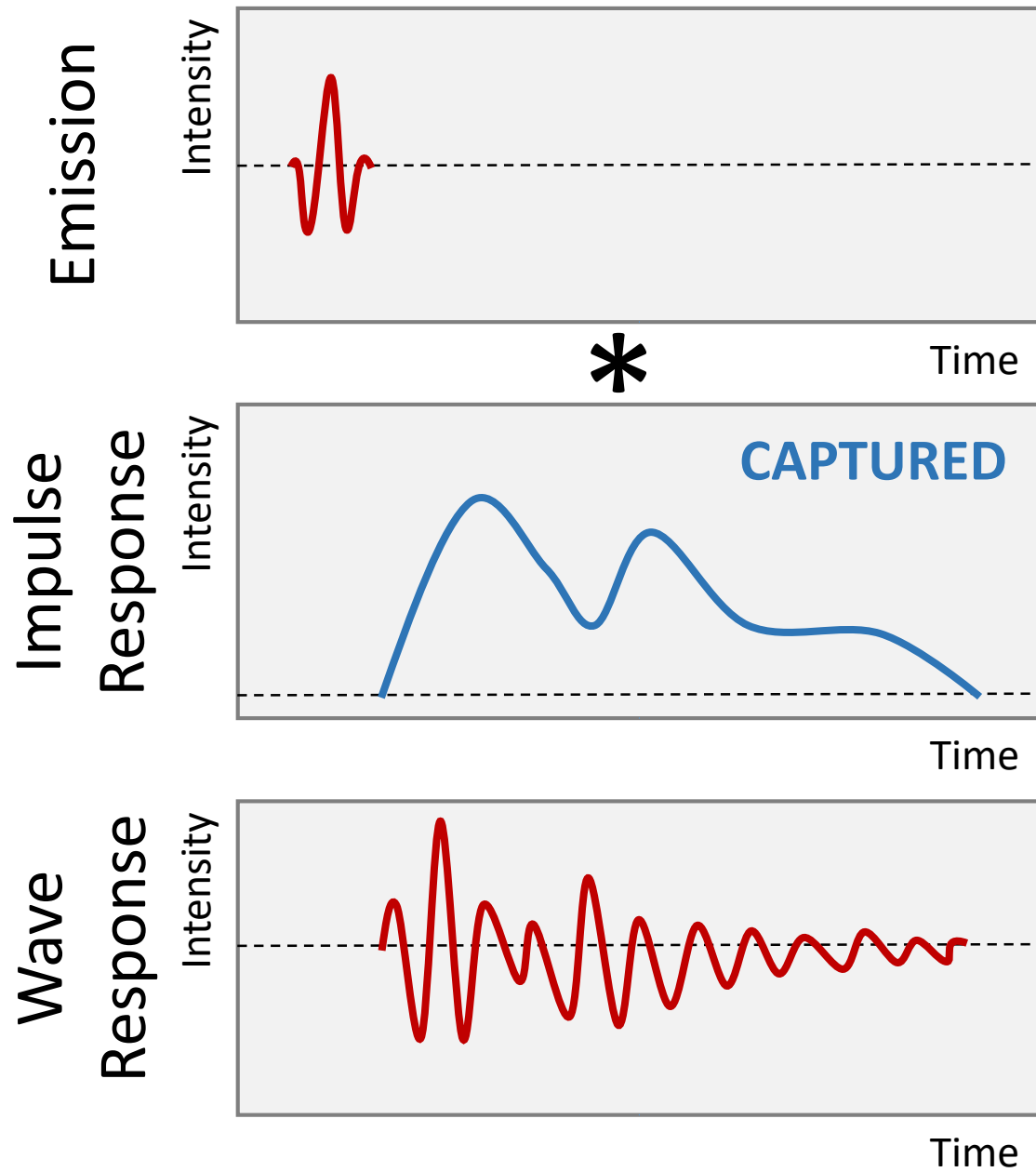
Emission



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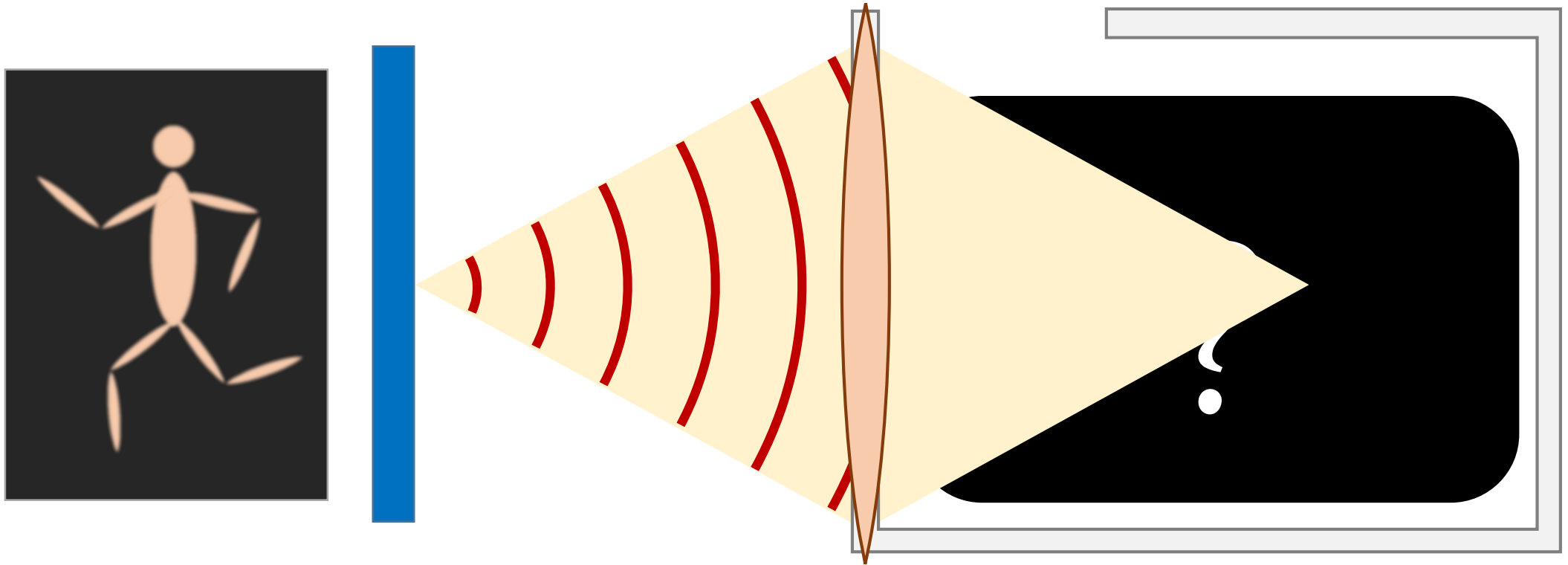


**VIRTUAL WAVE
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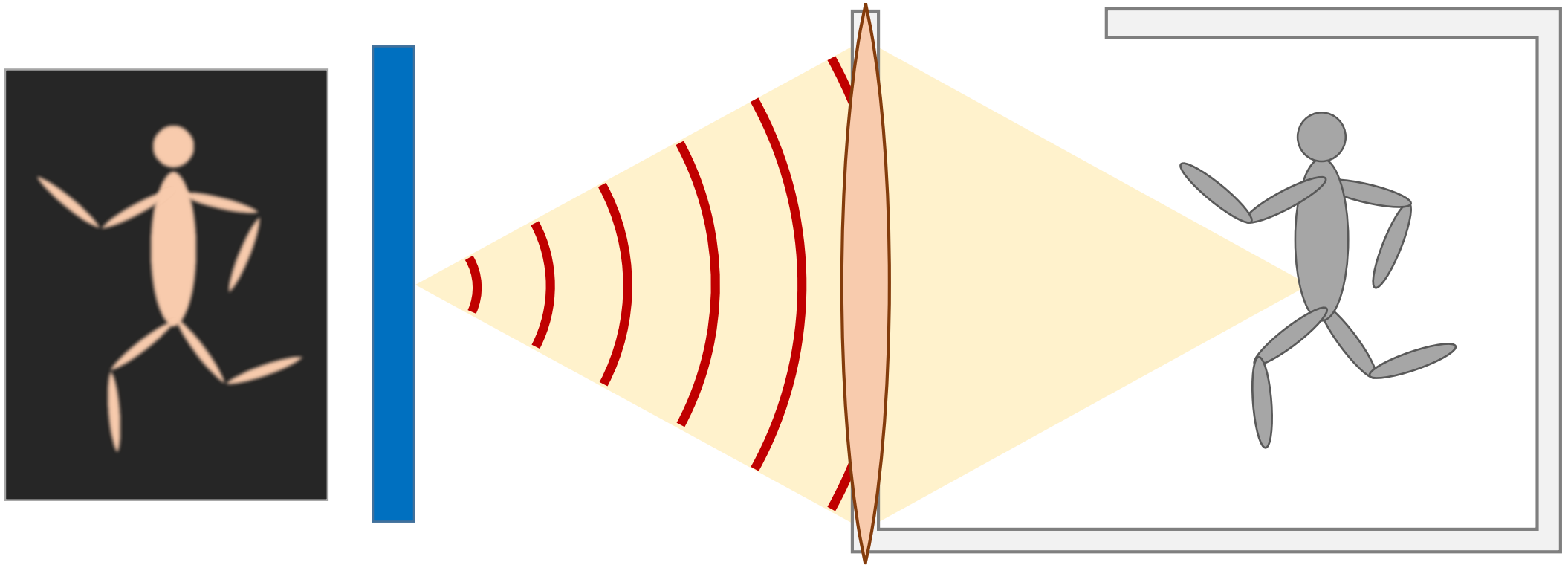
VIRTUAL WAVE PROPAGATION MODELS

Use forward propagators from classic optics to focus light onto a plane

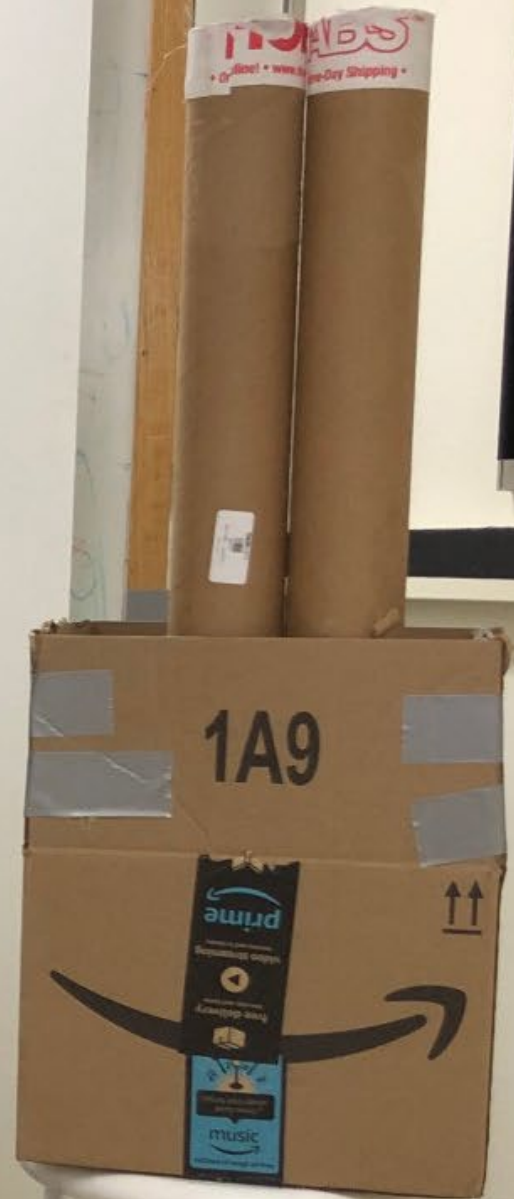


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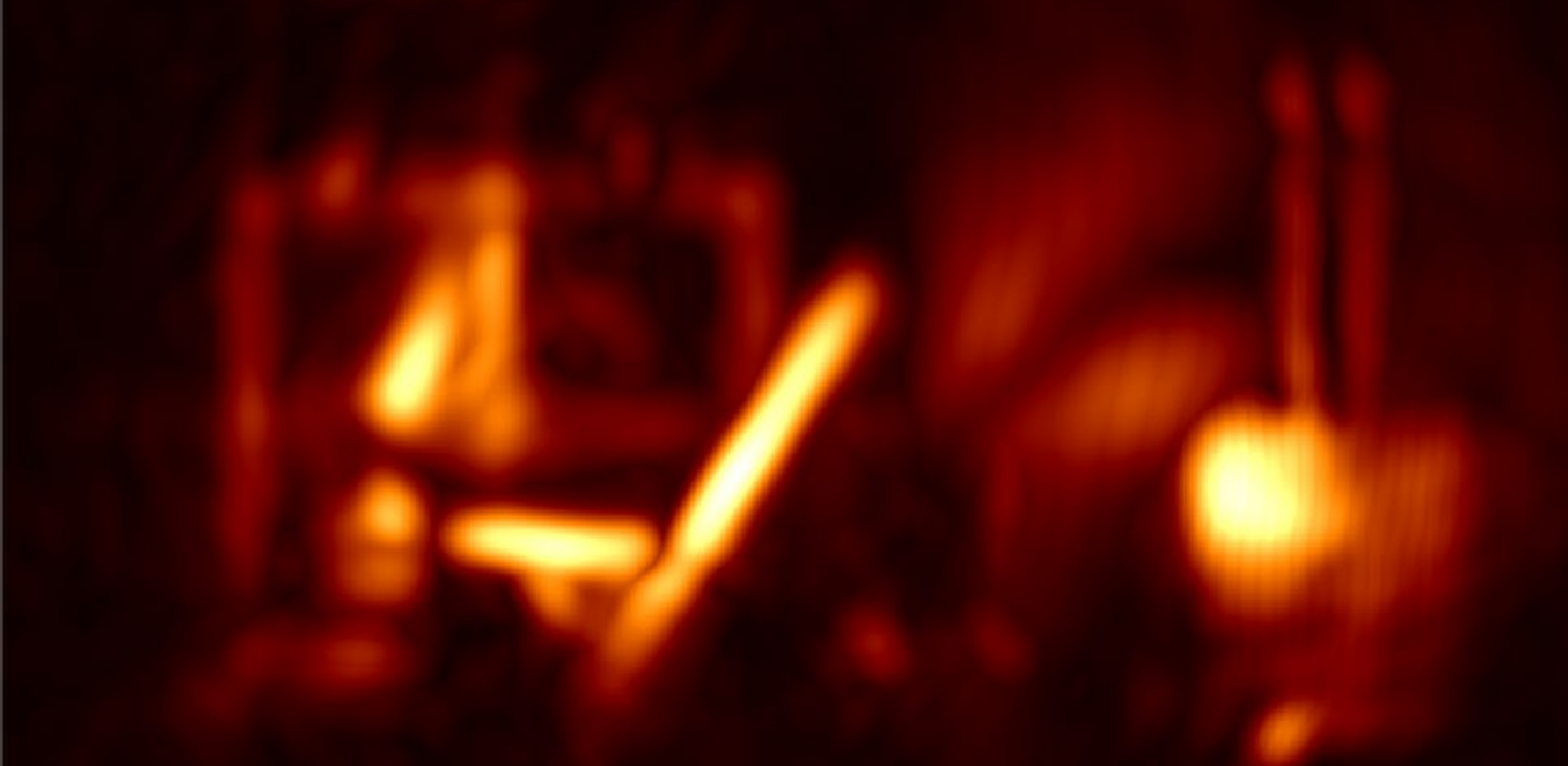
Use forward propagators from classic optics to focus light onto a plane



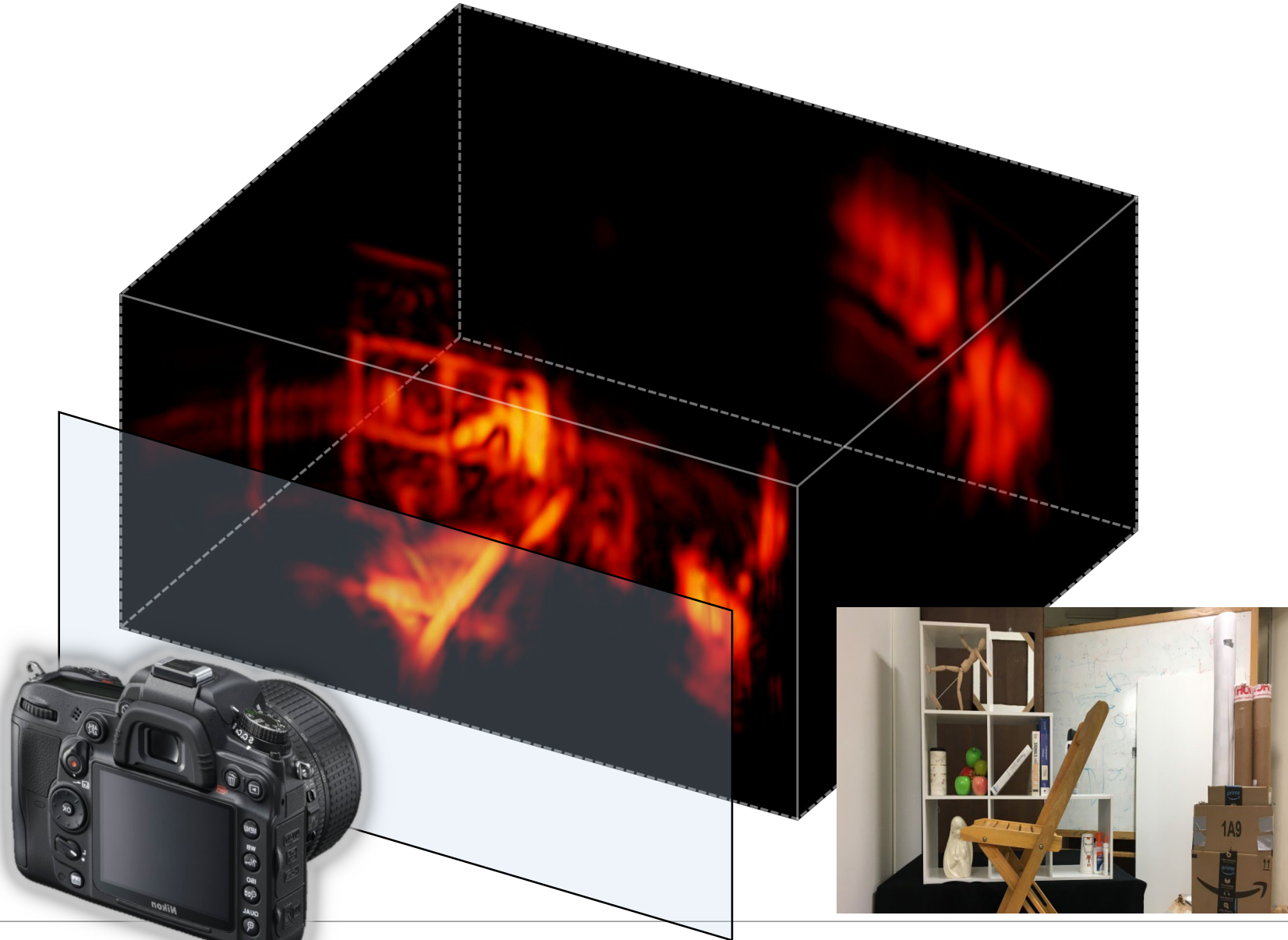
**VIRTUAL WAVE
PROPAGATION**



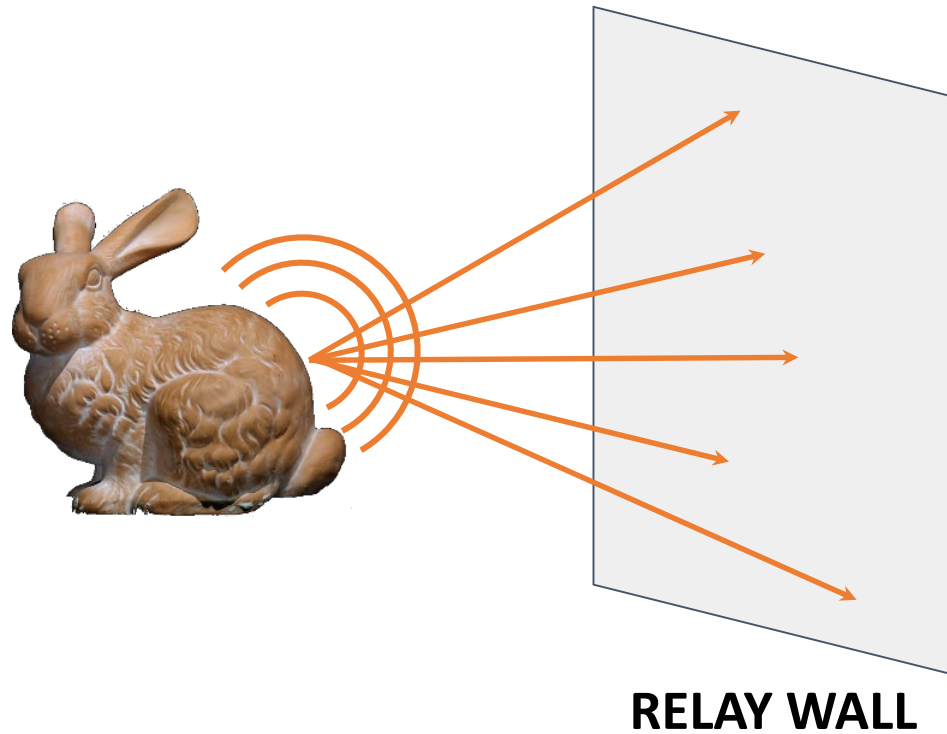
NLOS imaging: Phasor-fields [\[Liu et al. 2019\]](#)



NLOS imaging: Phasor-fields [\[Liu et al. 2019\]](#)

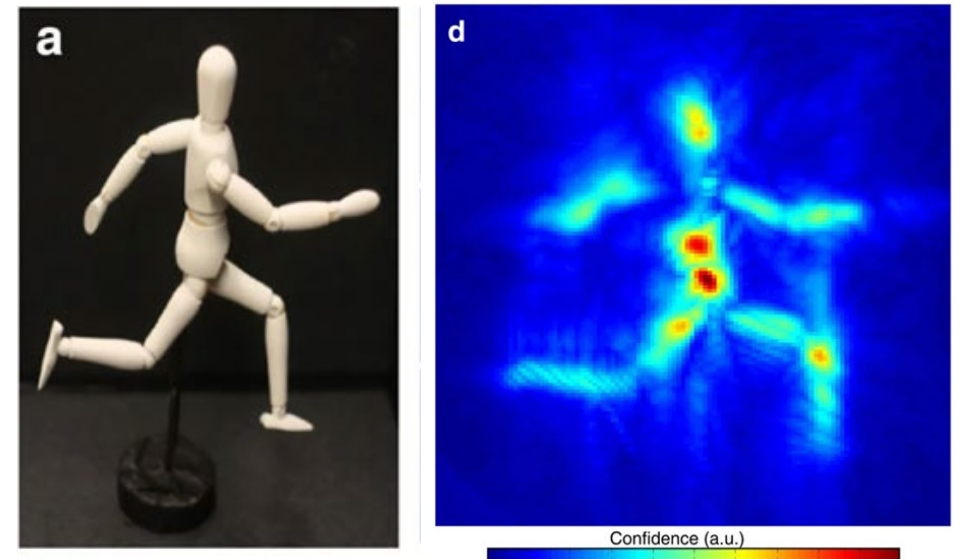


Backprojection & Lens analogy



$$f(\mathbf{x}_v) = \iint_S \int_L H'(\mathbf{x}_l, \mathbf{x}_s, t_{lv} + t_{sv}) d\mathbf{x}_l d\mathbf{x}_s$$

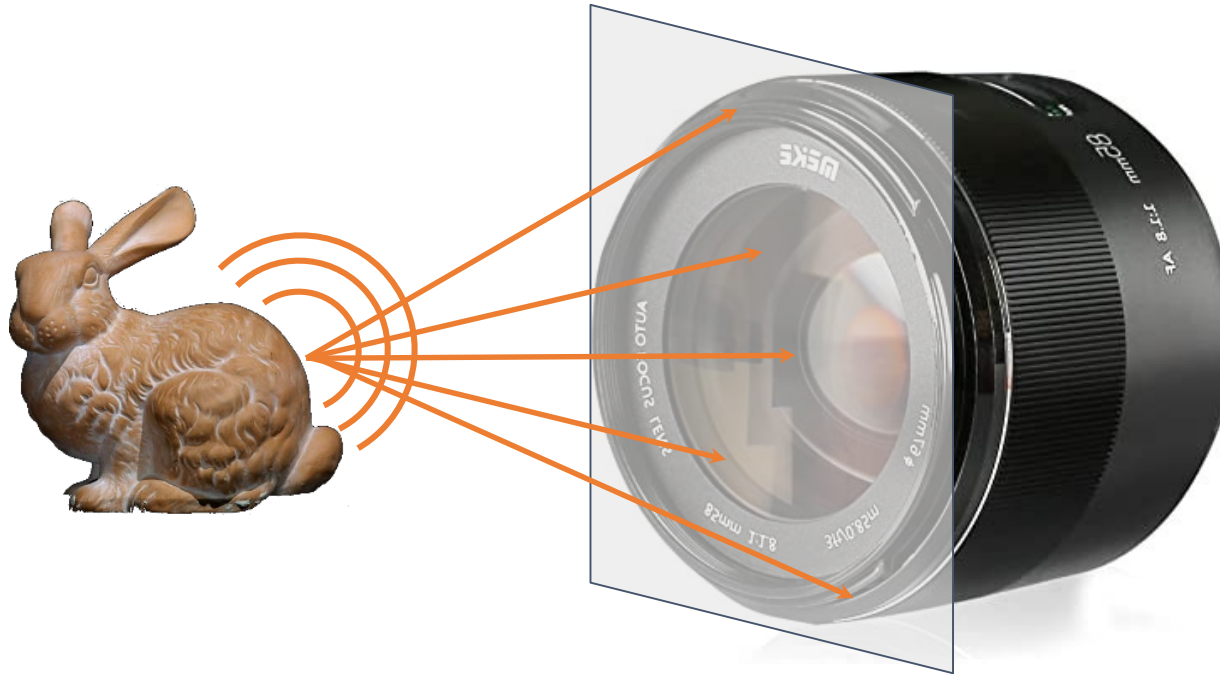
Looking around corners [Velten et al. 2012]



Backprojection & Lens analogy

Phasor fields
[Liu et al. 2019]

nature

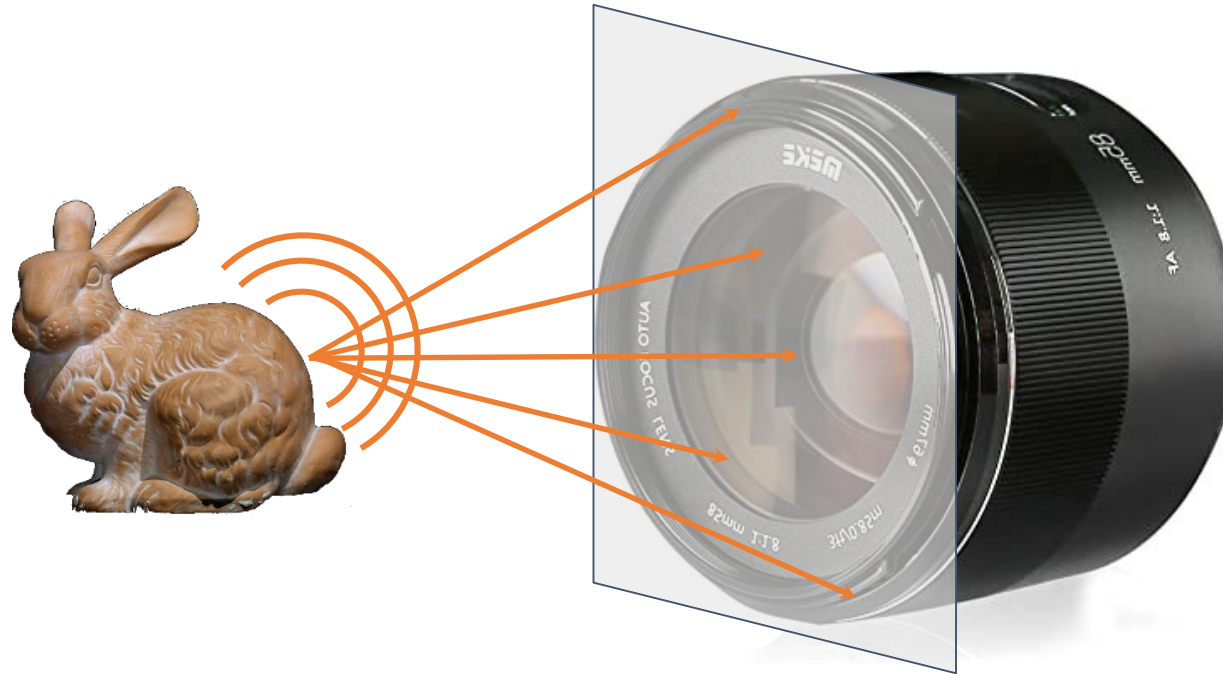


RELAY WALL = VIRTUAL LENS



$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

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Phasor fields
[Liu et al. 2019]

nature

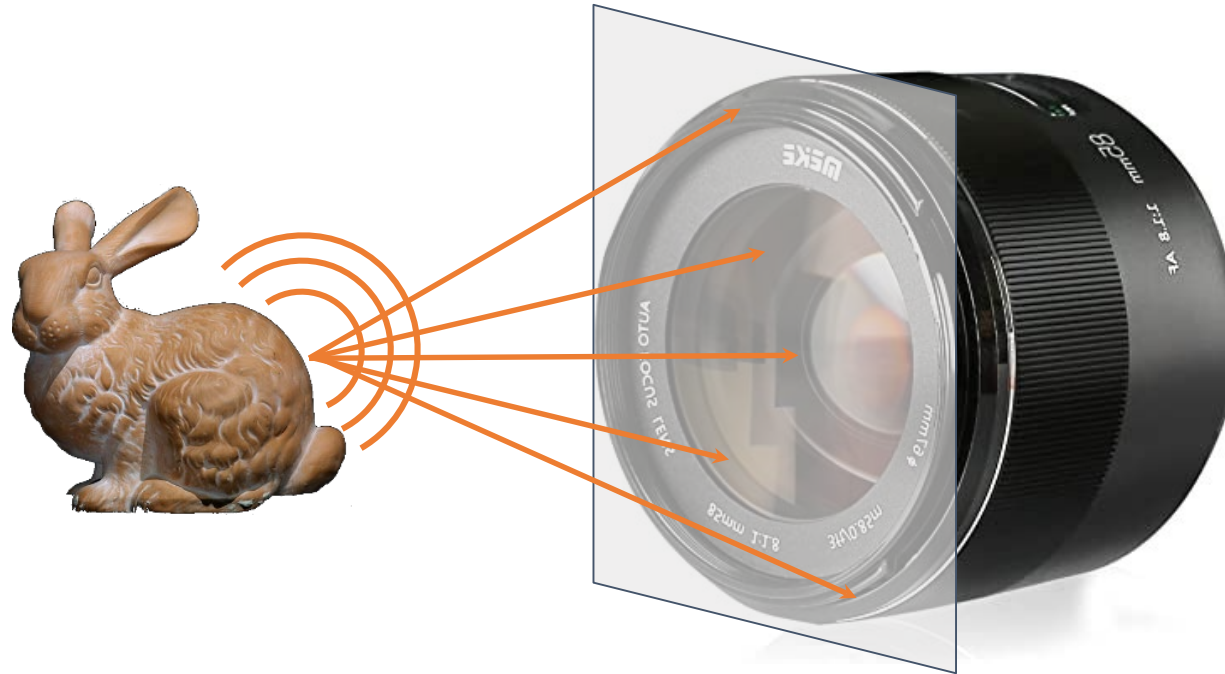


NLOS



VIRTUAL LOS

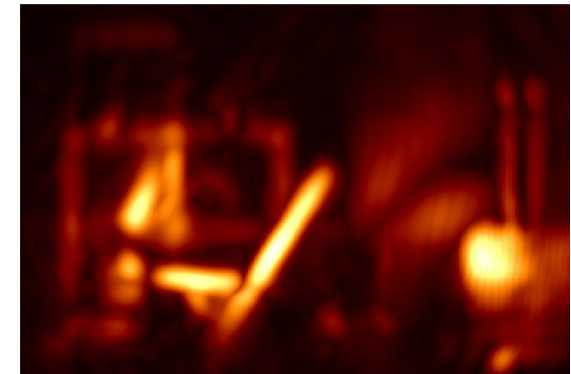
Backprojection & Lens analogy



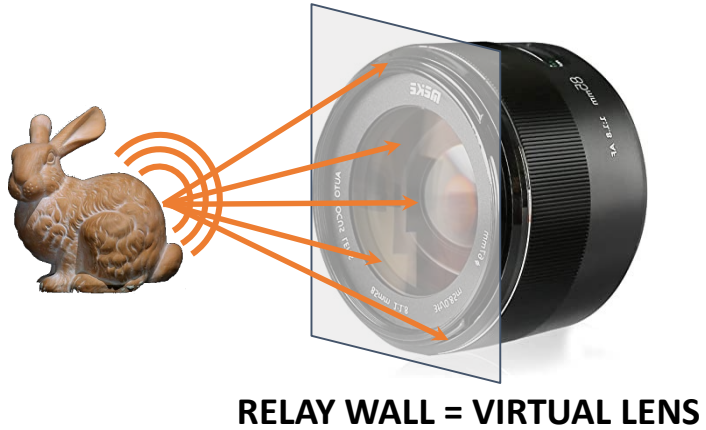
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Phasor fields
[Liu et al. 2019]



Why is the BP-Lens analogy useful?



Efficient optics-based solvers

[Liu et al. 2020] RSD-based, in seconds

[Nam et al. 2021] [5 frames per second](#)

[Liao et al 2021] Real-time on an FPGA



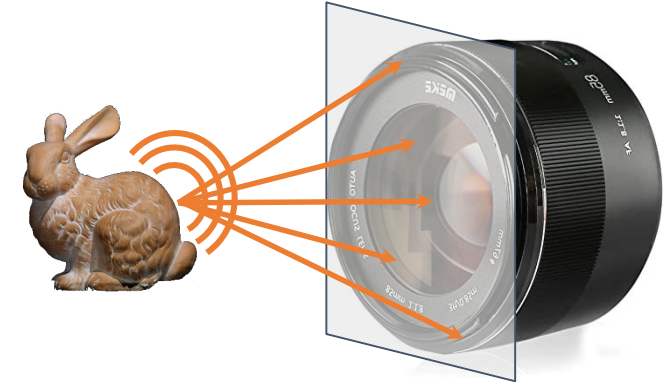
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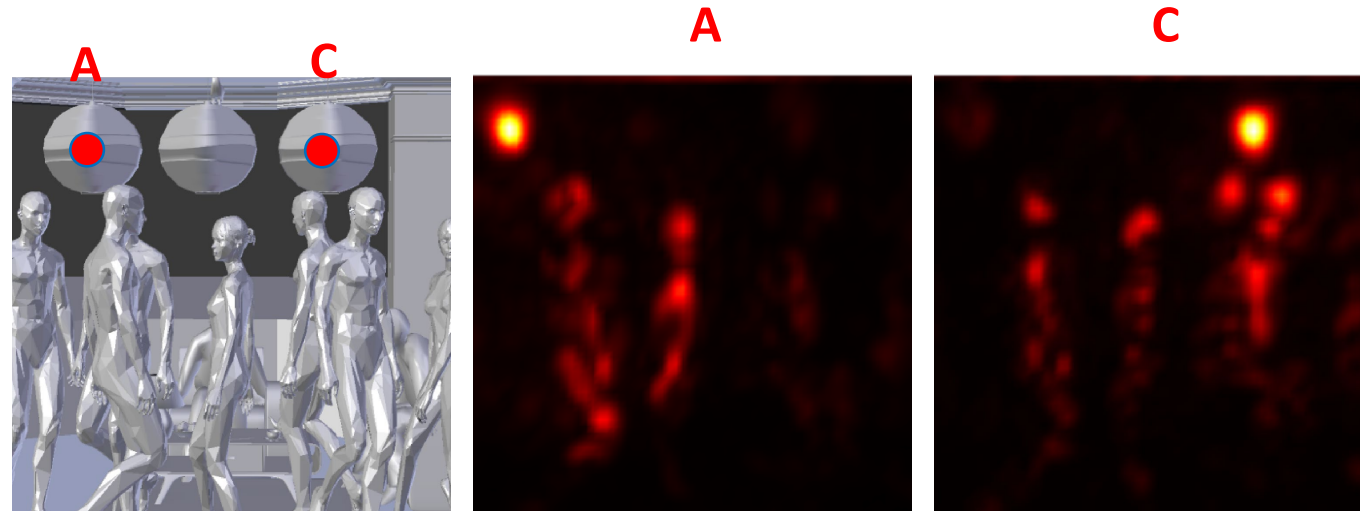
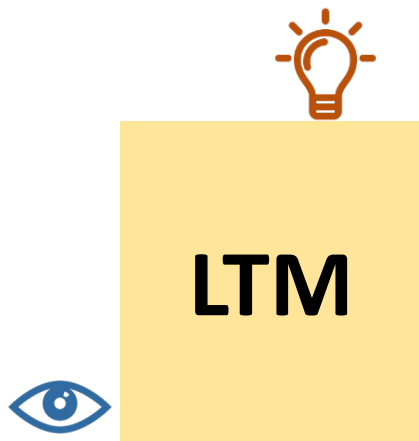
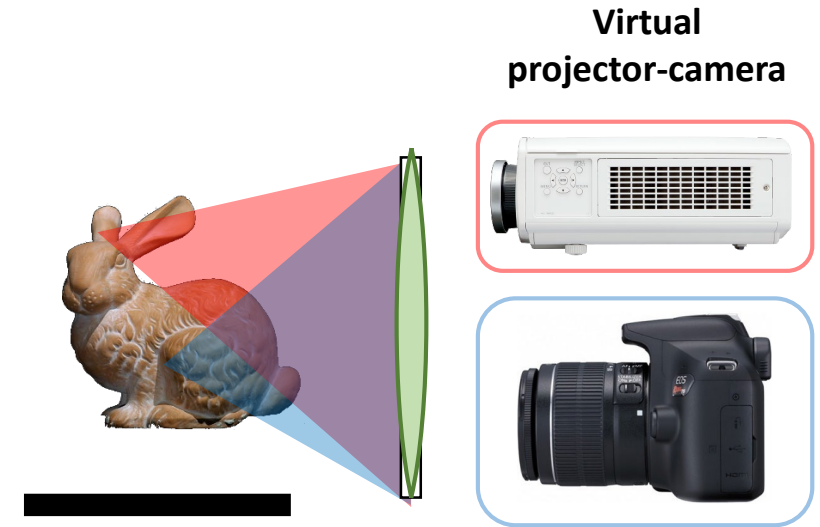
RELAY WALL = VIRTUAL LENS



Why is the BP-Lens analogy useful?

Design of new virtual imaging systems:

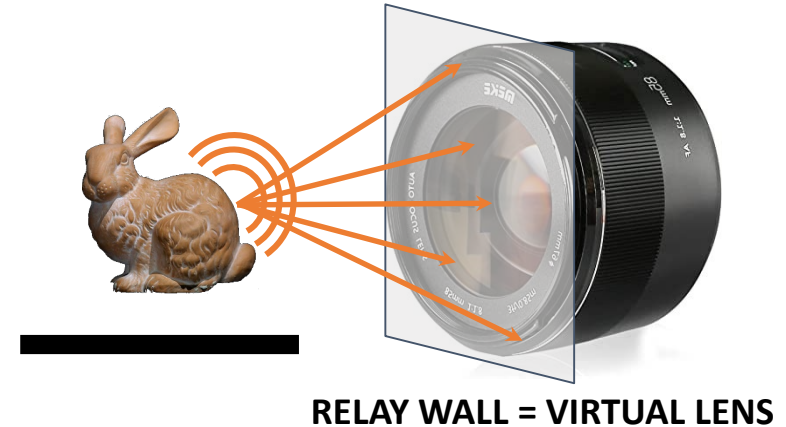
[Marco et al 2021] Virtual camera-projectors $\Rightarrow$
Virtual LTM, relighting, illumination separation



Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

- Explain behavior of light in NLOS scenes
- Analysis of visibility, resolution, SNR
- Reflectance $\rightarrow$ Virtual vs. real BRDF
- Application: **See after more than one corner**

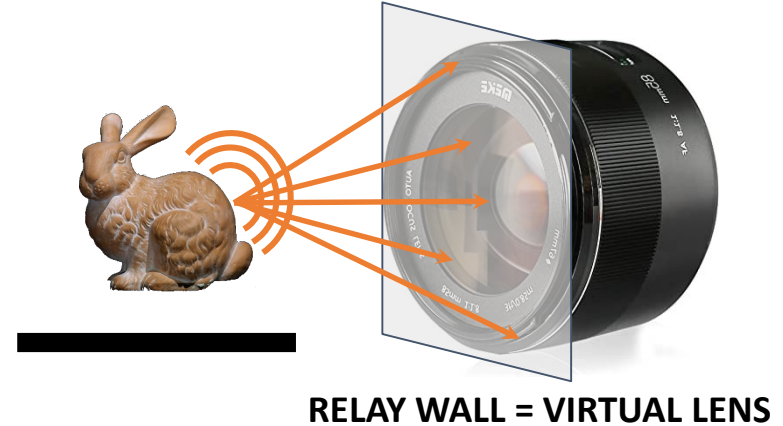


$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

Example - Reflectance in virtual space:

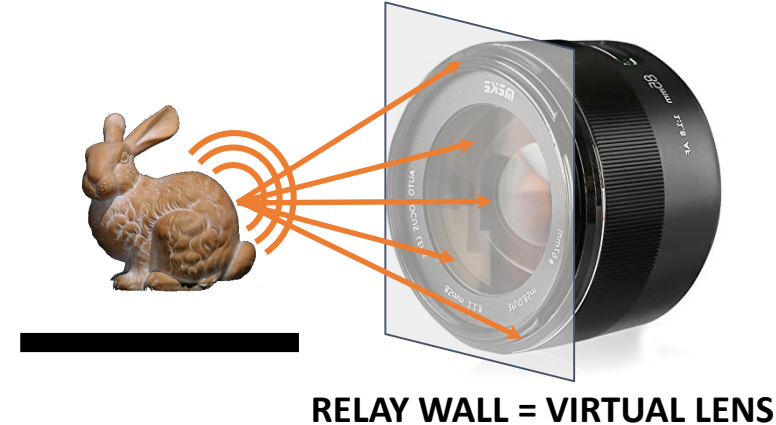


$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

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NLOS light (virtual optics) $\Rightarrow$ Wavelength of 5-8cm

but...

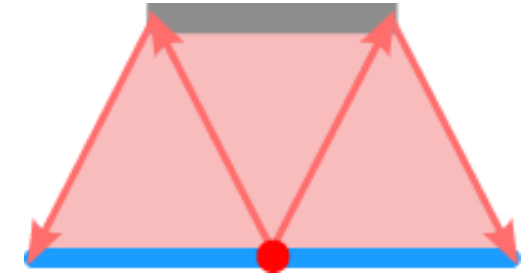
visible light (real optics) $\Rightarrow$ Wavelength of 320-700nm

Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

Example - Reflectance in virtual space:

- Planar surfaces == Mirrors
- BRDF behavior at *macro*-scale

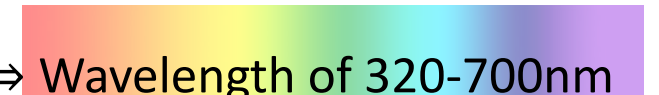


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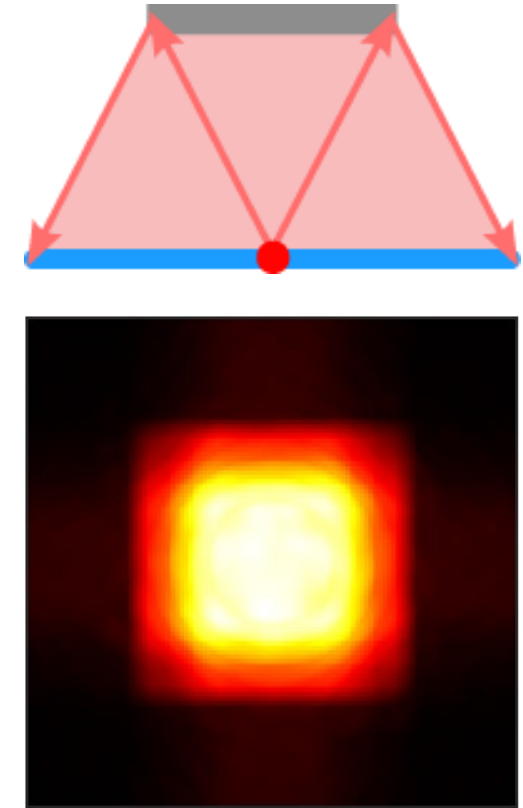
- Planar surfaces == Mirrors
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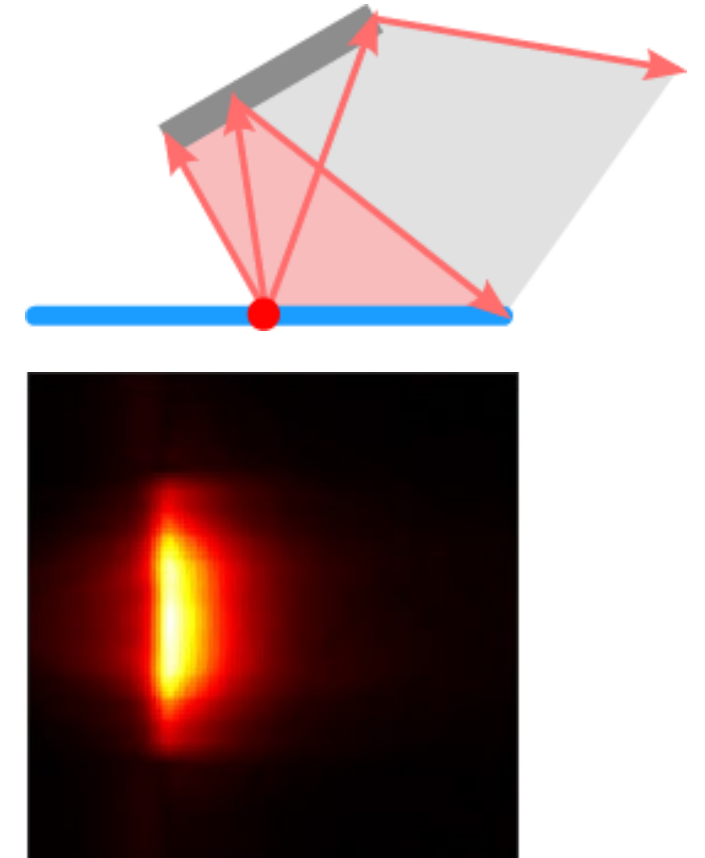
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Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

Example - Imaging resolution:

- Imaging wavelength $\Rightarrow$ Filtering
- Aperture
- Depth

$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

Virtual lenses

Rayleigh criteria, detector resolution

Wavelength λ *Focal length = Depth* d

$\Delta x = 1.22c \frac{\lambda d}{A}$ *Aperture = Relay wall* A

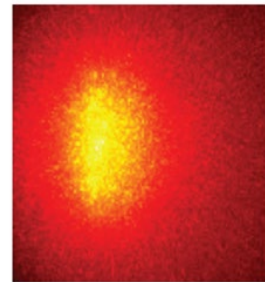
Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

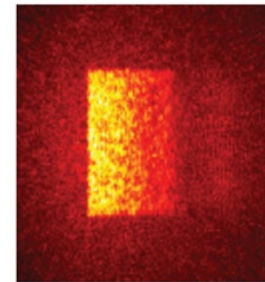
Example - Imaging resolution:

- **Imaging wavelength $\Rightarrow$ Filtering**
- Aperture
- Depth

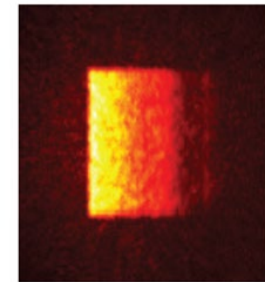
$\lambda_c = 1.5 \text{ cm}$



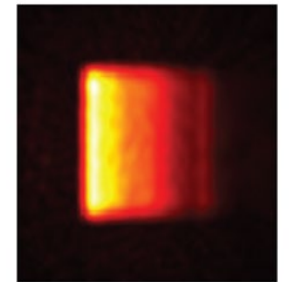
$\lambda_c = 6 \text{ cm}$



$\lambda_c = 9 \text{ cm}$



$\lambda_c = 18 \text{ cm}$



Wavelength

$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

*Virtual
lenses*

Rayleigh criteria, detector resolution

Wavelength

Focal length = Depth

$$\lambda = \frac{1}{\Omega}$$

$$\Delta x = 1.22c \frac{\lambda d}{A}$$

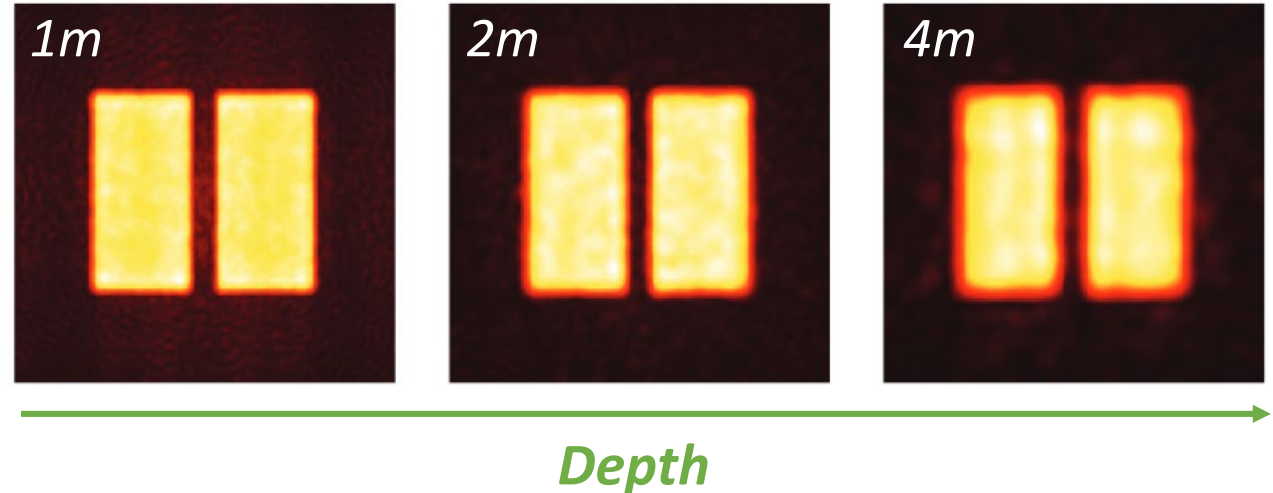
Aperture = Relay wall

Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

Example - Imaging resolution:

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- **Depth**



$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

Virtual lenses

Rayleigh criteria, detector resolution

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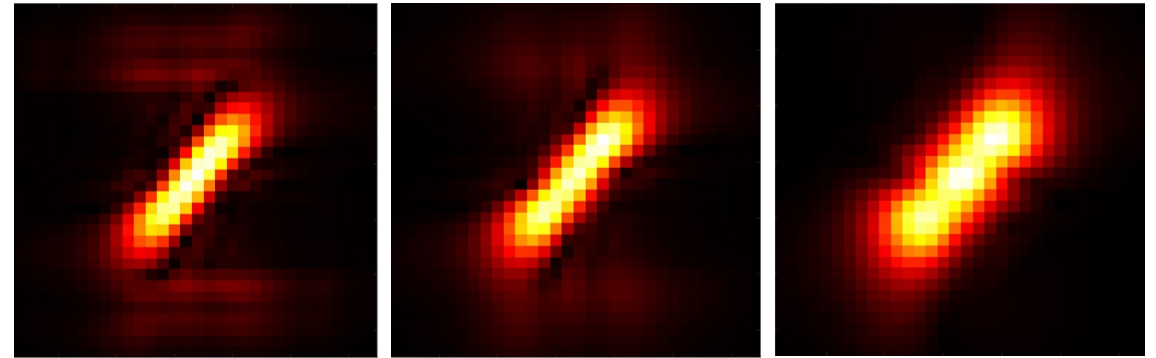
Aperture = Relay wall

Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

Example - Imaging resolution:

- Imaging wavelength $\Rightarrow$ Filtering
- **Aperture**
- Depth



Specularity $\Rightarrow$ Aperture size?

$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

*Virtual
lenses*

Rayleigh criteria, detector resolution

$$\Delta x = 1.22c \frac{\overset{\text{Wavelength}}{\lambda} \overset{\text{Focal length = Depth}}{d}}{\underset{\text{Aperture = Relay wall}}{A}}$$

Why is the BP-Lens analogy useful?

Analysis of NLOS light transport with classic optics:

Example - Imaging resolution:

- Imaging wavelength $\Rightarrow$ Filtering
- **Aperture**
- Depth



Diffuse $\Rightarrow$ Large aperture coverage

$$\hat{f}(\mathbf{x}_v, \Omega) = \int_S e^{i\Omega t_{sv}} \int_L e^{i\Omega t_{lv}} \hat{H}'(\mathbf{x}_l, \mathbf{x}_s, \Omega) d\mathbf{x}_l d\mathbf{x}_s$$

*Virtual
lenses*

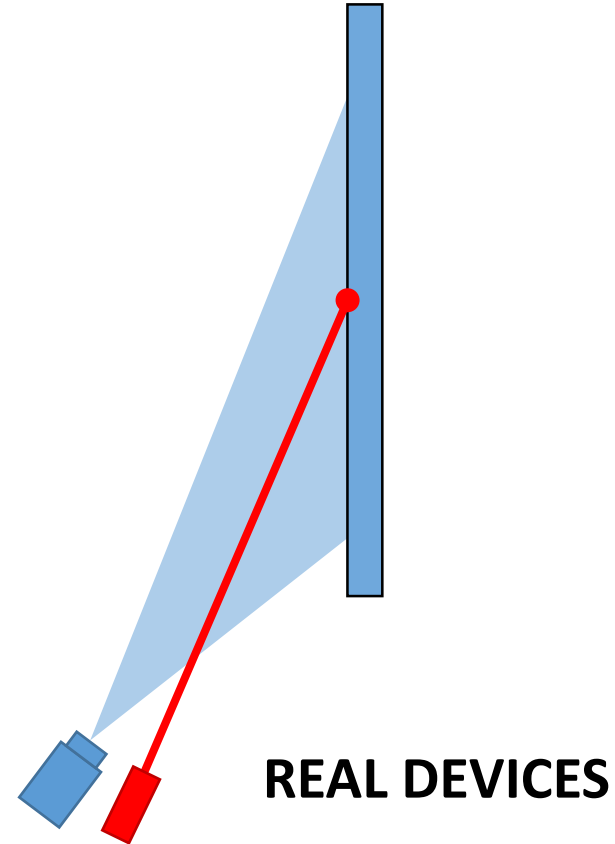
Rayleigh criteria, detector resolution

$$\Delta x = 1.22c \frac{\lambda d}{A}$$

Wavelength λ *Focal length = Depth* d *Aperture = Relay wall* A

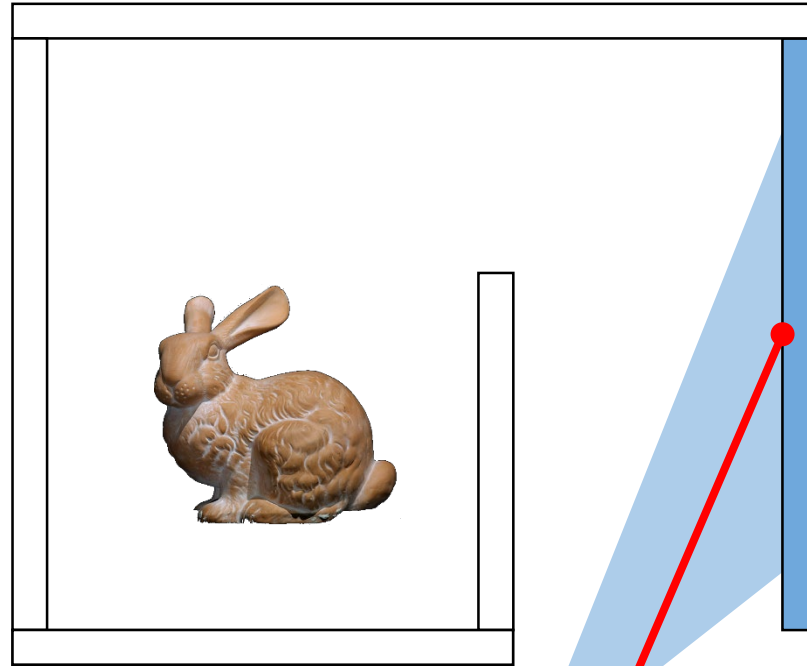
Applications: N*LOS, Beyond one corner

N*LOS: Looking around more than one corner?



Applications: N*LOS, Beyond one corner

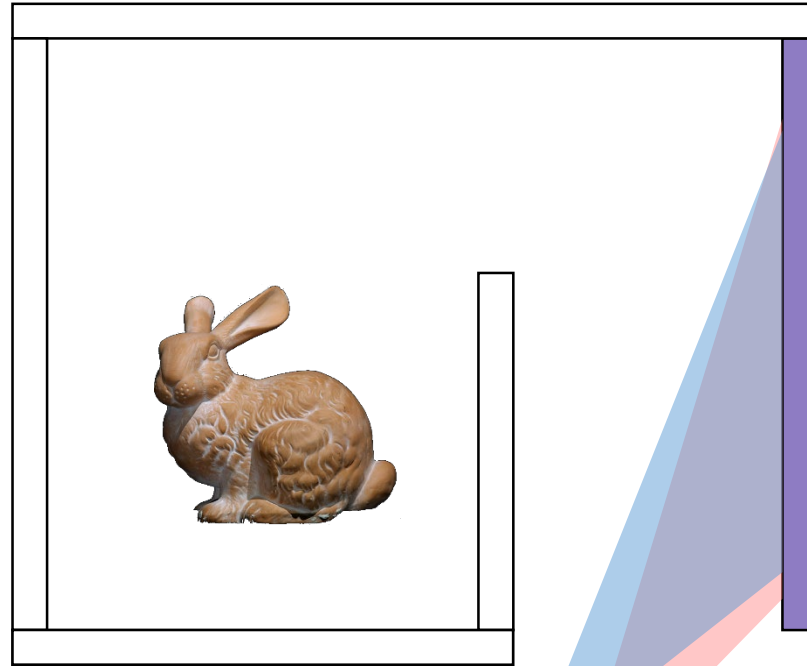
N*LOS: Looking around more than one corner?



REAL DEVICES

Applications: N*LOS, Beyond one corner

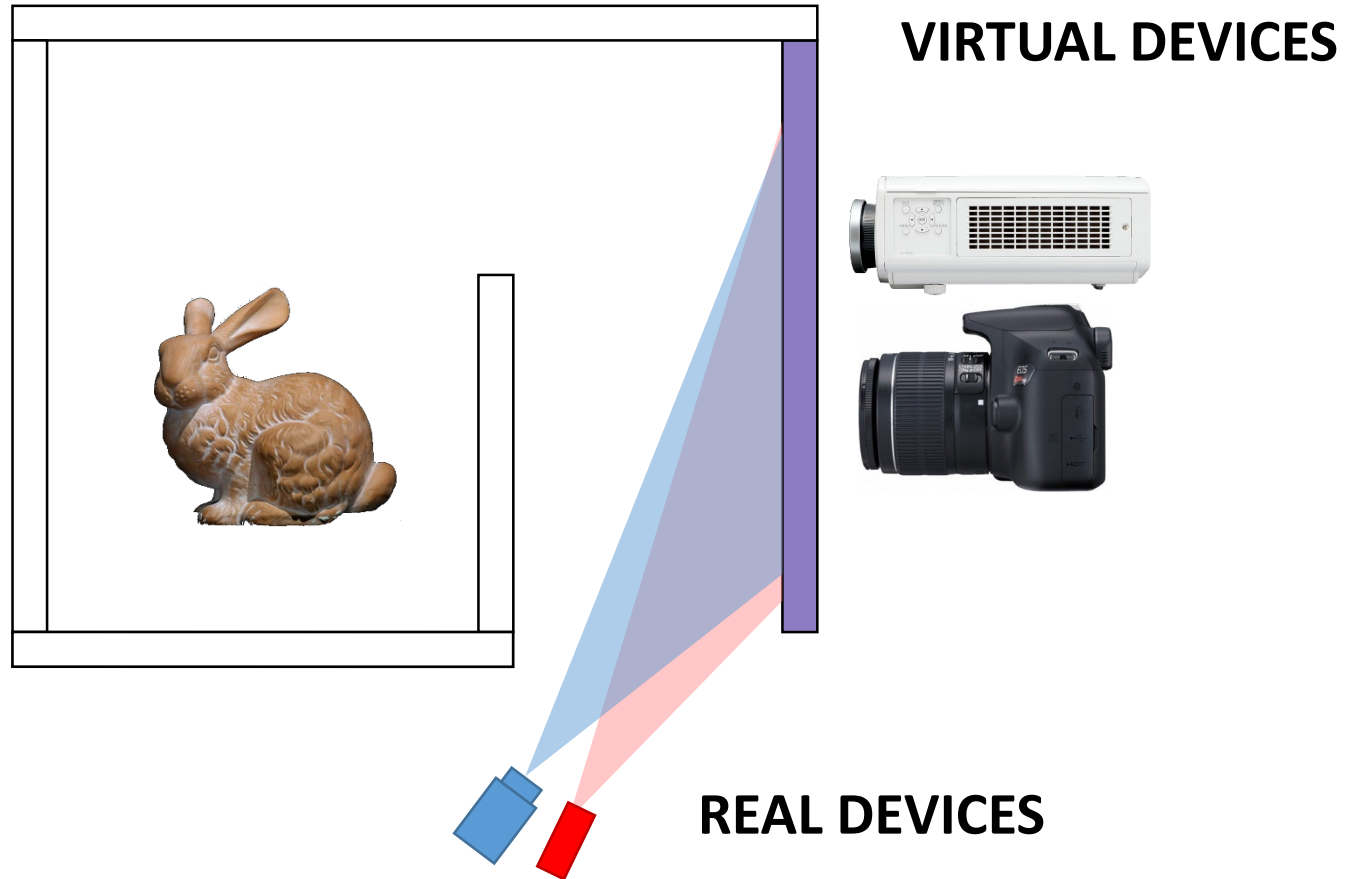
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REAL DEVICES

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Applications: N*LOS, Beyond one corner

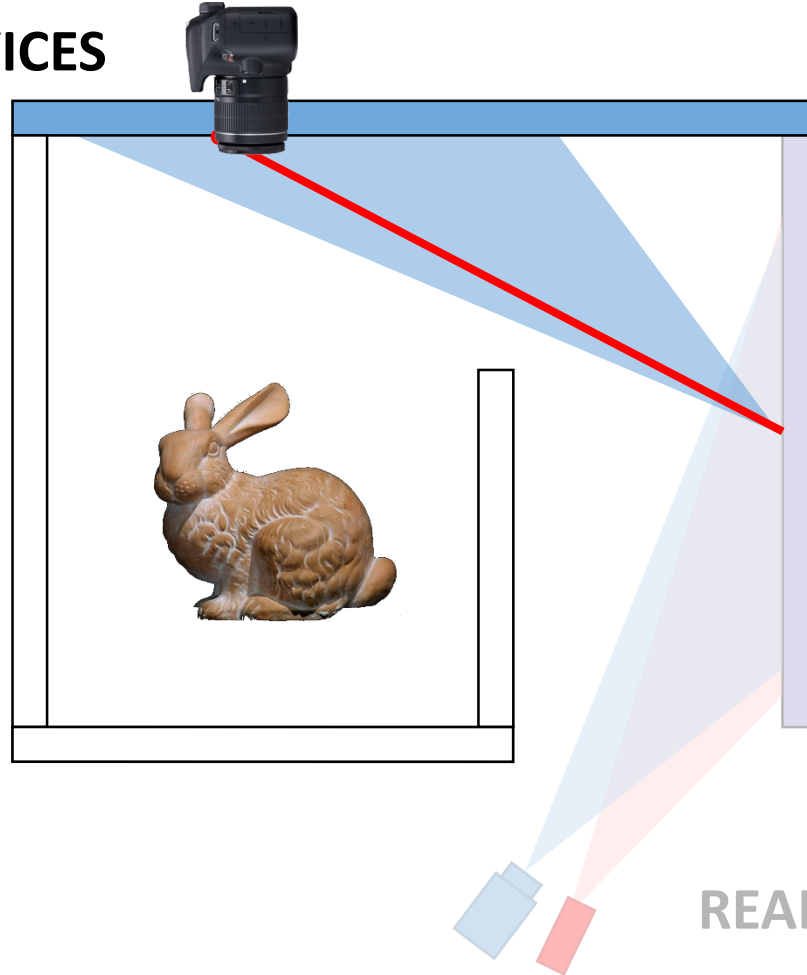
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Applications: N*LOS, Beyond one corner

N*LOS: Looking around more than one corner?

**VIRTUAL DEVICES
(2nd level)**



**VIRTUAL DEVICES
(1st level)**



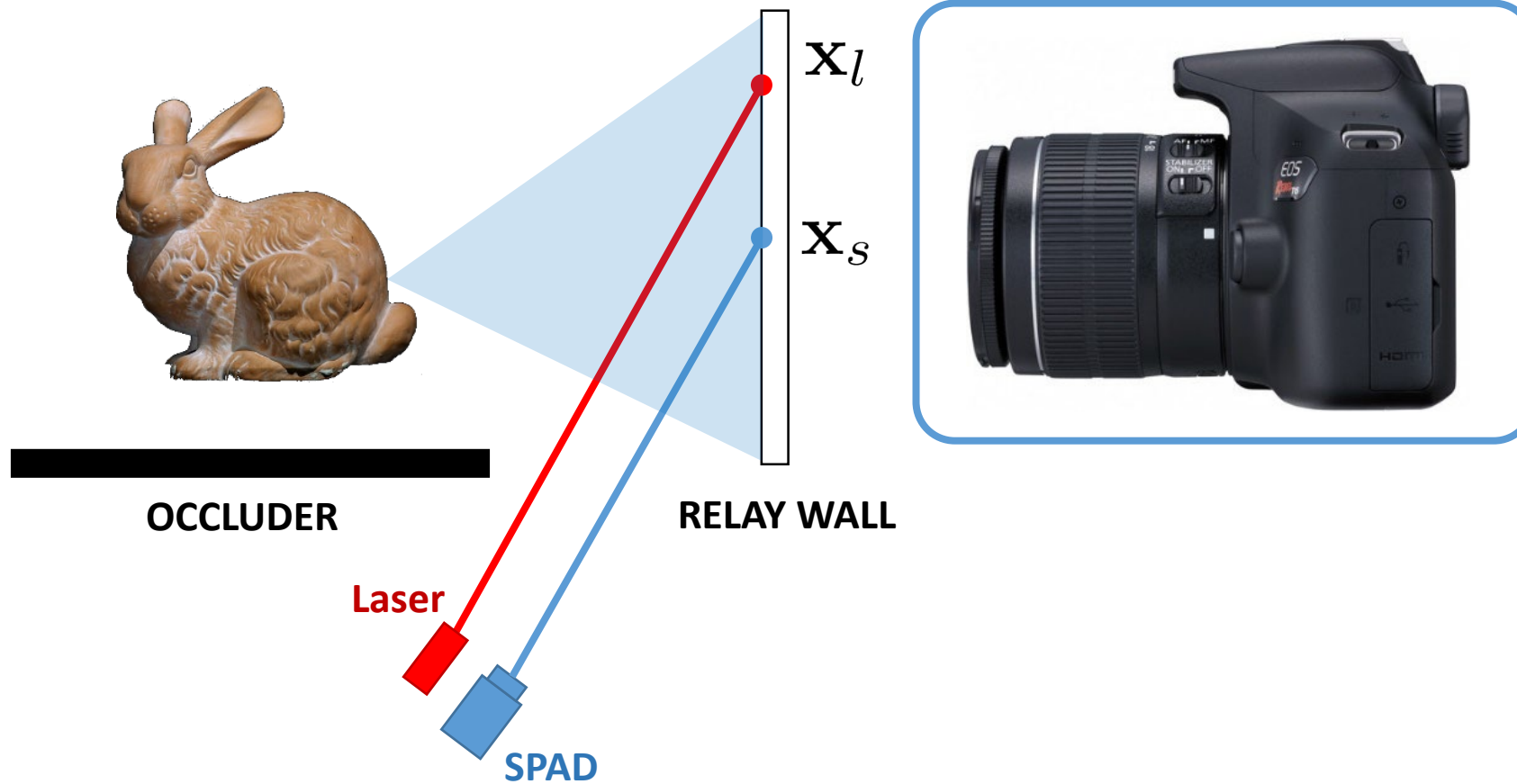
REAL DEVICES



NLOS imaging: Beyond phasor fields

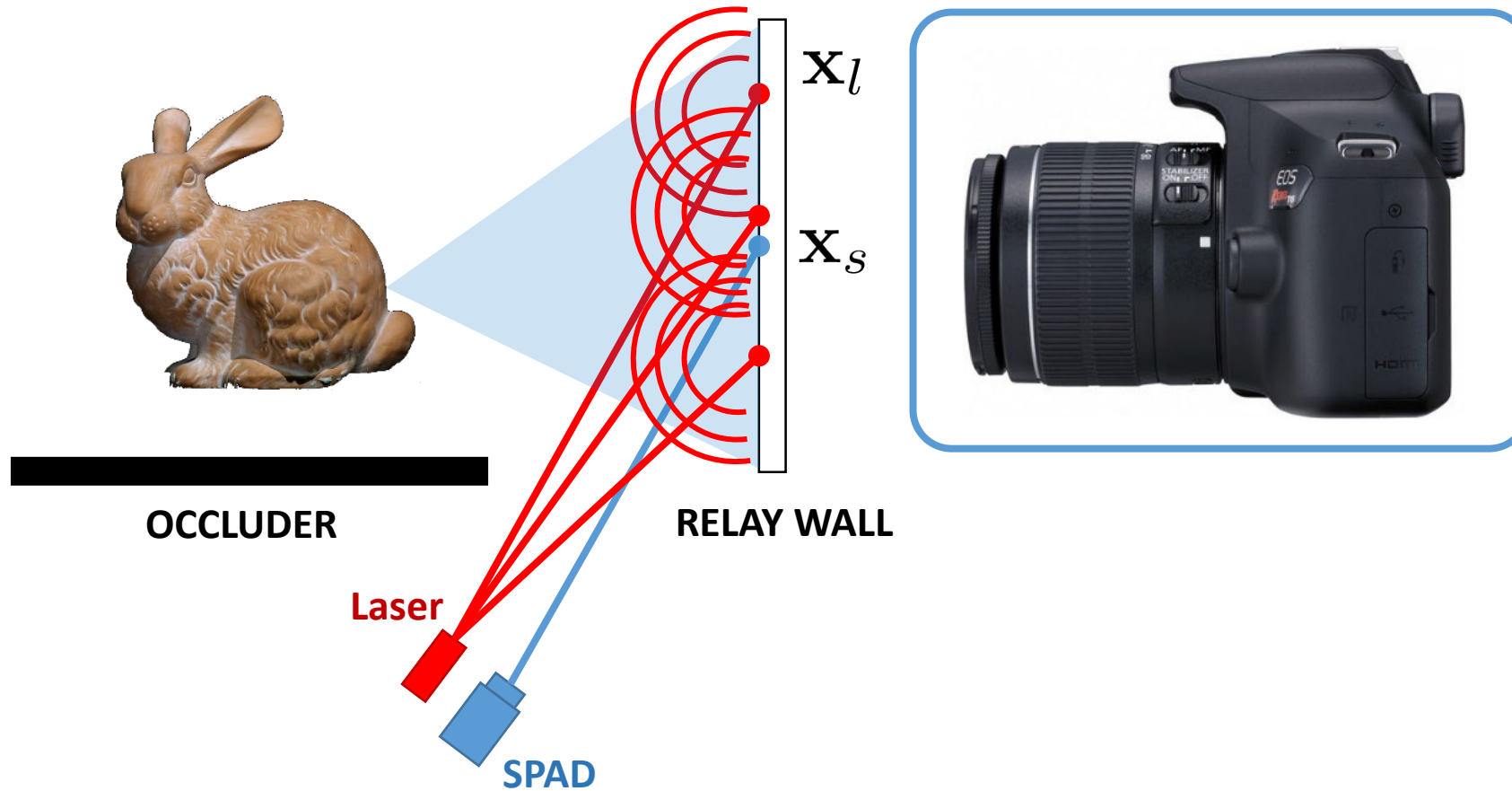
NLOS imaging: Virtual projector-camera setups

- We saw how to focus the image



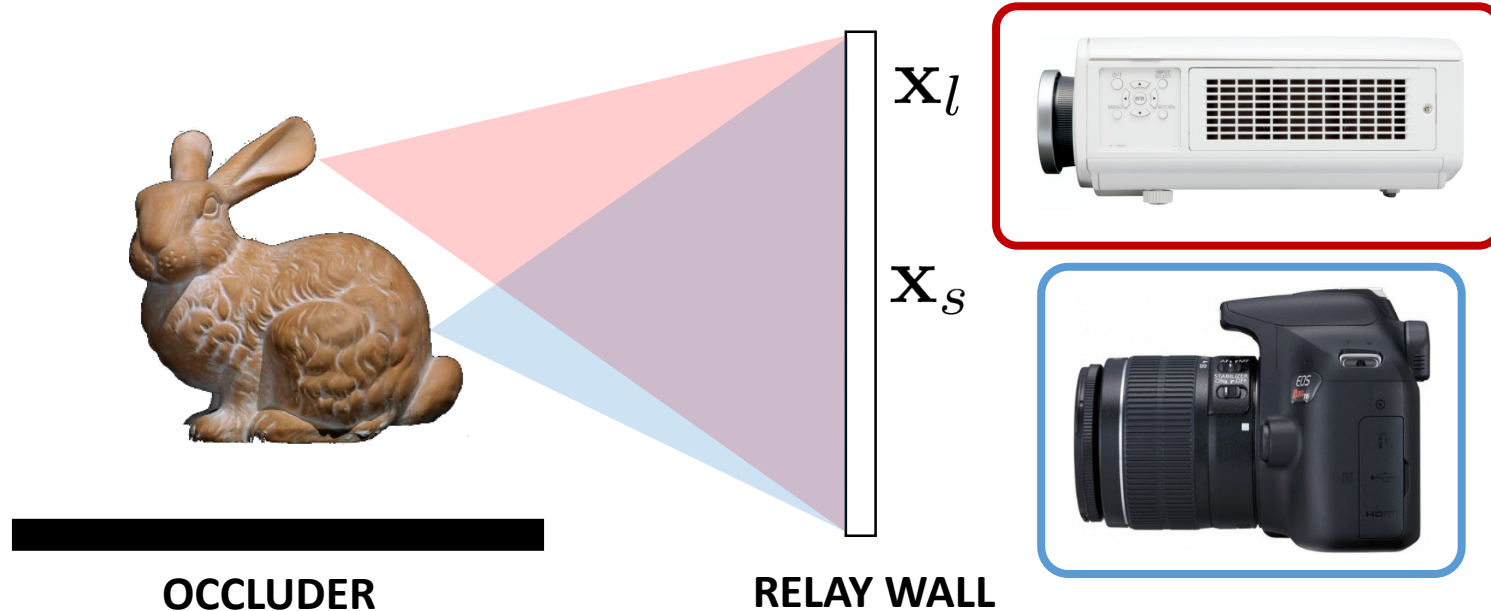
NLOS imaging: Virtual projector-camera setups

- We saw how to focus the image, **but light still scatters diffusely**



NLOS imaging: Virtual projector-camera setups

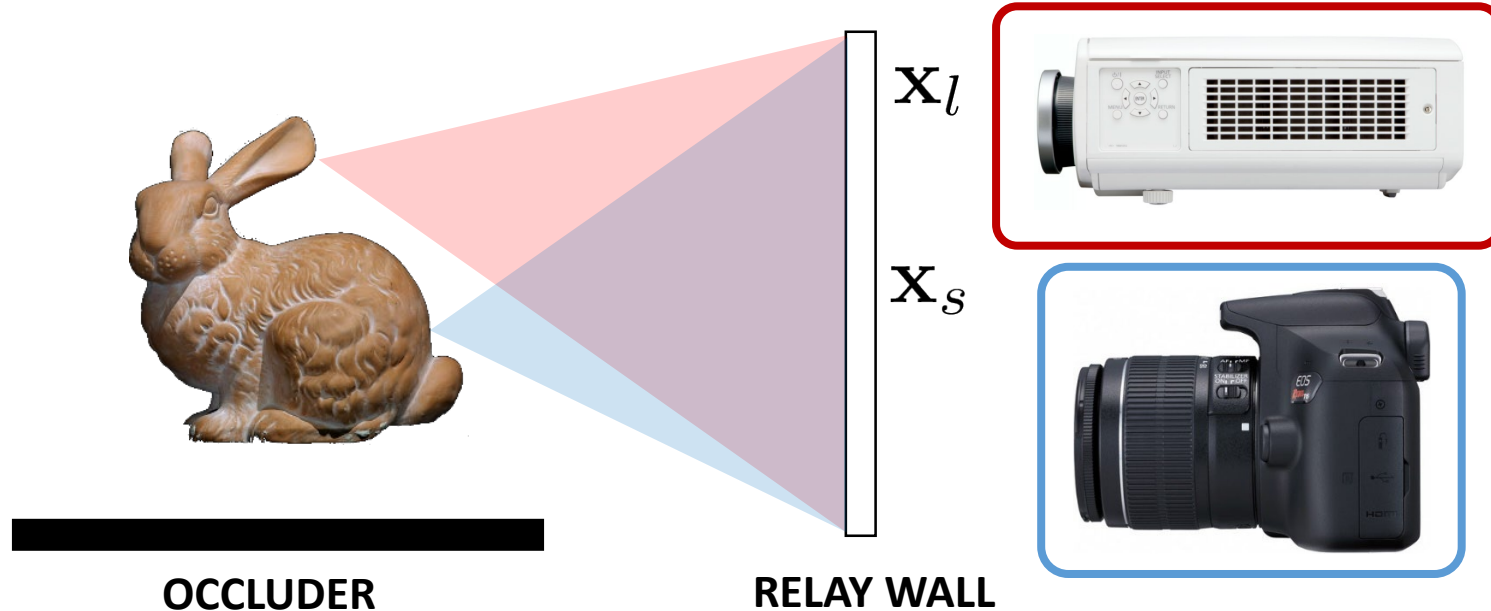
- What if we put not only a virtual camera, but also a virtual projector?



**Focus both light
and image onto
specific locations**

NLOS imaging: Virtual projector-camera setups

- What if we put not only a virtual camera, but also a virtual projector?



**Focus both light
and image onto
specific locations**

What for?

NLOS imaging: Virtual projector-camera setups

- What if we put not only a virtual camera, but also a virtual projector?

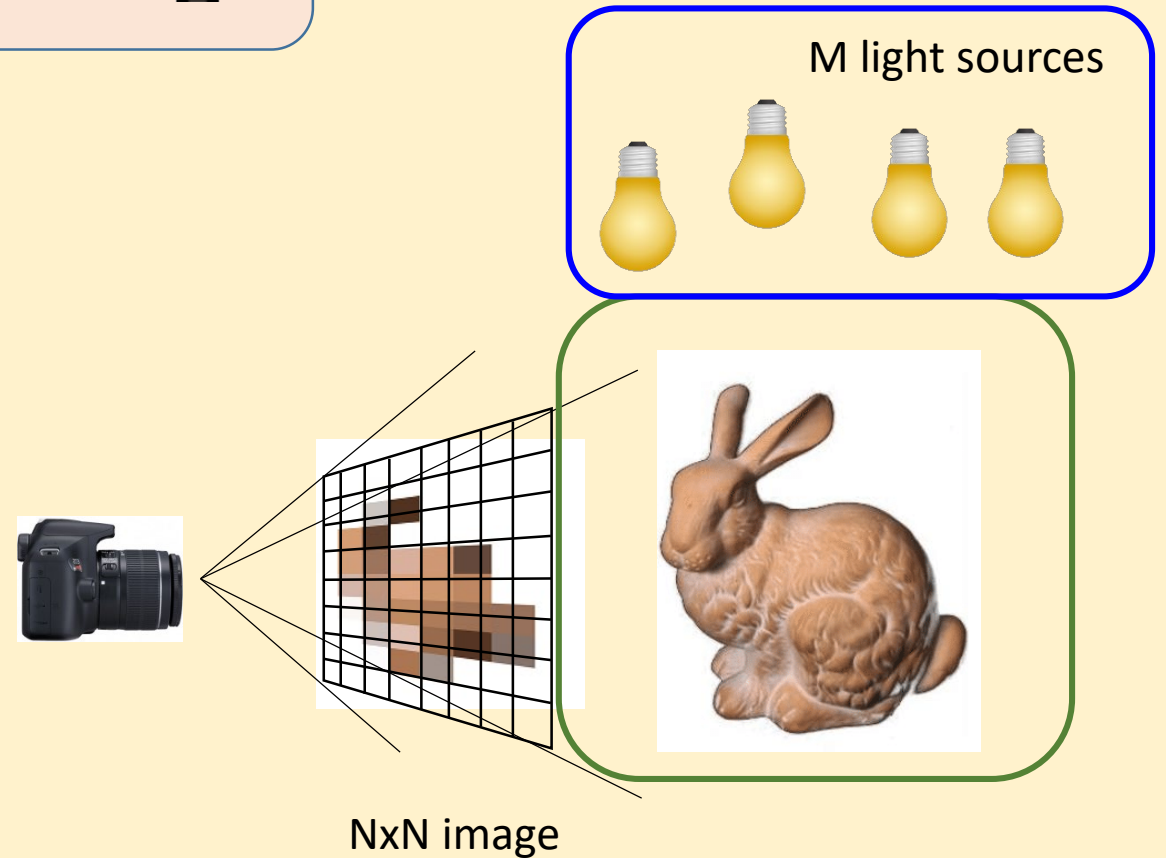
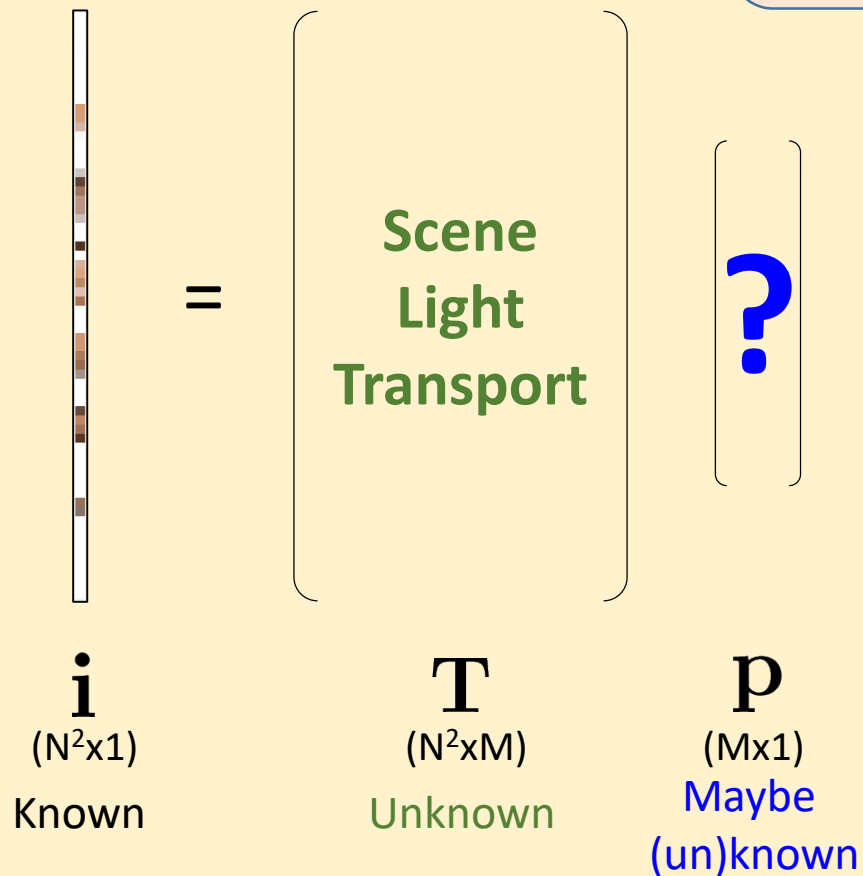


What for?

Light Transport Matrices
of the hidden scene!

Lecture 2 brush up – *The Light Transport Matrix*

$$\mathbf{i} = \mathbf{T}\mathbf{p}$$



Lecture 7 brush up – *LTM Applications*

1) Relighting

2) Descattering

If you can descatter, you can also obtain the info of the medium!

3) Making regions “invisible”

4) Direct-Indirect Separation

Question: Do we need the full LTM for direct-indirect separation?

Lecture 7 brush up – *LTM Applications*

1) Relighting

2) Descattering

If you can descatter, you can also obtain the

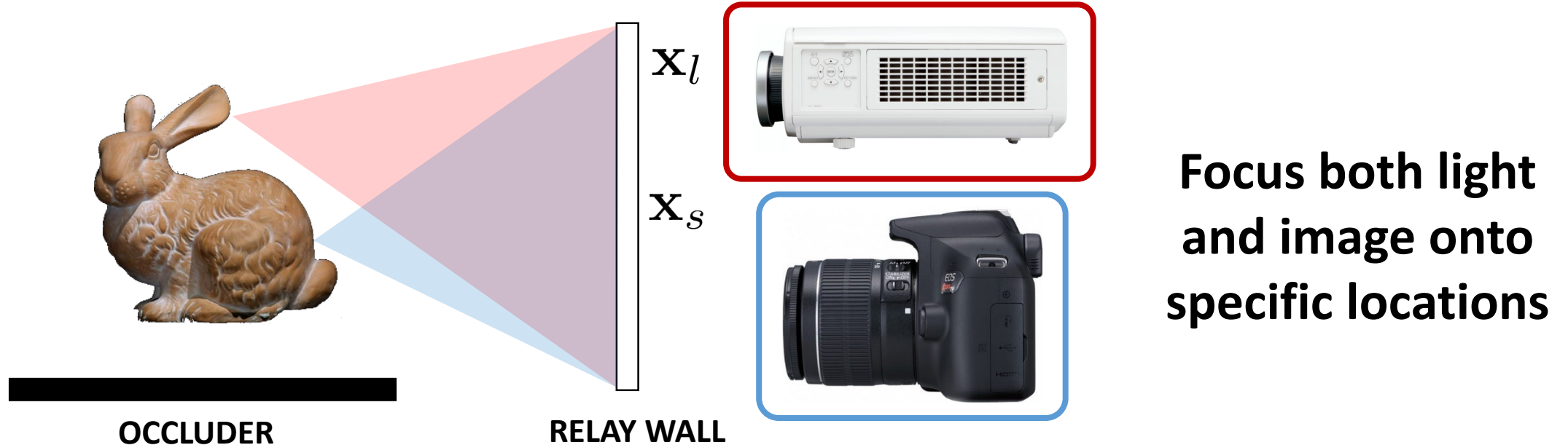
3) Making regions “invisible”
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NLOS imaging: Virtual projector-camera setups

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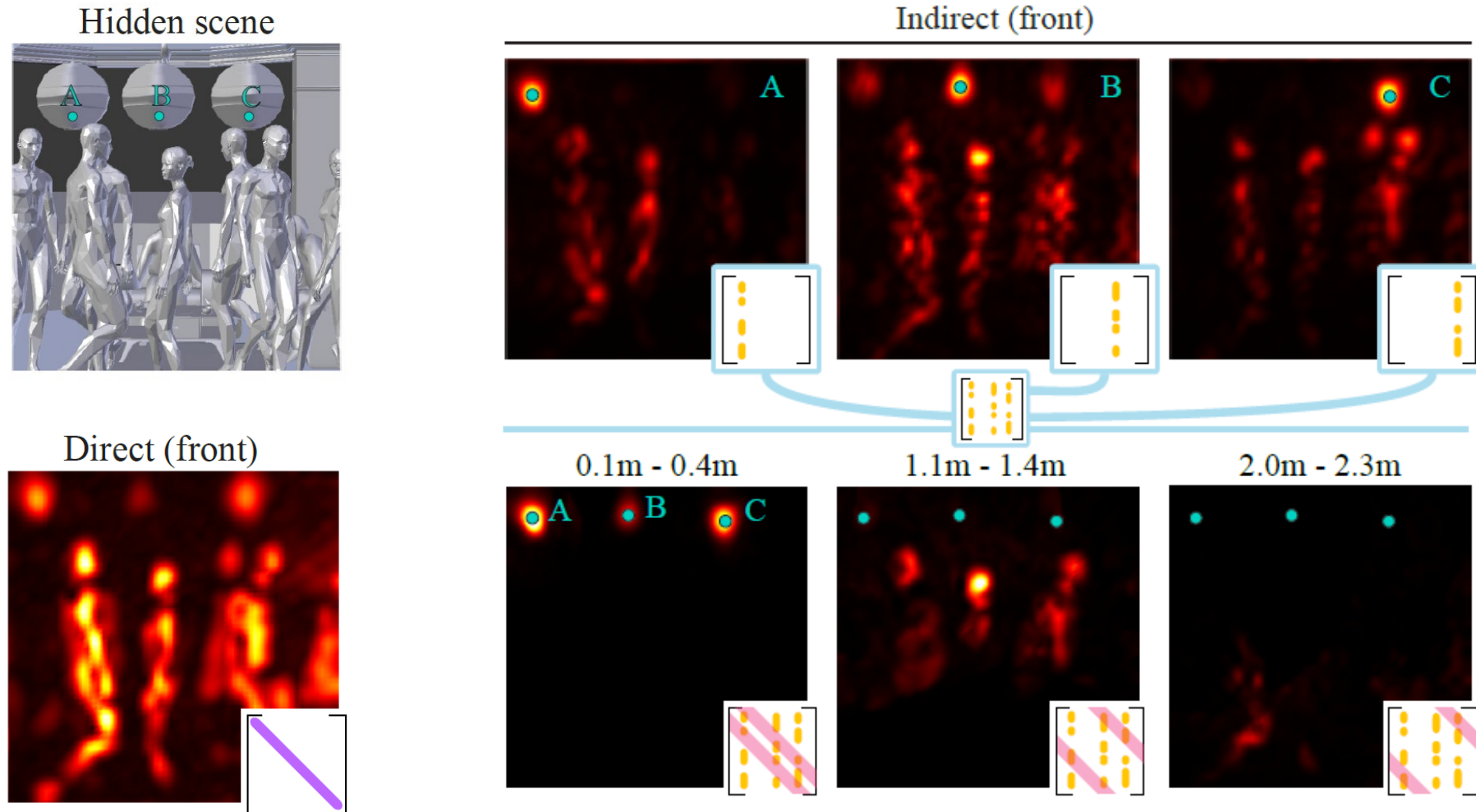


What for?

Hidden scene understanding
(e.g. relighting, direct-indirect separation)

NLOS imaging: Virtual projector-camera setups

- [\[Marco et al. 2017\]](#) Preprint on arxiv



References

- [\[Velten et al. 2012\]](#) Recovering three-dimensional shape around a corner using ultrafast time-of-flight imaging
- [\[Laurenzis and Velten 2014\]](#) Non-line-of-sight active imaging of scattered photons
- [\[Buttafava et al. 2015\]](#) Non-line-of-sight imaging using a time-gated single photon avalanche diode
- [\[Arellano et al. 2017\]](#) Fast Back-Projection for Non-Line of Sight Reconstruction
- [\[La Manna et al. 2018\]](#) Error Backprojection Algorithms for Non-Line-of-Sight Imaging
- [\[O'Toole et al. 2018\]](#) Confocal non-line-of-sight imaging based on the light-cone transform
- [\[Liu et al. 2019\]](#) Non-Line-of-Sight Imaging using Phasor Field Virtual Wave Optics
- [\[Lindell et al. 2019\]](#) Wave-Based Non-Line-of-Sight Imaging using Fast fk Migration
- [\[Xin et al. 2019\]](#) A Theory of Fermat Paths for Non-Line-of-Sight Shape Reconstruction
- [\[Ahn et al. 2019\]](#) Convolutional Approximations to the General Non-Line-of-Sight Imaging Operator
- [\[Tsai et al. 2019\]](#) The geometry of first-returning photons for non-line-of-sight imaging
- [\[Liu et al. 2020a\]](#) Analysis of Feature Visibility in Non-Line-of-Sight Measurements
- [\[Liu et al. 2020b\]](#) Phasor Field Diffraction Based Reconstruction for Fast Non-Line-of-Sight Imaging Systems
- [\[Guillén et al. 2020\]](#) On the Effect of Reflectance on Phasor Field Non-Line-of-Sight imaging
- [\[Young et al. 2020\]](#) Non-line-of-sight surface reconstruction using the directional light-cone transform
- [\[Iseringhausen et al. 2020\]](#) Non-Line-of-Sight Reconstruction using Efficient Transient Rendering

Questions